

Native Completed-Weil Geometry and a Five-Dimensional Endpoint Criterion for the Riemann Hypothesis

Prime-Torus Arithmetic, Bilateral Recognition, and Single-Sector Seam Inertia

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Abstract

We construct a native multiplicative realization of the completed Weil form and place its arithmetic, spectral and topological reductions on one completed carrier. Inversion and Euler-scale flow produce a bounded self-adjoint completed-Weil representative. Parity and Feshbach reduction give compact relative packets whose strict threshold multiplicity is a finite seam integer, identified with one-sided spectral flow, Morse index, threshold spectral-zeta values and a canonical threshold holonomy. A bilateral completion theorem retains visible, blind and cross-seam information. Independently, the arithmetic Bohr transform intertwines the logarithmic clock on $\ell^2(\mathbb{N}_{>0})$ with a weighted first-order generator on the Hardy space of the prime torus, carries multiplication to monomial shifts and rules out a scalar single-circle logarithmic model. The prime-torus compression and the native Feshbach compression are unitary representations of the same rank-at-most-five Birman–Schwinger matrix.

The eta-normalization family is exactly path independent. An outward Arb certificate proves nonnegativity of the unshifted common two-sheet pure source and the floor $S_{\eta,\pm} \geq \eta I$ in both parities. The even boundary augmentation is positive, so the even completed restriction is nonnegative and contributes no adverse seam charge. At $\eta = 0.01$ the remaining odd five-source packet satisfies

$$\lambda_{\max}(B_{0.01}) \leq 0.8388901637561339 < 1,$$

with reserve 0.1611098362438661. Exact Schur certificates prove $A_3 > 0$, eliminating an eta-zero accepted kernel; the apparent finite critical wall is a cutoff shadow rather than a native endpoint. A weighted X_3 first-moment estimate gives $\text{Ran } \mathcal{W}_\Sigma \subset \text{Dom}(S_{0,-}^{-1/2})$, and the odd five-label resolvent and gap matrices converge in norm at eta zero.

The bilateral carrier is representation complete while the terminal sign decision is single-sector. The global seam integer equals the odd five-label threshold count. Under the classical completed explicit-formula identification and Weil criterion, we prove the exact equivalence

$$\boxed{\text{RH} \iff \mathcal{K}_0 = \text{diag}(1, A_3) - (S_{0,-}^{-1/2} \mathcal{W}_\Sigma)^* (S_{0,-}^{-1/2} \mathcal{W}_\Sigma) \geq 0.}$$

The endpoint admits six independent certificate formulations: a direct outward Hermitian enclosure, a native source-readout projection defect, prime–Gamma parallel-reserve domination, a structured single-node square-root transfer, a rank-one seam-jet Schur criterion and, before accepted-space deflation, the full-odd boundary scalar

$$\beta_\partial = \left\| (S_{0,-}^{\text{full}})^{\dagger/2} L_\partial^* 1 \right\|^2, \quad W_3^- \geq 0 \iff \beta_\partial \leq 1,$$

under inverse-support regularity. The square-root theorem isolates a rank-one $O(\sqrt{\eta})$ seam correction and $O(\eta)$ regular and far remainders, reducing endpoint promotion to one outward 5×5 matrix. A synchronized six-node positive-shift packet is subcritical and approaches unit

saturation with reserve approximately linear in η ; the saturation is retained as numerical endpoint evidence only. These are alternative certificates for one condition, not cumulative terminal gates. The exact cut–joint exclusion and the finite native mixed-response intertwiner are also certified; their final use at the endpoint requires the distinct continuum placement of the actual obstruction in both recognition classes. The resulting framework completes the carrier, arithmetic transport, blindness removal, parity reduction, fixed-shift certification and eta-zero passage, and identifies the Riemann hypothesis with one five-dimensional Hermitian sign criterion.

Keywords. Riemann hypothesis; Weil criterion; completed-Weil operator; recognition-seam calculus; arithmetic Bohr transform; prime torus; target-faithful completion; compact response operator; integer seam charge; threshold holonomy; Birman–Schwinger inertia; five-source certificate; square-root endpoint transfer; boundary scalar; cut–joint memory; exact eta analyticity; Schur complement; Mellin transform; Feshbach reduction.

Part I

Cut-Derived Recognition-Seam Calculus

1 Introduction

This paper constructs a representation-complete operator geometry for the completed Weil form and identifies its sign with one Hermitian condition on a five-label event carrier. The multiplicative carrier, inversion cut, Euler-scale flow, boundary–Gamma–prime energy and target-faithful completion are fixed before the Mellin–Fourier and prime-torus charts are introduced. Those charts are therefore representations of an already defined native object, not sources of the object itself.

1.1 Principal achievements

Theorem 1.1 (Representation-complete completed-Weil reduction). *The native construction establishes the following theorem chain.*

- (i) *Inversion and Euler-scale flow define a multiplicative completed carrier with a bounded self-adjoint completed-Weil representative $W_\times$.*
- (ii) *The arithmetic Bohr transform is a unitary intertwiner from the logarithmic clock on $\ell^2(\mathbb{N}_{>0})$ to the weighted prime-torus generator. Multiplication is carried to monomial shifts, and a scalar single-circle logarithmic model is excluded.*
- (iii) *Bilateral completion retains both parity sectors, their cross seam and every target-relevant blind direction. The compact relative packet $\mathbb{K}_\eta$ defines the integer*

$$\mathfrak{k}_\Sigma(\eta) = \text{rank } \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta),$$

whose one-sided spectral-flow, Morse, threshold-zeta and threshold-holonomy representations are exact.

- (iv) *The prime-torus compression and native Feshbach compression are faithful unitary avatars of the same rank-at-most-five Birman–Schwinger matrix.*
- (v) *The eta-normalization family is exactly path independent. The passing unshifted Arb certificate gives*

$$S_{0,+} \geq 0, \quad S_{0,-} \geq 0, \quad S_{\eta,\pm} \geq \eta I \quad (\eta > 0).$$

The even boundary augmentation is positive, hence

$$W_3^+ \geq 0, \quad \mathfrak{k}_+(\eta) = 0,$$

and therefore

$$\boxed{\mathfrak{k}_\Sigma(\eta) = \mathfrak{k}_3(\eta) = N_+(B_\eta - I).}$$

(vi) At the closed native node,

$$\lambda_{\max}(B_{0.01}) \leq 0.8388901637561339 < 1,$$

with reserve 0.1611098362438661.

(vii) Exact Schur certificates give

$$A_3 > 0, \quad \ker A_3 = \{0\}.$$

The weighted X_3 first-moment theorem gives

$$\text{Ran } \mathcal{W}_\Sigma \subset \text{Dom}(S_{0,-}^{-1/2}),$$

and the five-label resolvent and gap converge in operator norm at eta zero.

(viii) The endpoint resolvent difference has an explicit structured transfer. Near the quadratic seam, its singular part is rank at most one and of order $\sqrt{\eta}$, while the regular Taylor and far-source contributions are of order η . One outward shifted 5×5 packet therefore yields a matrix $\mathcal{L}_0$ satisfying

$$\mathcal{K}_0 \geq \mathcal{L}_0.$$

Hence

$$\boxed{\mathcal{L}_0 \geq 0 \implies \mathcal{K}_0 \geq 0.}$$

(ix) Before accepted-space deflation, reflection oddness reduces the completion-boundary contribution to one negative rank-one form. If

$$L_\partial f = \sqrt{2} \hat{f}(i/2)$$

and $S_{0,-}^{\text{full}}$ is the full odd Gamma-prime source before the boundary subtraction, then

$$W_3^- = S_{0,-}^{\text{full}} - L_\partial^* L_\partial.$$

Under full-boundary inverse-support regularity,

$$\boxed{W_3^- \geq 0 \iff \beta_\partial := \left\| (S_{0,-}^{\text{full}})^\dagger / 2 L_\partial^* 1 \right\|^2 \leq 1.}$$

(x) Under the classical completed explicit-formula identification and Weil criterion,

$$\boxed{\text{RH} \iff \mathcal{K}_0 \geq 0,}$$

where

$$\mathcal{K}_0 = \text{diag}(1, A_3) - (S_{0,-}^{-1/2} \mathcal{W}_\Sigma)^* (S_{0,-}^{-1/2} \mathcal{W}_\Sigma).$$

Proof. The assertions are proved respectively by the native multiplicative carrier, prime-torus intertwiner, bilateral completion, compact threshold and inertia theorems, common-source Arb certificate, fixed-shift outward enclosure, accepted Schur certificate, inverse-half regularity and endpoint descent, the square-root transfer theorem of [Theorems 33.1 and 33.2](#), the full-odd boundary scalar reduction of [Theorem 32.6](#), and the exact completed-Weil criterion of [Theorems 31.5 and 31.7](#). $\square$

1.2 One endpoint, several certificates

The terminal object is the single matrix $\mathcal{K}_0$. The paper derives six independent certificate formulations:

- (a) a direct outward lower enclosure for $\lambda_{\min}(\mathcal{K}_0)$;
- (b) the native projection-defect identity

$$\mathcal{K}_0 = \mathcal{R}_\Sigma^*(I - P_{\text{src}})\mathcal{R}_\Sigma;$$

- (c) the prime–Gamma reserve domination

$$\Delta_\Gamma \leq D_{\Gamma|p};$$

- (d) the structured single-node square-root inequality

$$\mathcal{L}_0 \geq 0;$$

- (e) the rank-one seam-jet Schur inequality

$$\tau J_\Sigma A_{\text{reg}}^{-1} J_\Sigma^* \leq I;$$

- (f) the pre-Feshbach full-odd boundary condition

$$\beta_\partial \leq 1.$$

Each formulation closes the same endpoint. They are independent proof routes, not cumulative obligations. The state-locked packets calibrate the shifted matrix, seam row and boundary saturation. They acquire proof status only with the outward error data required by the corresponding certificate.

1.3 Scope and method

The paper distinguishes exact equivalences, sufficient certificates and numerical calibration. The framework, carrier and reduction are complete; the Riemann hypothesis is identified with the five-dimensional endpoint sign. Finite packets are used only with their declared outward error ledger, and no fitted response matrix, abstract dilation or extrapolated threshold vector is accepted as arithmetic provenance. The structural hierarchy and the excluded cut–joint inference are recorded explicitly in [Section 33](#).

2 Primitive cut-recognition quotient

This section records only the set-theoretic data later consumed by the sheet, dagger and completed-carrier constructions: recognition classes, seams and the involution induced by a recognition-compatible cut.

Definition 2.1 (Primitive cut-recognition datum). A primitive cut-recognition datum is a tuple

$$\mathfrak{G} = (X, \kappa, \chi, \mathcal{C}),$$

where X is the appearance space, $\kappa : X \rightarrow X$ is the distinguished first cut, $\chi : X \rightarrow X$ is idempotent,

$$\chi^2 = \chi,$$

and $\mathcal{C}$ is the declared family of admissible cuts containing Id_X and κ .

Definition 2.2 (Recognition quotient). For $x, y \in X$, set

$$x \sim_R y \iff \chi(x) = \chi(y),$$

and let

$$E : X \longrightarrow X/R, \quad E(x) = [x]_R,$$

be the quotient map.

Proposition 2.3 (Canonical recognized representative). *The relation $\sim_R$ is an equivalence relation, and*

$$[x]_R \longmapsto \chi(x)$$

defines a canonical bijection from X/R to $\text{Ran } \chi$.

Proof. Equality of recognized images is an equivalence relation. Idempotence gives $\chi(\chi(x)) = \chi(x)$, so each class has the canonical representative $\chi(x)$, and two such representatives agree exactly when the classes agree. $\square$

Definition 2.4 (Recognition seam). For $C \in \mathcal{C}$, define

$$S_C = \{x \in X : \chi(Cx) = \chi(x)\}.$$

Thus the seam is the equalizer of $\chi \circ C$ and χ : applying the cut does not change the recognized state there.

Definition 2.5 (Recognition-compatible involutive cut). A cut $C : X \rightarrow X$ is recognition compatible if

$$x \sim_R y \implies Cx \sim_R Cy,$$

and involutive modulo recognition if

$$C^2x \sim_R x \quad (x \in X).$$

Theorem 2.6 (Derived quotient involution). *If C is recognition compatible and involutive modulo recognition, then*

$$\overline{C} : X/R \longrightarrow X/R, \quad \overline{C}([x]_R) = [Cx]_R,$$

is a well-defined involution.

Proof. Recognition compatibility makes the definition independent of the chosen representative. Moreover,

$$\overline{C}^2([x]_R) = [C^2x]_R = [x]_R.$$

$\square$

Corollary 2.7 (Canonical Hilbert cut). *On*

$$\mathcal{H}_R = \ell^2(X/R),$$

define

$$J_C e_{[x]} = e_{\overline{C}([x])}.$$

Then J_C is a unitary self-adjoint involution,

$$J_C^2 = I, \quad J_C^* = J_C,$$

with spectral projections

$$P_{\pm} = \frac{I \pm J_C}{2}.$$

Proof. The quotient involution permutes the canonical orthonormal basis and equals its own inverse. $\square$

A concrete RH realization must still prove that its represented cut factors through this quotient construction and that its completed target is faithful. No fixed-locus identification with the critical line is installed by definition.

3 Sheet rotation and target-faithful comparison

The recognition quotient records classes, but the operative geometry keeps the two local sheets labelled and compares them by reversible transport. Agreement after transport defines the seam; disagreement remains available to any target that depends on sheet origin or crossing history.

3.1 Two-sheet transport and rotational seam

Let X_+ and X_- be labelled copies of a native state space and set

$$\tilde{X} = X_+ \sqcup X_-.$$

Let

$$E_+ : X_+ \rightarrow Y, \quad E_- : X_- \rightarrow Y$$

be recognition maps into a common target.

Definition 3.1 (Sheet transport). A sheet transport is a bijection

$$U : X_+ \longrightarrow X_-$$

with inverse $U^{-1} : X_- \rightarrow X_+$.

Definition 3.2 (Rotational seam). The seam of U is

$$S_{\text{rot}}(U) = \{x_+ \in X_+ : E_-(Ux_+) = E_+(x_+)\}.$$

It is the equalizer of E_+ and $E_- \circ U$.

When Y is metric, the corresponding residue is

$$r_{\text{rot}}(x_+; U) = d_Y(E_-(Ux_+), E_+(x_+)),$$

whose zero set is the rotational seam.

3.2 Rotation versus gluing

A gluing map $g : X_+ \rightarrow X_-$ produces the quotient

$$X_g = (X_+ \sqcup X_-) / \sim_g, \quad x_+ \sim_g g(x_+).$$

The quotient replaces a labelled pair by one class. Sheet rotation retains the pair (x_+, Ux_+) and its inverse transport.

Theorem 3.3 (Target factorization through a quotient). *Let $q : X \rightarrow Q$ be surjective and let $\Pi : X \rightarrow Z$ be a target. The following are equivalent:*

- (i) there is a map $\widehat{\Pi} : Q \rightarrow Z$ with $\Pi = \widehat{\Pi} \circ q$;
 - (ii) Π is constant on every fibre of q ;
 - (iii) $q(x) = q(y)$ implies $\Pi(x) = \Pi(y)$.
- When it exists, $\widehat{\Pi}$ is unique.

Proof. Factorization makes equal quotient images have equal target values. Conversely, if Π is fibre-constant, define $\widehat{\Pi}(q(x)) = \Pi(x)$. Surjectivity gives uniqueness. $\square$

Thus gluing is lawful for targets constant on glued fibres and destructive for targets that distinguish sheet origin. A binary sheet label is the minimal repair on a two-point fibre.

3.3 Hilbert graph and anti-graph sectors

Suppose the sheets are represented by Hilbert spaces $\mathcal{H}_+$ and $\mathcal{H}_-$ and

$$U : \mathcal{H}_+ \longrightarrow \mathcal{H}_-$$

is unitary. On $\mathcal{H}_+ \oplus \mathcal{H}_-$ define

$$J_U = \begin{pmatrix} 0 & U^* \\ U & 0 \end{pmatrix}.$$

Proposition 3.4 (Graph and anti-graph decomposition). *The operator J_U is a self-adjoint involution. Its eigenspaces are*

$$\begin{aligned} \text{Ran } P_+ &= \left\{ \frac{1}{\sqrt{2}}(u, Uu) : u \in \mathcal{H}_+ \right\}, \\ \text{Ran } P_- &= \left\{ \frac{1}{\sqrt{2}}(u, -Uu) : u \in \mathcal{H}_+ \right\}, \quad P_{\pm} = \frac{I \pm J_U}{2}. \end{aligned}$$

Proof. Unitarity gives $U^* = U^{-1}$, hence $J_U^* = J_U$ and $J_U^2 = I$. Direct substitution gives the two eigenspaces. $\square$

The graph sector becomes a recognized seam sector only after

$$\mathbf{E}_-(Uu) = \mathbf{E}_+(u)$$

is proved on the declared domain. Operator matching and recognition agreement are therefore distinct, and the later bilateral completion theorem must retain both sectors and their cross terms on the completed target.

4 Finite cut and dagger calibration

The circle calibration in the next section consumes one finite fibre algebra. This section records exactly that algebra and distinguishes the Hilbert adjoint from the fundamental-symmetry dagger.

4.1 Two-class cut representation

Assume the recognized quotient has two classes exchanged by the derived involution. On $\mathbb{C}^2$ define

$$R_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad K = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then

$$R_0^2 = -I, \quad K^2 = I, \quad R_0 K = -K R_0.$$

The mixed cut channel is

$$J_{\text{cut}} = R_0 K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Proposition 4.1 (Cut involution). *The operator J_{cut} is a self-adjoint involution,*

$$J_{\text{cut}}^* = J_{\text{cut}}, \quad J_{\text{cut}}^2 = I,$$

and

$$R_0 J_{\text{cut}} = -J_{\text{cut}} R_0.$$

Proof. The identities follow by direct multiplication from the displayed relations for R_0 and K . $\square$

Its spectral projections are

$$P_{\pm} = \frac{I \pm J_{\text{cut}}}{2}.$$

The names seam and anti-seam are representation relative: a concrete application must prove that this split is induced by its recognition seam.

4.2 Fundamental-symmetry dagger

The matrix K defines the indefinite Hermitian form

$$[u, v]_K = \langle Ku, v \rangle.$$

For $A \in M_2(\mathbb{C})$ define

$$A^{\sharp} = K A^* K.$$

Theorem 4.2 (Dagger algebra). *For all $A, B \in M_2(\mathbb{C})$,*

$$(A^{\sharp})^{\sharp} = A, \quad (AB)^{\sharp} = B^{\sharp} A^{\sharp},$$

and $\sharp$ is conjugate linear. Thus it is an involutive anti-automorphism.

Proof. Since $K^* = K$ and $K^2 = I$,

$$(A^{\sharp})^{\sharp} = K(K A^* K)^* K = A,$$

and the product and scalar rules follow from the Hilbert adjoint. $\square$

Proposition 4.3 (Dagger versus Hilbert adjoint). *The rotation generator satisfies*

$$R_0^* = -R_0, \quad R_0^{\sharp} = R_0.$$

Hence dagger self-adjointness is not ordinary Hilbert self-adjointness.

Proof. The first identity follows from skew symmetry. Anticommutation gives $K R_0 K = -R_0$, hence

$$R_0^{\sharp} = K R_0^* K = R_0.$$

$\square$

4.3 Block preservation

Let J be any self-adjoint involution on a Hilbert space, with $P_{\pm} = (I \pm J)/2$. For a bounded operator A , the following are equivalent:

$$[A, J] = 0, \quad P_-AP_+ = P_+AP_- = 0, \quad A \text{ is block diagonal in } \text{Ran } P_+ \oplus \text{Ran } P_-.$$

This elementary criterion is the finite calibration for the parity and leakage statements used later. It does not by itself identify either block with an RH obstruction.

Part II

Exact Index and Carrier Calibrations

5 Circle spectral flow, winding and Fredholm index

The finite dagger algebra becomes analytically meaningful only after a densely defined operator with a declared domain is supplied. The circle family provides an exact unbounded calibration in which topology, spectrum and Fredholm index carry the same integer. It is a calibration of the index convention, not an arithmetic representation theorem.

5.1 The shifted circle operator

Let

$$\mathcal{H} = L^2(S^1), \quad \text{Dom}(D_a) = H^1(S^1), \quad a \in \mathbb{R},$$

with periodic boundary conditions, and define

$$D_a = -i \frac{d}{d\theta} + a.$$

For the normalized Fourier basis

$$e_n(\theta) = (2\pi)^{-1/2} e^{in\theta}, \quad n \in \mathbb{Z},$$

one has

$$D_a e_n = (n + a) e_n.$$

Theorem 5.1 (Self-adjoint circle carrier and compact resolvent). *For every real a , the operator D_a is densely defined, closed and self-adjoint. Its spectrum is*

$$\text{Spec}(D_a) = \{n + a : n \in \mathbb{Z}\},$$

with simple eigenvalues, and $(D_a - z)^{-1}$ is compact whenever $z \notin \text{Spec}(D_a)$.

Proof. The Fourier transform sends D_a to the maximal real multiplication operator

$$(M_a c)_n = (n + a) c_n$$

on

$$\left\{ c \in \ell^2(\mathbb{Z}) : \sum_{n \in \mathbb{Z}} (1 + n^2) |c_n|^2 < \infty \right\}.$$

This proves self-adjointness and the spectrum. The resolvent is diagonal with coefficients $(n + a - z)^{-1} \rightarrow 0$, so finite-coordinate truncations converge to it in operator norm. $\square$

5.2 Circle amplification of the finite cut and dagger

Let R_0 , K and $J_{\text{cut}} = R_0 K$ be the finite matrices of the preceding section. On

$$\tilde{\mathcal{H}} = L^2(S^1) \otimes \mathbb{C}^2$$

define

$$\tilde{D}_a = D_a \otimes I_2, \quad \text{Dom}(\tilde{D}_a) = H^1(S^1) \otimes \mathbb{C}^2,$$

and amplify the finite fibre operators by

$$\mathcal{R}_{S^1} = I \otimes R_0, \quad \mathcal{K}_{S^1} = I \otimes K, \quad \boxed{\mathcal{J}_{S^1} = I \otimes J_{\text{cut}}} = \mathcal{R}_{S^1} \mathcal{K}_{S^1}.$$

Thus the finite cut acts fibrewise on the doubled circle carrier; it does not act on the scalar space $L^2(S^1)$ before this amplification is declared.

Proposition 5.2 (Domain invariance and exact commutation). *The amplified cut is a bounded self-adjoint involution satisfying*

$$\mathcal{J}_{S^1} \text{Dom}(\tilde{D}_a) = \text{Dom}(\tilde{D}_a), \quad [\tilde{D}_a, \mathcal{J}_{S^1}] = 0$$

on $\text{Dom}(\tilde{D}_a)$. Moreover,

$$\tilde{D}_a^* = \tilde{D}_a, \quad \tilde{D}_a^\sharp = \tilde{D}_a, \quad A^\sharp = \mathcal{K}_{S^1} A^* \mathcal{K}_{S^1}.$$

Proof. All three amplified fibre operators are bounded and preserve $H^1(S^1) \otimes \mathbb{C}^2$. Since $\tilde{D}_a$ acts on the first tensor factor and $\mathcal{J}_{S^1}$, $\mathcal{K}_{S^1}$ act on the second, they commute on the declared domain. Ordinary self-adjointness follows from the tensor product with the finite identity, and the dagger identity follows from commutation with $\mathcal{K}_{S^1}$. $\square$

Let

$$u_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad u_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \tilde{P}_\pm = \frac{I \pm \mathcal{J}_{S^1}}{2}.$$

Define isometries

$$V_\pm : L^2(S^1) \longrightarrow \text{Ran } \tilde{P}_\pm, \quad V_\pm f = f \otimes \nu_\pm.$$

Theorem 5.3 (Sectorwise circle calibration). *Each cut sector reduces $\tilde{D}_a$ and*

$$V_\pm^* (\tilde{D}_a|_{\text{Ran } \tilde{P}_\pm}) V_\pm = D_a.$$

Consequently, for the path $a(t) = \delta + tq$,

$$\text{sf}(\tilde{D}_{a(t)}|_{\text{Ran } \tilde{P}_\pm}; 0) = q,$$

whereas the full doubled path has

$$\text{sf}(\tilde{D}_{a(t)}; 0) = 2q.$$

Proof. The vectors $\nu_\pm$ span the ± 1 eigenspaces of J_{cut} . Since $\tilde{D}_a = D_a \otimes I_2$, each sector is unitarily equivalent to the scalar circle carrier. Spectral flow is therefore q on each sector and additive under the orthogonal direct sum, giving $2q$ on the full doubled carrier. $\square$

Remark 5.4 (No arithmetic-to-circle intertwiner is consumed). No map

$$\Phi_N : \ell^2(\{1, \dots, N\}) \longrightarrow L^2(S^1)$$

or from the radical/prime-shell module to the circle carrier is asserted or used in the RH sign route. In particular, D_a is not an analytic continuation of the finite logarithmic clock H_N , and $\mathcal{J}_{S^1}$ is not identified with an arithmetic cut by analogy. The circle family fixes the normalization

$$\text{sf} = \text{Wind} = \text{ind}$$

for an explicit model. The arithmetic module enters the later native multiplicative completed-Weil construction through its own source and transfer theorems.

5.3 Endpoint gauge relation

For $q \in \mathbb{Z}$, let $U_q = M_{z^q}$ on $L^2(S^1)$. Then U_q is unitary and

$$\boxed{U_q^* D_\delta U_q = D_{\delta+q}.$$

Indeed, $U_q e_n = e_{n+q}$. Thus the endpoint shift and the winding loop belong to one gauge-covariant construction rather than merely producing equal integers by separate calculations.

5.4 Exact spectral flow

Fix

$$0 < \delta < 1, \quad q \in \mathbb{Z},$$

and define

$$D_t = D_{\delta+tq}, \quad 0 \leq t \leq 1.$$

The endpoints are invertible and the eigenvalue branches are

$$\lambda_n(t) = n + \delta + tq.$$

Theorem 5.5 (Integer spectral flow). *With upward crossings counted positively,*

$$\text{sf}(D_t; 0) = q.$$

Proof. If $q > 0$, exactly the modes $n = -q, -q+1, \dots, -1$ cross zero, each upward. If $q = -r < 0$, exactly the modes $n = 0, 1, \dots, r-1$ cross downward. The signed count is q in both cases. No finite truncation is used. $\square$

5.5 Winding representative

Let

$$u_q : S^1 \rightarrow U(1), \quad u_q(z) = z^q.$$

Then

$$\text{Wind}(\nu_q) = \frac{1}{2\pi i} \oint_{S^1} \nu_q^{-1} d\nu_q = q.$$

5.6 Toeplitz Fredholm realization

Let $H^2(S^1)$ be the Hardy space with basis $1, z, z^2, \dots$, and let Π_H denote the Hardy projection. Define

$$C_q = \Pi_H M_{z^{-q}} \Pi_H|_{H^2(S^1)}.$$

The notation Π_H is kept distinct from the seam projections $P_{\pm}$.

Theorem 5.6 (Toeplitz index). *For every $q \in \mathbb{Z}$, the operator C_q is Fredholm and*

$$\text{ind}(C_q) = q.$$

Proof. For $q \geq 0$,

$$C_q z^n = \begin{cases} z^{n-q}, & n \geq q, \\ 0, & 0 \leq n < q, \end{cases}$$

so $\dim \text{Ker } C_q = q$ and the cokernel is zero. For $q = -r < 0$, C_q is multiplication by z^r ; it is injective and its cokernel has basis $1, z, \dots, z^{r-1}$, so the index is $-r = q$. $\square$

Theorem 5.7 (Unified circle calibration). *For $0 < \delta < 1$ and $q \in \mathbb{Z}$,*

$$\boxed{\text{sf}(D_{\delta+q}; 0) = \text{Wind}(z^q) = \text{ind}(\Pi_H M_{z^{-q}} \Pi_H) = q.}$$

The endpoints also satisfy $U_q^ D_{\delta} U_q = D_{\delta+q}$.*

Proof. The spectral-flow, winding and Toeplitz calculations above give the same integer, and the endpoint gauge identity was proved directly on the Fourier basis. $\square$

Remark 5.8 (Relation to general index theory). The general equality between spectral flow and index belongs to classical index theory; see [APS75, Phi96]. The present model is retained because every statement can be verified explicitly from Fourier and Hardy-space bases.

Remark 5.9 (Separation of the calibration and arithmetic chains). The logical dependencies are

$$\text{finite cut algebra} \longrightarrow \mathcal{J}_{S^1} \longrightarrow \text{circle index calibration},$$

and separately

$$\text{finite arithmetic module} \longrightarrow \text{native source/transfer calculus} \longrightarrow \text{completed-Weil carrier}.$$

The two chains meet only through proved later representation theorems and the normalization of the integer obstruction; no undeclared operator map from the arithmetic module to D_a is used.

Remark 5.10 (Arithmetic nonpromotion). Nothing in this calibration identifies the integer q with a prime, a circle crossing with a zero of ξ , or the Toeplitz index with an RH obstruction. Those identifications require separate transfer, faithfulness and completion theorems.

6 Arithmetic logarithmic clock and the prime-torus intertwiner

The preceding circle family calibrates spectral flow, winding and Toeplitz index, but its affine lattice spectrum cannot carry the logarithmic arithmetic clock. The exact arithmetic replacement is not one circle. It is the product of one circle for each prime, with unique factorization supplying the Fourier multi-index.

6.1 No exact single-circle logarithmic intertwiner

Let

$$\mathcal{A} = \ell^2(\mathbb{N}_{>0}), \quad H_{\log} e_n = (\log n) e_n,$$

with domain

$$\text{Dom}(H_{\log}) = \left\{ c : \sum_{n \geq 1} (\log n)^2 |c_n|^2 < \infty \right\}.$$

Proposition 6.1 (Single-circle no-go). *For no real a is there an isometry*

$$\Phi : \text{span}\{e_1, e_2\} \longrightarrow L^2(S^1)$$

whose range lies in $H^1(S^1)$ and which satisfies

$$D_a \Phi = \Phi H_{\log}.$$

Consequently the arithmetic logarithmic clock does not intertwine exactly with any shifted scalar circle Dirac operator $D_a = -i d/d\theta + a$.

Proof. If such an isometry existed, the nonzero vectors Φe_1 and Φe_2 would be eigenvectors of D_a with eigenvalues 0 and $\log 2$. Since

$$\text{Spec}(D_a) = a + \mathbb{Z},$$

their difference would belong to $\mathbb{Z}$. This is impossible because

$$0 < \log 2 < 1.$$

□

Thus any exact arithmetic bridge must retain the incommensurable prime logarithms rather than force them onto one periodic frequency lattice.

6.2 The Hardy space of the prime torus

Let $\mathbb{P}$ denote the set of primes and let

$$\mathbb{T}^{\mathbb{P}} = \prod_{p \in \mathbb{P}} S^1$$

with product Haar probability measure. For a finitely supported multi-index

$$\alpha = (\alpha_p)_{p \in \mathbb{P}} \in \mathbb{N}^{(\mathbb{P})},$$

write

$$z^\alpha = \prod_{p \in \mathbb{P}} z_p^{\alpha_p}.$$

Let $H^2(\mathbb{T}^{\mathbb{P}})$ be the closed span of these monomials in $L^2(\mathbb{T}^{\mathbb{P}})$.

For $n \geq 1$, let

$$\nu(n) = (v_p(n))_{p \in \mathbb{P}}$$

be its prime-exponent vector. Unique factorization gives a bijection

$$n \longleftrightarrow \nu(n)$$

from $\mathbb{N}_{>0}$ to $\mathbb{N}^{(\mathbb{P})}$.

Definition 6.2 (Arithmetic Bohr transform). Define

$$\mathcal{B} : \mathcal{A} \longrightarrow H^2(\mathbb{T}^{\mathbb{P}}), \quad \mathcal{B}e_n = z^{\nu(n)}.$$

Theorem 6.3 (Prime-torus arithmetic intertwiner). *The map $\mathcal{B}$ is unitary. Define the weighted prime-torus generator by*

$$D_{\mathbb{P}} z^{\alpha} = \left(\sum_{p \in \mathbb{P}} \alpha_p \log p \right) z^{\alpha}$$

on the maximal domain

$$\text{Dom}(D_{\mathbb{P}}) = \left\{ f = \sum_{\alpha} c_{\alpha} z^{\alpha} : \sum_{\alpha} \left(\sum_p \alpha_p \log p \right)^2 |c_{\alpha}|^2 < \infty \right\}.$$

Then $D_{\mathbb{P}}$ is self-adjoint with compact resolvent and

$$\boxed{D_{\mathbb{P}} \mathcal{B} = \mathcal{B} H_{\log}.$$

Equivalently,

$$\mathcal{B} e^{-s H_{\log}} \mathcal{B}^* = e^{-s D_{\mathbb{P}}}$$

whenever the functional calculus is defined. For $\Re s > 1$,

$$\boxed{\text{Tr}(e^{-s D_{\mathbb{P}}}) = \sum_{n \geq 1} n^{-s} = \zeta(s).$$

Proof. The monomials z^{α} form an orthonormal basis of the prime-torus Hardy space. Unique factorization identifies this basis bijectively with the arithmetic basis e_n , so $\mathcal{B}$ is unitary. Moreover,

$$\sum_p v_p(n) \log p = \log n,$$

which proves the intertwining identity on the algebraic basis and hence on the maximal domains. The operator is unitarily equivalent to $H_{\log}$, so it is self-adjoint. Its eigenvalues are $\log n$ with multiplicity one and tend to infinity, hence the resolvent is compact. The trace formula follows from diagonal functional calculus and absolute convergence for $\Re s > 1$. $\square$

6.3 Multiplicative covariance

For $m \geq 1$, define the arithmetic multiplication isometry

$$S_m e_n = e_{mn}$$

and the torus monomial isometry

$$M_m f = z^{\nu(m)} f.$$

Proposition 6.4 (Exact multiplicative covariance). *For every $m \geq 1$,*

$$\boxed{\mathcal{B} S_m = M_m \mathcal{B}.$$

Furthermore,

$$[H_{\log}, S_m] = (\log m) S_m, \quad [D_{\mathbb{P}}, M_m] = (\log m) M_m$$

on the algebraic cores.

Proof. Unique factorization gives

$$\nu(mn) = \nu(m) + \nu(n).$$

Therefore

$$\mathcal{B}S_m e_n = z^{\nu(mn)} = z^{\nu(m)} z^{\nu(n)} = M_m \mathcal{B}e_n.$$

The commutator identities follow from $\log(mn) = \log m + \log n$. $\square$

This is the exact operator form of the statement that integer multiplication becomes addition of prime-circle Fourier indices.

6.4 Arithmetic time as a nonperiodic Kronecker flow

Define

$$U_t f(z) = f((p^{-it} z_p)_{p \in \mathbb{P}}).$$

Then

$$U_t z^\alpha = e^{-it \sum_p \alpha_p \log p} z^\alpha,$$

so $D_{\mathbb{P}}$ is the self-adjoint generator of this flow.

Proposition 6.5 (No period collapse). *The flow U_t has no nonzero period. In particular, the exact arithmetic evolution does not descend to a single circle action.*

Proof. If $U_T = I$ for some $T \neq 0$, then

$$T \log p \in 2\pi\mathbb{Z}$$

for every prime p . Hence every ratio $\log p / \log q$ would be rational. This contradicts unique factorization: a rational relation $r \log p = s \log q$ with nonzero integers r, s would give $p^r = q^s$ for distinct primes. $\square$

The one-parameter arithmetic flow is therefore real and exact, but its compact frequency carrier is the prime torus rather than one periodic circle.

6.5 Cut and dagger covariance

On the doubled arithmetic and prime-torus carriers set

$$\begin{aligned} \tilde{\mathcal{A}} &= \mathcal{A} \otimes \mathbb{C}^2, & \tilde{\mathcal{T}} &= H^2(\mathbb{T}^{\mathbb{P}}) \otimes \mathbb{C}^2, \\ \tilde{H}_{\log} &= H_{\log} \otimes I_2, & \tilde{D}_{\mathbb{P}} &= D_{\mathbb{P}} \otimes I_2, \\ J_{\text{ar}} &= I_{\mathcal{A}} \otimes J_{\text{cut}}, & J_{\mathbb{P}} &= I_{H^2} \otimes J_{\text{cut}}, \end{aligned}$$

and

$$\tilde{\mathcal{B}} = \mathcal{B} \otimes I_2.$$

Theorem 6.6 (Arithmetic-cut intertwining). *The unitary $\tilde{\mathcal{B}}$ satisfies*

$$\boxed{\tilde{D}_{\mathbb{P}} \tilde{\mathcal{B}} = \tilde{\mathcal{B}} \tilde{H}_{\log}, \quad J_{\mathbb{P}} \tilde{\mathcal{B}} = \tilde{\mathcal{B}} J_{\text{ar}}.}$$

Both J_{ar} and $J_{\mathbb{P}}$ preserve the respective operator domains and commute with the logarithmic generators.

Proof. The first identity is the tensor amplification of [Theorem 6.3](#). The second follows because $\tilde{\mathcal{B}}$ acts on the arithmetic/torus factor while the cut acts on the common two-dimensional fibre. Domain invariance and commutation follow from the same tensor-factor separation. $\square$

For a finite cutoff N , restriction to the span of e_n , $1 \leq n \leq N$, gives the finite intertwiner

$$\Phi_N e_n = z^{\nu(n)}$$

into the finite prime torus generated by primes $p \leq N$.

6.6 Role in the completed-Weil architecture

The theorem supplies the exact bridge missing from a naive arithmetic-to-circle roadmap:

$$\boxed{\text{logarithmic arithmetic clock} \xrightarrow{\mathcal{B}} \text{weighted prime-torus generator.}}$$

It does not identify the spectrum of $D_{\mathbb{P}}$ with nontrivial zeros of ξ , analytically continue the heat trace past $\Re s > 1$, or replace the native completed-Weil sign carrier. Its role is narrower and exact: it proves that the arithmetic module has a canonical multicircle realization compatible with multiplication, the finite cut and logarithmic evolution. The scalar circle family of the preceding section remains the one-dimensional index calibration.

7 Reflected diagonal carriers and the damped no-go theorem

The scalar circle calibrates an integer index but does not realize the arithmetic logarithmic clock. The prime torus supplies the exact arithmetic representation. For completeness, we now record why the earlier paired diagonal prime carrier cannot serve as a zero-bearing completed-zeta carrier.

Let

$$\mathcal{H}_{\mathcal{P}} = \ell^2(\mathcal{P} \times \{R, L\}),$$

and let J exchange the two sectors. For $\lambda \geq 0$ and $s \in \mathbb{C}$, define

$$K_{\lambda}(s)e_{p,R} = e^{-\lambda(\log p)^2} p^{-s} e_{p,R}, \quad K_{\lambda}(s)e_{p,L} = e^{-\lambda(\log p)^2} p^{s-1} e_{p,L}.$$

Then

$$JK_{\lambda}(s)J = K_{\lambda}(1-s), \quad K_{\lambda}(s)^* = K_{\lambda}(\bar{s}).$$

Thus the paired carrier has the desired reflection covariance. Covariance, however, does not supply a critical-strip spectral event.

Theorem 7.1 (Damped diagonal no-go). *For every $\lambda > 0$, the map $s \mapsto K_{\lambda}(s)$ is entire with values in the trace class, locally uniformly in trace norm. Its Fredholm determinant is*

$$D_{\lambda}(s) = \prod_{p \in \mathcal{P}} \left(1 - e^{-\lambda(\log p)^2} p^{-s}\right) \left(1 - e^{-\lambda(\log p)^2} p^{s-1}\right).$$

Its zeros are exactly

$$s = -\lambda \log p - \frac{2\pi i n}{\log p}, \quad s = 1 + \lambda \log p + \frac{2\pi i n}{\log p}, \quad p \in \mathcal{P}, \quad n \in \mathbb{Z}.$$

Consequently

$$\boxed{D_{\lambda}(s) \neq 0 \quad \text{whenever} \quad 0 \leq \Re s \leq 1,}$$

and

$$D_{\lambda}\left(\frac{1}{2} + it\right) > 0.$$

Proof. On every compact s -set, the trace norm and all complex derivatives are dominated by a convergent sum of the form

$$\sum_{p \in \mathcal{P}} e^{-\lambda(\log p)^2} (p^M + p^{1+M}).$$

Hence the family is trace-class holomorphic and its diagonal determinant is the displayed product. A factor vanishes only at the two listed families of points, whose real parts are respectively negative and greater than one. On the critical line the two factors belonging to each prime are conjugates, so their product is a strictly positive squared modulus. $\square$

The damping therefore does more than improve convergence: it moves the entire divisor outside the closed critical strip. This is an exact structural exclusion, not a failed numerical approximation. The paired diagonal carrier is retained only as a no-go calibration; the zero-bearing sign problem is carried instead by the native completed-Weil form developed below.

8 Undamped diagonal endpoint and zero-blindness

At $\lambda = 0$ the paired diagonal family becomes

$$K_0(s)e_{p,R} = p^{-s}e_{p,R}, \quad K_0(s)e_{p,L} = p^{s-1}e_{p,L}.$$

The ordinary Fredholm determinant is no longer available in the critical strip, but the complete operator classification remains explicit.

Theorem 8.1 (Undamped diagonal classification). *Let $\sigma = \Re s$. Then $K_0(s)$ is bounded precisely for $0 \leq \sigma \leq 1$, and in that strip*

$$\|K_0(s)\| = \max\{2^{-\sigma}, 2^{\sigma-1}\}.$$

It is compact precisely for $0 < \sigma < 1$. Hence

$$\boxed{\|K_0(s)\| < 1 \quad \text{throughout the open critical strip.}}$$

Moreover, for $0 < q < \infty$,

$$K_0(s) \in \mathcal{S}_q \iff \frac{1}{q} < \Re s < 1 - \frac{1}{q}.$$

In particular the family is nowhere trace class and nowhere Hilbert–Schmidt in the open critical strip, while on the critical line it belongs to $\mathcal{S}_q$ exactly for $q > 2$.

Proof. The moduli of the diagonal entries are $p^{-\sigma}$ and $p^{\sigma-1}$. Boundedness, the norm formula and compactness follow directly from these two sequences. The Schatten criterion is equivalent to convergence of

$$\sum_{p \in \mathcal{P}} (p^{-q\sigma} + p^{-q(1-\sigma)}),$$

and the prime sum $\sum_p p^{-a}$ converges exactly for $a > 1$. $\square$

For every integer $m \geq 3$, put

$$\Omega_m = \left\{ s : \frac{1}{m} < \Re s < 1 - \frac{1}{m} \right\}.$$

Then $K_0(s) \in \mathcal{S}_m$ on Ω_m . Let

$$P_{\mathcal{P}}(z) = \sum_{p \in \mathcal{P}} p^{-z}, \quad \Re z > 1,$$

denote the prime zeta function in its absolute-convergence half-plane.

Theorem 8.2 (Zero-free regularized endpoint). *For $s \in \Omega_m$,*

$$\boxed{\det_m(I - K_0(s)) = \exp \left[- \sum_{n=m}^{\infty} \frac{P_{\mathcal{P}}(ns) + P_{\mathcal{P}}(n(1-s))}{n} \right]}.$$

The series is locally uniformly convergent, so the determinant is holomorphic and nowhere zero on Ω_m .

Proof. Since $\|K_0(s)\| < 1$ in the open strip, the logarithm of the m th regularized determinant is

$$- \sum_{n=m}^{\infty} \frac{\text{Tr}(K_0(s)^n)}{n}.$$

Diagonal evaluation gives

$$\text{Tr}(K_0(s)^n) = P_{\mathcal{P}}(ns) + P_{\mathcal{P}}(n(1-s)).$$

Both arguments lie in the absolute-convergence half-plane for $n \geq m$ and $s \in \Omega_m$. The displayed determinant is therefore the exponential of a holomorphic function and cannot vanish. $\square$

Corollary 8.3 (Diagonal-carrier exclusion). *Throughout the open critical strip, $I - K_0(s)$ is invertible. Every licensed ordinary or regularized determinant attached to this diagonal noninvertibility event is zero-free on its domain. Thus on a domain containing a nontrivial zero of ξ there is no identity*

$$\det_m(I - K_0(s)) = g(s)\xi(s)$$

with g holomorphic and nowhere zero.

The diagonal route is therefore not repaired by stronger damping, a higher Schatten determinant or an Archimedean prefactor. It lacks an interior spectral event. The native completed-Weil carrier below replaces the topology and sign mechanism rather than continuing to polish this zero-blind model.

9 Finite arithmetic calibration and defect decomposition

The prime-torus intertwiner gives the exact infinite arithmetic representation. This section retains only the finite arithmetic identities that explain the prime source and calibrate the seam/anti-seam defect split.

For $N \geq 1$, let

$$\mathcal{H}_N = \ell^2(\{1, \dots, N\}), \quad H_N e_n = (\log n) e_n.$$

Then

$$Z_N(s) = \text{Tr}(e^{-sH_N}) = \sum_{n=1}^N n^{-s}$$

is an entire Dirichlet polynomial and is used only as a finite clock.

Theorem 9.1 (Möbius pressure extractor). *Let $\ell(n) = \log n$. Then*

$$\boxed{\ell * \mu = \Lambda.}$$

Hence the extracted arithmetic stream is supported exactly on prime powers.

Proof. Möbius inversion applied to

$$\log n = \sum_{d|n} \Lambda(d)$$

gives the identity. Thus $(\ell * \mu)(p^k) = \log p$ and the value vanishes away from prime powers. $\square$

Let

$$\mathcal{Q}_N = \{q \leq N : q \text{ is square-free}\}$$

and define the finite radical pushforward

$$(T_N f)(q) = \sum_{\substack{n \leq N \\ \text{rad}(n)=q}} f(n).$$

Write Q_+ for the projection onto prime radical classes and Q_- for the projection onto square-free classes with at least two prime factors.

Theorem 9.2 (Finite prime-power support calibration). *For every prime $p \leq N$, put*

$$\nu_p(N) = \left\lfloor \frac{\log N}{\log p} \right\rfloor.$$

Then

$$\boxed{T_N \Lambda = \sum_{p \leq N} \nu_p(N) \log(p) e_p.}$$

In particular,

$$\boxed{Q_- T_N \Lambda = 0.}$$

Proof. Every nonzero value of Λ occurs at a prime power p^k , where it equals $\log p$ and has radical p . Summing over the powers below N gives the coefficient, and no prime power enters a composite-radical fibre. $\square$

This finite vanishing statement is a support calibration, not an RH theorem. It shows that Möbius extraction removes mixed-prime radical content. It does not replace the Bohr unitary, the completed boundary–Gamma–prime form or the continuum sign analysis.

For a general finite source map A , arithmetic action D , represented action U and seam projections $P_\pm$, define

$$\Delta = UA - AD, \quad S_{\text{def}} = P_+ \Delta, \quad J_{\text{def}} = P_- \Delta.$$

Proposition 9.3 (Orthogonal two-channel defect identity). *One has*

$$\Delta = S_{\text{def}} + J_{\text{def}}, \quad S_{\text{def}}^* J_{\text{def}} = 0,$$

and therefore

$$\boxed{\Delta^* \Delta = S_{\text{def}}^* S_{\text{def}} + J_{\text{def}}^* J_{\text{def}}.}$$

Exact covariance holds if and only if both defect channels vanish.

Proof. Use $P_+ + P_- = I$ and $P_+P_- = 0$. □

The distinction matters: anti-seam vanishing alone does not exclude a common symmetric transport error. In the completed argument this bookkeeping is absorbed by the target-faithful bilateral carrier and by transporting the actual source operator and source columns. The exact infinite arithmetic realization is the prime torus, not the finite radical quotient.

10 The RH representation interface

The recognition carrier, arithmetic prime torus and critical line live in different mathematical spaces. The completed-Weil construction distinguishes functional symmetry, reality symmetry, arithmetic representation and the final Hermitian sign criterion.

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s).$$

It is entire and satisfies

$$\xi(s) = \xi(1-s), \quad \xi(\bar{s}) = \overline{\xi(s)};$$

see [Tit, Edw74]. Put

$$\tau(s) = 1-s, \quad c(s) = \bar{s}, \quad \rho(s) = 1-\bar{s}.$$

Then

$$\text{Fix}(\tau) = \left\{\frac{1}{2}\right\}, \quad \text{Fix}(\rho) = \left\{s : \Re s = \frac{1}{2}\right\}.$$

The holomorphic functional transformation and the anti-holomorphic critical-line reflection are distinct operations.

Theorem 10.1 (Functional, reality and dagger covariance). *Let J be a unitary involution and let $K(s)$ satisfy*

$$JK(s)J = K(1-s), \quad K(s)^* = K(\bar{s}).$$

Then

$$\boxed{JK(s)^*J = K(1-\bar{s}).}$$

*Equivalently, for $A^\sharp = JA^*J$,*

$$K(s)^\sharp = K(\rho(s)).$$

Proof. Apply the first covariance identity at $\bar{s}$ and use the adjoint identity. □

Proposition 10.2 (Equivariance and localization). *Let X carry an involution J_X and let $\pi : X \rightarrow \mathbb{C}$ satisfy*

$$\pi(J_X x) = \rho(\pi(x)).$$

Then

$$\pi(\text{Fix } J_X) \subseteq \text{Fix}(\rho).$$

A fixed-fibre theorem supplies the reverse inclusion when required. The paper does not infer zero localization from covariance alone; localization enters through completed-Weil positivity.

Proof. If $J_X x = x$, then $\pi(x) = \rho(\pi(x))$. The reverse inclusion is exactly the fixed-fibre assertion. □

10.1 Three carriers with three distinct roles

The scalar circle family proves the calibration identity

$$\text{sf}(D_{\delta+ tq}; 0) = \text{Wind}(z^q) = \text{ind}(C_q) = q.$$

It calibrates the sign of the integer index. Its affine lattice spectrum does not realize the logarithmic arithmetic clock.

The arithmetic Bohr transform gives the exact representation

$$\mathcal{B} : \ell^2(\mathbb{N}_{>0}) \longrightarrow H^2(\mathbb{T}^{\mathbb{P}}), \quad D_{\mathbb{P}}\mathcal{B} = \mathcal{B}H_{\log}.$$

Thus the prime torus is the exact multicircle carrier of integer multiplication and logarithmic flow.

The native completed-Weil carrier bears the sign. The prime torus is its arithmetic chart, while the scalar circle calibrates the integer index. These roles are complementary and are never identified by notation alone.

10.2 Terminal representation theorem

The classical Weil criterion converts the Riemann hypothesis into nonnegativity of the completed Hermitian explicit-formula form. The native construction derives its parity and Feshbach reductions and represents every shifted adverse direction by

$$\mathfrak{k}_{\Sigma}(\eta) = \text{rank } \mathbf{1}_{(1,\infty)}(\mathbb{K}_{\eta}).$$

The common-source theorem and favorable even boundary give

$$\mathfrak{k}_{\Sigma}(\eta) = \mathfrak{k}_3(\eta) = N_+(B_{\eta} - I).$$

The prime-torus compression, native matrix, seam integer and threshold holonomy are faithful representations of this same obstruction.

At eta zero, the completed-Weil sign is exactly the five-dimensional condition

$$\mathcal{K}_0 = \text{diag}(1, A_3) - (S_{0,-}^{-1/2}\mathcal{W}_{\Sigma})^*(S_{0,-}^{-1/2}\mathcal{W}_{\Sigma}) \geq 0.$$

Under the classical explicit-formula identification and Weil criterion,

$$\boxed{\text{RH} \iff \mathcal{K}_0 \geq 0.}$$

This is the terminal representation interface used throughout the paper.

11 The zero-identification boundary

Modeled points, spectral characteristic values, determinant zeros and actual zeros of ξ are different mathematical objects. The arrows between them must be proved for the same completed carrier. This section closes the spectral-to-determinant arrow when analytic Schatten theory applies and records that, for the present diagonal paired-prime carrier, both sets are empty in the critical strip.

11.1 Four zero layers

Definition 11.1 (Modeled points). Let Z_{model} be a finite or formal collection introduced to test reflection, seam projection, crossing or leakage rules. Membership is a modeling choice.

Definition 11.2 (Spectral characteristic values). For a declared operator family $H(s)$, let Z_{spectral} be the parameters at which the declared spectral event occurs, such as noninvertibility, an eigenvalue crossing or loss of Fredholmness.

Definition 11.3 (Determinant zeros). If a determinant F is analytically licensed on Ω , define

$$Z_{\text{det}} = \{s \in \Omega : F(s) = 0\}.$$

The determinant type and normalization are part of the definition.

Definition 11.4 (Completed-zeta zeros). Let

$$Z_{\xi} = \{s \in \mathbb{C} : \xi(s) = 0\},$$

with multiplicity.

A carrier-dependent route has the form

$$Z_{\text{model}} \dashrightarrow Z_{\text{spectral}} \dashrightarrow Z_{\text{det}} \dashrightarrow Z_{\xi}.$$

The first arrow is a representation theorem, the second is an analytic determinant theorem, and the third is a divisor-identification theorem.

11.2 Analytic Schatten determinant bridge

Theorem 11.5 (Spectral event equals regularized determinant zero). *Let Ω be a domain, let $m \geq 1$ be an integer, and let*

$$K : \Omega \longrightarrow \mathcal{S}_m$$

be holomorphic. Define

$$F_m(s) = \det_m(I - K(s)).$$

Then F_m is holomorphic and

$$\boxed{F_m(s) = 0 \iff I - K(s) \text{ is not invertible} \iff 1 \in \text{Spec } K(s).}$$

Thus, for this declared event,

$$Z_{\text{spectral}} = Z_{\text{det}}$$

as sets. Multiplicity requires the standard analytic characteristic-value convention on both sides.

Proof. The regularized determinant is holomorphic on analytic $\mathcal{S}_m$ families. Its zero set is exactly the set where $I - K(s)$ fails to be invertible, with algebraic multiplicity governed by analytic Fredholm theory. $\square$

Corollary 11.6 (Strict contraction is zero-blind). *If $\|K(s)\| < 1$ throughout a domain, then*

$$Z_{\text{spectral}} = Z_{\text{det}} = \emptyset$$

there.

Proof. The Neumann series makes $I - K(s)$ invertible; apply the theorem. $\square$

11.3 Finite paired determinant as a normalization test

Let $0 < \lambda_1 \leq \dots \leq \lambda_m$ be declared levels and define

$$D_N(E) = \prod_{j=1}^m \left(1 - \frac{E^2}{\lambda_j^2}\right).$$

Then $D_N(0) = 1$, $D_N(E) = D_N(-E)$, its zeros are the prescribed signed levels, and

$$\frac{D'_N(E)}{D_N(E)} = \sum_{j=1}^m \frac{2E}{E^2 - \lambda_j^2}$$

away from them. If H_N has spectrum $\{\pm\lambda_j\}$, then its characteristic polynomial is the corresponding unnormalized product.

These are correct finite identities, but the zero agreement is built into the chosen spectrum. The scaffold tests normalization and multiplicity bookkeeping; it supplies no arithmetic correspondence unless the levels are derived independently from a completed carrier.

11.4 Nonvanishing-factor promotion

Lemma 11.7 (Zero and multiplicity transfer). *Let F , g and ξ be holomorphic on a domain Ω . If*

$$F(s) = g(s)\xi(s)$$

and g is nowhere zero, then F and ξ have the same divisor on Ω .

Proof. At every point, g is a unit in the local ring of holomorphic germs, so multiplication by g preserves the order of vanishing. $\square$

The lemma is easy because its hypothesis already contains the decisive zero theorem. Constructing a lawful F and proving the identity is the hard step.

11.5 Diagonal paired-prime carrier exclusion

The preceding determinant audit proves:

- (i) for every $\lambda > 0$, the damped Fredholm determinant is zero-free on the closed critical strip;
- (ii) for every integer $m \geq 3$, the undamped regularized determinant is zero-free on

$$\Omega_m = \left\{s : \frac{1}{m} < \Re(s) < 1 - \frac{1}{m}\right\};$$

- (iii) throughout the open critical strip,

$$\|K_0(s)\| = \max\{2^{-\Re(s)}, 2^{\Re(s)-1}\} < 1.$$

Theorem 11.8 (Carrier-exclusion theorem). *For the declared diagonal paired-prime family,*

$$Z_{\text{spectral}} = Z_{\text{det}} = \emptyset$$

in every licensed critical-strip domain. Consequently there is no identity

$$\det_m(I - K_0(s)) = g(s)\xi(s)$$

on a domain containing a zero of ξ when g is holomorphic and nowhere zero. The same conclusion holds for the damped Fredholm determinant on the closed strip.

Proof. The norm estimate gives invertibility in the open strip, and the exact damped divisor lies outside the closed strip. The analytic Schatten bridge therefore gives empty determinant zero sets. A nowhere-zero multiple of ξ must vanish at every zero of ξ , which these determinants do not. $\square$

The direct diagonal route is therefore not waiting for a more elaborate normalization. It lacks the required interior spectral event. The native completed-Weil form changes both topology and terminal mechanism rather than decorating this excluded carrier.

11.6 Zero-correspondence obstruction classes

A completed audit should distinguish:

- Z1.** extra determinant zeros;
- Z2.** missing target zeros;
- Z3.** multiplicity mismatch;
- Z4.** archimedean zeros or poles introduced by incomplete Γ and π treatment;
- Z5.** trivial-zero contamination;
- Z6.** spectral pollution in a limiting approximation;
- Z7.** domain or analytic-sheet mismatch;
- Z8.** regulator dependence of the divisor.

A clean zero theorem must eliminate every relevant class on the same completed carrier.

11.7 Guarded promotion theorem

Theorem 11.9 (Completed divisor promotion). *Suppose a completed carrier provides a holomorphic Schatten family K and $F_m = \det_m(I - K)$ on Ω . Assume:*

- (i) *the determinant is licensed by the operator ideal and completion;*
- (ii) *regulators have been removed without changing the divisor;*
- (iii) *archimedean and trivial factors are accounted for;*
- (iv) *spectral pollution and domain mismatch are absent;*
- (v) *there is a holomorphic nowhere-zero g with*

$$F_m = g\xi.$$

Then

$$Z_{\text{spectral}} = Z_{\text{det}} = Z_{\xi} \cap \Omega$$

with the declared multiplicity convention.

Proof. The analytic Schatten bridge gives the first equality. The nonvanishing-factor lemma gives the equality of divisors, and the remaining assumptions ensure that the objects belong to the completed carrier rather than to a regulator or finite approximation. $\square$

This is a promotion theorem and claim ledger. It states what a successful carrier would imply; it does not reduce the difficulty of constructing that carrier.

11.8 Forbidden direct identifications

Neither

$$Z_{\text{model}} \longrightarrow Z_{\xi}$$

nor

$$Z_{\text{spectral}} \longrightarrow Z_{\xi}$$

is lawful without the intervening representation and divisor theorems. A modeled point may calibrate a sign rule without being a zero. A self-adjoint operator may have real spectrum without representing the ordinates of zeta zeros.

Likewise,

$$\rho(s) = s \quad \Longleftrightarrow \quad \Re(s) = \frac{1}{2}$$

identifies a reflection-fixed locus, not a zero set. Seam membership and zero membership remain logically independent until a completed target-faithful theorem connects them.

Part III

Native Multiplicative and X_3 Completed-Weil Realization

12 Native multiplicative carrier, EMK cut and Euler-scale flow

The completed operator theory is most naturally organized on the multiplicative scale line before logarithmic or Fourier coordinates are introduced. The guiding order is

native scale carrier $\longrightarrow$ EMK cut and scale flow $\longrightarrow$ native energy completion $\longrightarrow$ classical representation charts
--

The logarithmic, Mellin and Fourier descriptions used later are faithful computational representations of this source structure. They are not used to repair a carrier that is blind to the declared target.

Throughout this section, Hilbert inner products are linear in the first variable.

12.1 The multiplicative core and inversion cut

Let

$$\mathcal{D}_\times = C_c^\infty(0, \infty), \quad d^\times r = \frac{dr}{r}.$$

The measure $d^\times r$ is invariant under dilation and inversion.

Definition 12.1 (EMK inversion cut and scale flow). For $F \in \mathcal{D}_\times$ and $t \in \mathbb{R}$, define

$$(J_\times F)(r) = F(r^{-1}), \quad (U_t F)(r) = F(e^{-t}r).$$

The distinguished seam is the fixed scale $r = 1$; the two open sheets are $0 < r < 1$ and $r > 1$.

Theorem 12.2 (Native cut-flow law). *On $L^2((0, \infty), d^\times r)$, $J_\times$ is a unitary self-adjoint involution, $(U_t)_{t \in \mathbb{R}}$ is a strongly continuous unitary group, and*

$$\boxed{J_\times U_t J_\times = U_{-t}.$$

Its self-adjoint generator is the closure E of

$$E_0 = -ir \frac{d}{dr} \quad \text{on } \mathcal{D}_\times,$$

with

$$U_t = e^{-itE}, \quad \boxed{J_\times E J_\times = -E.}$$

Proof. Invariance of $d^\times r$ gives

$$\|J_\times F\|_2 = \|F\|_2, \quad \|U_t F\|_2 = \|F\|_2.$$

The identities $J_\times^2 = I$ and

$$(J_\times U_t J_\times F)(r) = F(e^t r) = (U_{-t} F)(r)$$

are direct. Strong continuity is first checked on $\mathcal{D}_\times$ and then extended by density. Stone's theorem gives a self-adjoint generator. Differentiation at $t = 0$ on the core gives $-ir d/dr$, and conjugating the unitary group identity gives the generator identity. $\square$

12.2 Euler-harmonic representation

Use the Mellin convention

$$(\mathcal{M}F)(\xi) = \int_0^\infty F(r) r^{-i\xi} d^\times r.$$

Mellin-Plancherel gives

$$\|F\|_{L^2(d^\times r)}^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\mathcal{M}F(\xi)|^2 d\xi.$$

On the core,

$$\mathcal{M}(EF)(\xi) = \xi \mathcal{M}F(\xi), \quad \mathcal{M}(U_t F)(\xi) = e^{-i\xi t} \mathcal{M}F(\xi),$$

and

$$\mathcal{M}(J_\times F)(\xi) = \mathcal{M}F(-\xi).$$

For $\xi \in \mathbb{R}$, let

$$\chi_\xi(r) = r^{i\xi}$$

be the generalized multiplicative character, interpreted distributionally against $\mathcal{D}_\times$.

Theorem 12.3 (Euler-harmonic representation theorem). *The native Euler-scale structure determines the classical harmonic characters as follows.*

(i) *The generalized characters diagonalize the Euler generator and its flow:*

$$\boxed{E_0 \chi_\xi = \xi \chi_\xi, \quad U_t \chi_\xi = e^{-i\xi t} \chi_\xi.}$$

Equivalently,

$$\boxed{\mathcal{M} E \mathcal{M}^{-1} = M_\xi, \quad \mathcal{M} U_t \mathcal{M}^{-1} = M_{e^{-i\xi t}}.}$$

(ii) *The EMK cut reverses the spectral character:*

$$\boxed{J_\times \chi_\xi = \chi_{-\xi}.$$

Its even and odd combinations are

$$\boxed{\frac{\chi_\xi + \chi_{-\xi}}{2} = \cos(\xi \log r), \quad \frac{\chi_\xi - \chi_{-\xi}}{2i} = \sin(\xi \log r).}$$

Thus cosine and sine are respectively the seam-even and seam-odd spectral channels of the native scale flow.

(iii) Under the logarithmic coordinate $x = \log r$,

$$\boxed{\chi_\xi(e^x) = e^{i\xi x},}$$

so the Mellin spectral representation becomes the ordinary Fourier representation of translation.
(iv) Let

$$H_\times = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad H_\times^2 = I.$$

For $a > 0$ and $x = \log r$,

$$\boxed{\exp((aH_\times - i\xi I)x) = \begin{pmatrix} r^{-a-i\xi} & 0 \\ 0 & r^{a-i\xi} \end{pmatrix} = e^{-i\xi x}(\cosh(ax)I + \sinh(ax)H_\times).}$$

The two entries are exactly the kernels of the weighted Mellin boundary channels in their declared order. Hyperbolic sheet separation and circular Euler phase therefore commute in one two-sheet representation.

Proof. The first character identity follows from

$$-ir \frac{d}{dr} r^{i\xi} = \xi r^{i\xi},$$

and the flow identity follows from

$$\chi_\xi(e^{-t}r) = e^{-i\xi t} \chi_\xi(r).$$

The Mellin multiplication formulas are the corresponding unitary spectral representation. Inversion gives $(r^{-1})^{i\xi} = r^{-i\xi}$, and the sine/cosine formulas are the symmetric and antisymmetric combinations. The logarithmic identity is immediate. Finally, $H_\times$ commutes with iI and satisfies $H_\times^2 = I$, so its exponential is $\cosh(ax)I + \sinh(ax)H_\times$; multiplication by the circular phase gives the displayed matrix. $\square$

Corollary 12.4 (Classical harmonic analysis as the Euler shadow). *Within the declared completed-Weil realization, the chain*

$$\boxed{\text{Euler-scale flow} \longrightarrow \text{self-adjoint generator} \longrightarrow \text{Mellin characters} \longrightarrow \text{sine/cosine cut sectors} \longrightarrow \text{Fourier translation}}$$

is derived from the native carrier and its inversion cut. The additional weights $r^{\pm a}$ produce the commuting hyperbolic two-sheet factor. Classical harmonic analysis is therefore a faithful spectral shadow of the native scale dynamics in this realization, not an independent source axiom.

The corollary is deliberately scoped. It does not claim to derive the completed zeta explicit formula, the Weil criterion or the terminal no-crossing estimate; those are respectively a classical input and open sign obligations.

12.3 Native energy and completed cut

Let

$$G(\xi) = \operatorname{Re} \psi\left(\frac{1}{4} + \frac{i\xi}{2}\right) - \log \pi, \quad w_\Gamma(\xi) = 1 + |G(\xi)|.$$

Definition 12.5 (Native scale energy). For $F \in \mathcal{D}_\times$, set

$$\|F\|_{\times,3}^2 = \int_0^\infty \max(r^3, r^{-3}) |F(r)|^2 d^\times r + \frac{1}{2\pi} \int_{\mathbb{R}} w_\Gamma(\xi) |\mathcal{M}F(\xi)|^2 d\xi.$$

Let $\mathcal{X}_{\times,3}$ be the completion of $\mathcal{D}_\times$ in this norm.

The native Hilbert carrier is therefore produced by completing the declared scale and archimedean energy. The inversion cut is isometric because both weights are even under $r \mapsto r^{-1}$ and $\xi \mapsto -\xi$.

Theorem 12.6 (Completed EMK cut). *The inversion cut extends uniquely to a unitary self-adjoint involution on $\mathcal{X}_{\times,3}$, again denoted $J_\times$. Hence*

$$\mathcal{X}_{\times,3} = \mathcal{X}_{\times,3}^+ \oplus \mathcal{X}_{\times,3}^-,$$

where

$$\mathcal{X}_{\times,3}^\pm = \ker(J_\times \mp I).$$

Proof. For core vectors, a change of variables gives equality of both terms in the native norm after inversion. The extension follows by density, and the orthogonal decomposition is the spectral decomposition of a self-adjoint involution. $\square$

12.4 Direct native compactness

Let

$$\iota_\times : \mathcal{X}_{\times,3} \longrightarrow L^2((0, \infty), d^\times r)$$

be the natural inclusion.

Theorem 12.7 (Native compact embedding). *The inclusion $\iota_\times$ is compact.*

Proof. Let $\mathcal{K}$ be bounded in the native energy norm. For $R > 1$,

$$\int_{r>R} |F(r)|^2 d^\times r + \int_{0<r<R^{-1}} |F(r)|^2 d^\times r \leq R^{-3} \|F\|_{\times,3}^2,$$

so $\mathcal{K}$ has uniform multiplicative spatial tails. Put

$$m(\Omega) = \inf_{|\xi|>\Omega} w_\Gamma(\xi).$$

Since $m(\Omega) \rightarrow \infty$,

$$\int_{|\xi|>\Omega} |\mathcal{M}F(\xi)|^2 d\xi \leq \frac{2\pi}{m(\Omega)} \|F\|_{\times,3}^2.$$

Finally,

$$\|U_t F - F\|_2^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |e^{-i\xi t} - 1|^2 |\mathcal{M}F(\xi)|^2 d\xi.$$

The high-frequency part is uniformly small by the preceding estimate, and the bounded-frequency part is uniformly small as $t \rightarrow 0$. The multiplicative Kolmogorov–Riesz criterion, equivalently the ordinary criterion after the Haar isomorphism $r = e^x$, gives relative compactness. $\square$

12.5 The logarithmic representation theorem

Define

$$(\mathcal{L}F)(x) = F(e^x).$$

Since $d^\times r = dx$ and

$$\max(r^3, r^{-3}) = e^{3|x|},$$

this map carries the native spatial channel to the exponential spatial channel used in X_3 .

Theorem 12.8 (Native-to- X_3 unitary chart). *The logarithmic map extends to a unitary*

$$\boxed{\mathcal{L} : \mathcal{X}_{\times,3} \longrightarrow X_3.}$$

It intertwines the native and logarithmic structures:

$$\boxed{\mathcal{L}J_\times\mathcal{L}^{-1} = J_3, \quad \mathcal{L}E\mathcal{L}^{-1} = -i\frac{d}{dx}, \quad \mathcal{L}U_t\mathcal{L}^{-1} = T_t.}$$

Consequently

$$\mathcal{L}\mathcal{X}_{\times,3}^\pm = X_3^\pm.$$

Proof. The norm identity follows from the change of variables $r = e^x$. With the Mellin convention above,

$$\mathcal{M}F(\xi) = \widehat{\mathcal{L}F}(-\xi).$$

The frequency weight is even, so the sign reversal does not change the norm. The intertwining identities are direct on the core and extend by continuity. $\square$

This theorem fixes the proof direction: X_3 is a faithful logarithmic representation of the native scale carrier. It is not a separate source axiom.

12.6 Native completed-Weil form

For $F \in \mathcal{D}_\times$, define

$$\ell_+^\times(F) = \int_0^\infty F(r)r^{-1/2}d^\times r, \quad \ell_-^\times(F) = \int_0^\infty F(r)r^{1/2}d^\times r.$$

Cauchy–Schwarz and the spatial energy give

$$|\ell_\pm^\times(F)|^2 \leq \frac{3}{4}\|F\|_{\times,3}^2.$$

Define the native boundary, Gamma and prime forms by

$$\mathfrak{B}_\times(F, H) = \ell_+^\times(F)\overline{\ell_-^\times(H)} + \ell_-^\times(F)\overline{\ell_+^\times(H)},$$

$$\mathfrak{A}_{\Gamma,\times}(F, H) = \frac{1}{2\pi} \int_{\mathbb{R}} G(\xi)\mathcal{M}F(\xi)\overline{\mathcal{M}H(\xi)}d\xi,$$

and

$$\mathfrak{P}_\times(F, H) = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (\langle F, U_{\log n} H \rangle + \langle F, U_{-\log n} H \rangle).$$

The spatial channel gives the native correlation estimate

$$|\langle F, U_t H \rangle| \leq e^{-3|t|/2} \|F\|_{\times,3} \|H\|_{\times,3}.$$

Hence the prime form is absolutely bounded by

$$2 \sum_{n \geq 2} \frac{\Lambda(n)}{n^2} \|F\|_{\times,3} \|H\|_{\times,3}.$$

Definition 12.9 (Native completed-Weil form). Set

$$\mathfrak{W}_{\times}(F, H) = \mathfrak{B}_{\times}(F, H) + \mathfrak{A}_{\Gamma, \times}(F, H) - \mathfrak{P}_{\times}(F, H).$$

Theorem 12.10 (Native bounded Weil representative). *The form $\mathfrak{W}_{\times}$ extends uniquely to a bounded Hermitian form on $\mathcal{X}_{\times,3}$. There is a unique bounded self-adjoint operator*

$$W_{\times} \in \mathcal{B}(\mathcal{X}_{\times,3})$$

such that

$$\mathfrak{W}_{\times}(F, H) = \langle F, W_{\times} H \rangle_{\times,3}.$$

Moreover,

$$\mathcal{L} W_{\times} \mathcal{L}^{-1} = W_3, \quad [W_{\times}, J_{\times}] = 0.$$

Proof. The three form estimates above prove boundedness. Hermiticity follows from the paired boundary terms, the real Gamma symbol and the adjoint pair of dilation operators. Riesz representation gives $W_{\times}$. Under $x = \log r$, the boundary, Gamma and prime terms become exactly the corresponding terms in $\mathfrak{W}_{\xi}$; uniqueness gives unitary equivalence. Inversion exchanges the two boundary channels, reflects the even Gamma symbol and interchanges the two prime dilations, so the form is inversion invariant and the representative commutes with the cut. $\square$

12.7 Two-sheet prime seam operator

For $a > 0$, define the boundary map

$$\mathcal{B}_a^{\times} F(\xi) = \begin{pmatrix} \mathcal{M}(r^{-a} F)(\xi) \\ \mathcal{M}(r^a F)(\xi) \end{pmatrix}.$$

Mellin–Plancherel gives

$$\frac{1}{2\pi} \|\mathcal{B}_a^{\times} F\|_2^2 = \int_0^{\infty} (r^{-2a} + r^{2a}) |F(r)|^2 d^{\times} r.$$

At $a = 3/2$, this norm is equivalent, with constants 1 and 2, to the native spatial channel.

For $a > 1/2$, put

$$q_a(\xi) = \sum_{n \geq 2} \frac{\Lambda(n)}{n^{a+1/2}} e^{i\xi \log n} = -\frac{\zeta'}{\zeta} \left(a + \frac{1}{2} - i\xi \right)$$

and

$$\mathcal{Q}_a(\xi) = \begin{pmatrix} 0 & q_a(\xi) \\ \overline{q_a(\xi)} & 0 \end{pmatrix}.$$

This matrix is self-adjoint for real ξ .

Theorem 12.11 (Native prime seam operator). *The prime operator has the two equivalent native forms*

$$B_{\text{prime}}^{\times} = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \iota_{\times}^* (\mathbf{U}_{\log n} + \mathbf{U}_{-\log n}) \iota_{\times}$$

and

$$B_{\text{prime}}^{\times} = (\mathcal{B}_{3/2}^{\times})^* M_{\mathcal{Q}_{3/2}} \mathcal{B}_{3/2}^{\times}.$$

It is compact and self-adjoint on $\mathcal{X}_{\times,3}$, and

$$\|\mathcal{Q}_{3/2}\|_{L^{\infty}} \leq -\frac{\zeta'(2)}{\zeta(2)}.$$

Proof. Each dilation atom is compact after sandwiching by the compact inclusion $\iota_{\times}$. The correlation estimate gives

$$\|\iota_{\times}^* (\mathbf{U}_t + \mathbf{U}_{-t}) \iota_{\times}\| \leq 2e^{-3|t|/2},$$

so the series converges absolutely in operator norm. The two-sheet formula follows from weighted Mellin–Plancherel. At $a = 3/2$, the seam factor converts $\Lambda(n)/\sqrt{n}$ into $\Lambda(n)/n^2$, permitting interchange of sum and integral. The norm bound follows from the absolutely convergent Dirichlet series. $\square$

The first formula supplies native compactness; the second displays the hyperbolic two-sheet coupling and circular Euler spectrum. Neither formula requires a spatial grid interpolation.

13 The native X_3 completed-Weil carrier

The recognition-seam framework becomes analytically substantive only after a native completed carrier is constructed. In the MP development this carrier is not postulated as a Hilbert–Pólya spectrum. It is obtained by completing the explicit-formula test core in a two-channel norm that controls both spatial escape and archimedean frequency escape. The resulting completed Weil form has a bounded self-adjoint representative, admits a derived even/odd Hilbert decomposition, and supports the exact parity/Feshbach reduction used in the terminal recognition-seam geometry.

The purpose of this section is therefore foundational. The Riemann hypothesis enters only after the carrier theorem, through the classical Weil positivity criterion.

13.1 The two-channel Hilbert envelope

Fix the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{i\xi x} dx, \quad \|f\|_{L^2}^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{f}(\xi)|^2 d\xi.$$

Let

$$\mathcal{D} = C_c^{\infty}(\mathbb{R})$$

and put

$$G(\xi) = \operatorname{Re} \psi\left(\frac{1}{4} + \frac{i\xi}{2}\right) - \log \pi, \quad w_{\Gamma}(\xi) = 1 + |G(\xi)|,$$

where $\psi = \Gamma'/\Gamma$ is the digamma function.

Definition 13.1 (Native X_3 envelope). Define X_3 to be the completion of $\mathcal{D}$ in the norm

$$\|f\|_{X_3}^2 = \int_{\mathbb{R}} e^{3|x|} |f(x)|^2 dx + \frac{1}{2\pi} \int_{\mathbb{R}} w_{\Gamma}(\xi) |\hat{f}(\xi)|^2 d\xi.$$

The first term is the spatial channel and the second is the completed archimedean frequency channel.

The definition makes X_3 a Hilbert space. Since $e^{3|x|} \geq 1$, the native inclusion

$$\iota : X_3 \longrightarrow L^2(\mathbb{R})$$

is bounded.

Lemma 13.2 (Weighted correlation decay). *Let $T_t h(x) = h(x - t)$. For $f, h \in X_3$ and $t \in \mathbb{R}$,*

$$|\langle f, T_t h \rangle_{L^2}| \leq e^{-3|t|/2} \|f\|_{X_3} \|h\|_{X_3}.$$

Proof. For core vectors, insert the spatial weights and use $|x| + |x - t| \geq |t|$:

$$\begin{aligned} |\langle f, T_t h \rangle| &\leq \int e^{-\frac{3}{2}(|x|+|x-t|)} (e^{\frac{3}{2}|x|} |f(x)|) (e^{\frac{3}{2}|x-t|} |h(x-t)|) dx \\ &\leq e^{-3|t|/2} \|e^{3|x|/2} f\|_2 \|e^{3|x|/2} h\|_2. \end{aligned}$$

Density gives the result on X_3 . □

Theorem 13.3 (Native position-frequency compactness). *The inclusion*

$$\iota : X_3 \hookrightarrow L^2(\mathbb{R})$$

is compact.

Proof. Let (f_j) be bounded in X_3 , with squared norm at most M . The spatial channel gives

$$\int_{|x|>R} |f_j(x)|^2 dx \leq M e^{-3R}.$$

Since $|G(\xi)| \rightarrow \infty$ as $|\xi| \rightarrow \infty$, if

$$m(\Omega) = \inf_{|\xi|>\Omega} w_{\Gamma}(\xi),$$

then $m(\Omega) \rightarrow \infty$ and

$$\int_{|\xi|>\Omega} |\hat{f}_j(\xi)|^2 d\xi \leq \frac{2\pi M}{m(\Omega)}.$$

Finally, Plancherel gives

$$\|f_j(\cdot + h) - f_j\|_2^2 = \frac{1}{2\pi} \int |e^{-ih\xi} - 1|^2 |\hat{f}_j(\xi)|^2 d\xi.$$

After splitting at $|\xi| = \Omega$, the high-frequency part is uniformly small by the preceding tail and the low-frequency part is uniformly small for sufficiently small h . The Kolmogorov–Riesz criterion now gives relative compactness in $L^2(\mathbb{R})$. □

13.2 The completed critical Weil form

For $f \in \mathcal{D}$, define the completion-pole channels

$$\ell_+(f) = \widehat{f}(i/2) = \int_{\mathbb{R}} f(x) e^{-x/2} dx, \quad \ell_-(f) = \widehat{f}(-i/2) = \int_{\mathbb{R}} f(x) e^{x/2} dx.$$

They satisfy

$$|\ell_{\pm}(f)|^2 \leq \frac{3}{4} \|f\|_{X_3}^2.$$

Define

$$\begin{aligned} \mathfrak{B}(f, h) &= \ell_+(f) \overline{\ell_-(h)} + \ell_-(f) \overline{\ell_+(h)}, \\ \mathfrak{A}_{\Gamma}(f, h) &= \frac{1}{2\pi} \int_{\mathbb{R}} G(\xi) \widehat{f}(\xi) \overline{\widehat{h}(\xi)} d\xi, \end{aligned}$$

and

$$\mathfrak{P}_{1/2}(f, h) = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (\langle f, T_{\log n} h \rangle + \langle f, T_{-\log n} h \rangle).$$

On $\mathcal{D}$ the prime sum is finite, because the correlation $f * h^{\sharp}$ has compact support. On X_3 , [Theorem 13.2](#) gives

$$|\mathfrak{P}_{1/2}(f, h)| \leq 2 \left(\sum_{n \geq 2} \frac{\Lambda(n)}{n^2} \right) \|f\|_{X_3} \|h\|_{X_3}.$$

The scalar series equals $-\zeta'(2)/\zeta(2)$ and is finite.

Definition 13.4 (Completed critical Weil form). On $\mathcal{D}$, set

$$\boxed{\mathfrak{W}_{\xi}(f, h) = \mathfrak{B}(f, h) + \mathfrak{A}_{\Gamma}(f, h) - \mathfrak{P}_{1/2}(f, h).}$$

Theorem 13.5 (Bounded native Weil representative). *The form $\mathfrak{W}_{\xi}$ extends uniquely to a bounded Hermitian form on X_3 . Consequently there is a unique bounded self-adjoint operator*

$$\boxed{W_3 \in \mathcal{B}(X_3), \quad \mathfrak{W}_{\xi}(f, h) = \langle f, W_3 h \rangle_{X_3}.}$$

Proof. The pole functionals are continuous by the displayed 3/4 estimate. The Gamma form is bounded because $|G| \leq w_{\Gamma}$ and the frequency term is part of the native norm. The prime form is bounded by the preceding correlation estimate. Thus $\mathfrak{W}_{\xi}$ is bounded. It is Hermitian because the pole form is Hermitian, G is real, and the prime translations occur in adjoint pairs. The Riesz representation theorem gives W_3 , and Hermiticity gives $W_3^* = W_3$. $\square$

Remark 13.6 (Classical explicit-formula membrane). The identification of $\mathfrak{W}_{\xi}$ with the symmetrically summed completed-zeta explicit formula, and the equivalence

$$\text{RH} \iff \mathfrak{W}_{\xi}(f, f) \geq 0 \quad (f \text{ in the admissible Weil core}),$$

are classical analytic-number-theory inputs. The paper's new framework theorem begins after this membrane: it constructs the native Hilbert representation, its recognition-compatible parity split, and the reduced anti-seam obstruction. It does not claim a new proof of the explicit formula or of the Weil criterion.

13.3 The cut-to- X_3 completion theorem

On the explicit-formula core $\mathcal{D}$, define the reflection cut

$$(\kappa_0 f)(x) = f(-x).$$

For the faithful analytic realization used here, recognition on the core is equality. The primitive fixed and anti-fixed sectors are therefore

$$\mathcal{D}^+ = \{f \in \mathcal{D} : \kappa_0 f = f\}, \quad \mathcal{D}^- = \{f \in \mathcal{D} : \kappa_0 f = -f\}.$$

Thus $\mathcal{D}^+$ is the core seam of the reflection cut and $\mathcal{D}^-$ is its oriented anti-fixed sector. The analytic target determines the two-channel metric; the cut does not manufacture the Gamma weight or the exponent 3. Its exact role is to require that the chosen completion preserve the declared reflection distinction.

Theorem 13.7 (Cut-to- X_3 completion). *The reflection cut κ_0 is isometric for the native norm and extends uniquely to a unitary self-adjoint involution*

$$J_3 : X_3 \longrightarrow X_3.$$

Its fixed and anti-fixed Hilbert sectors are

$$X_3^+ = \ker(J_3 - I) = \overline{\mathcal{D}^+}^{X_3}, \quad X_3^- = \ker(J_3 + I) = \overline{\mathcal{D}^-}^{X_3},$$

and

$$X_3 = X_3^+ \oplus X_3^-.$$

Proof. Both weights in [Theorem 13.1](#) are even. Moreover,

$$\widehat{\kappa_0 f}(\xi) = \widehat{f}(-\xi).$$

A change of variables therefore gives

$$\|\kappa_0 f\|_{X_3} = \|f\|_{X_3} \quad (f \in \mathcal{D}).$$

Hence κ_0 extends uniquely by density to a unitary J_3 on X_3 . Since $\kappa_0^2 = I$ on the core, $J_3^2 = I$ on the completion; a unitary involution is self-adjoint. The projections

$$P_{\pm} = \frac{I \pm J_3}{2}$$

are complementary orthogonal projections. On the core, $P_{\pm}\mathcal{D} = \mathcal{D}^{\pm}$. Density of $\mathcal{D}$ and continuity of $P_{\pm}$ imply

$$\text{Ran } P_{\pm} = \overline{\mathcal{D}^{\pm}}^{X_3}.$$

The spectral decomposition of the self-adjoint involution gives the direct sum. □

Theorem 13.8 (Cut-reduced Weil representative). *The bounded Weil representative commutes with the completed cut,*

$$W_3 J_3 = J_3 W_3.$$

Consequently

$$W_3 = W_3^+ \oplus W_3^-, \quad W_3 \geq 0 \iff W_3^+ \geq 0 \text{ and } W_3^- \geq 0.$$

Proof. Reflection interchanges ℓ_+ and ℓ_- , leaves the even Gamma multiplier invariant, and interchanges $T_{\log n}$ with $T_{-\log n}$. Therefore

$$\mathfrak{W}_\xi(J_3 f, J_3 h) = \mathfrak{W}_\xi(f, h).$$

By uniqueness of the Riesz representative, $W_3 J_3 = J_3 W_3$. Hence the two eigenspaces in [Theorem 13.7](#) reduce W_3 , and positivity is equivalent to positivity on both direct-sum sectors. $\square$

The even and odd Hilbert spaces are therefore derived from the primitive reflection cut after completion. They are not independently postulated carriers selected after the sign problem is inspected.

Define the parity-resolved boundary functionals

$$\ell_{\text{ev}} = \frac{\ell_+ + \ell_-}{\sqrt{2}}, \quad \ell_{\text{odd}} = \frac{\ell_- - \ell_+}{\sqrt{2}}.$$

Then

$$\ell_{\text{ev}}(f) = \sqrt{2} \int_{\mathbb{R}} f(x) \cosh(x/2) dx, \quad \ell_{\text{odd}}(f) = \sqrt{2} \int_{\mathbb{R}} f(x) \sinh(x/2) dx,$$

and

$$\mathfrak{B}(f, h) = \ell_{\text{ev}}(f) \overline{\ell_{\text{ev}}(h)} - \ell_{\text{odd}}(f) \overline{\ell_{\text{odd}}(h)}.$$

Let $r_{\text{ev}} \in X_3^+$ and $r_{\text{odd}} \in X_3^-$ be their Riesz vectors. Direct dual estimates give

$$\|r_{\text{ev}}\|_{X_3}^2 \leq \frac{17}{12}, \quad \|r_{\text{odd}}\|_{X_3}^2 \leq \frac{1}{12}, \quad \langle r_{\text{ev}}, r_{\text{odd}} \rangle_{X_3} = 0.$$

The boundary sector thus carries a positive even rank-one channel and a negative odd rank-one channel. This signed parity split is structural, not numerical.

13.4 Compact prime and Gamma defects

The compact embedding in [Theorem 13.3](#) converts the critical prime distribution into a compact native operator. For $a \in \mathbb{R}$, define

$$C_a = \iota^*(T_a + T_{-a})\iota.$$

Each C_a is compact and self-adjoint. Moreover, [Theorem 13.2](#) gives

$$\|C_a\| \leq 2e^{-3|a|/2}.$$

Consequently

$$B_{\text{prime}} = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} C_{\log n}$$

converges absolutely in operator norm, is compact and self-adjoint, and represents $\mathfrak{P}_{1/2}$. Its directed tail satisfies, for $N \geq 2$,

$$\left\| B_{\text{prime}} - \sum_{2 \leq n \leq N} \frac{\Lambda(n)}{\sqrt{n}} C_{\log n} \right\| \leq 2 \sum_{n > N} \frac{\Lambda(n)}{n^2} \leq \frac{2(\log N + 1)}{N}.$$

For the Gamma sector, define

$$C_\Gamma f = (2\pi)^{-1/2} \sqrt{w_\Gamma} \hat{f}, \quad s(\xi) = \frac{G(\xi)}{w_\Gamma(\xi)}.$$

Then

$$B_\Gamma = C_\Gamma^* M_s C_\Gamma$$

represents $\mathfrak{A}_\Gamma$. Put

$$A_\Gamma = C_\Gamma^* C_\Gamma, \quad D_\Gamma = A_\Gamma - B_\Gamma.$$

The scalar defect

$$m_\Gamma(\xi) = w_\Gamma(\xi) - G(\xi) = 1 + |G(\xi)| - G(\xi)$$

is bounded, nonnegative and even. Hence

$$D_\Gamma = \iota^* \mathcal{F}^{-1} M_{m_\Gamma} \mathcal{F} \iota$$

is compact, positive and self-adjoint.

13.5 The exact even carrier and its shifted compact route

Let $B_{\text{ev}} = r_{\text{ev}} \otimes r_{\text{ev}}$ on X_3^+ , where $(r \otimes r)h = \langle h, r \rangle r$. Restrict the prime and Gamma operators to X_3^+ and set

$$A_0 = A_{\Gamma,+} + B_{\text{ev}}, \quad C_+ = D_{\Gamma,+} + B_{\text{prime},+}.$$

Theorem 13.9 (Exact critical even split). *The even Weil representative satisfies*

$$\boxed{W_3^+ = A_0 - C_+}.$$

The operator A_0 is positive and injective, with

$$0 < A_0 \leq \frac{29}{12}I,$$

while C_+ is compact and self-adjoint.

Proof. On the even sector the pole form is $+B_{\text{ev}}$, while $B_{\Gamma,+} = A_{\Gamma,+} - D_{\Gamma,+}$. Substitution into

$$W_3^+ = B_{\text{ev}} + B_{\Gamma,+} - B_{\text{prime},+}$$

gives the split. The upper bound follows from $A_{\Gamma,+} \leq I$ and $\|B_{\text{ev}}\| \leq 17/12$. The frequency channel is injective, so A_0 has trivial kernel. Compactness of C_+ follows from the two compact summands. $\square$

Injectivity is not coercivity. In fact the exact critical even carrier has an essential zero threshold.

Theorem 13.10 (Essential-zero threshold). *There exists a singular sequence $(f_R) \subset X_3^+$ such that*

$$\|f_R\|_{X_3} = 1, \quad f_R \rightharpoonup 0, \quad \|W_3^+ f_R\|_{X_3} \rightarrow 0.$$

Consequently

$$\boxed{0 \in \sigma_{\text{ess}}(W_3^+)}.$$

Proof. Choose nonzero even $\phi \in C_c^\infty((-1/4, 1/4))$ and set

$$u_R(x) = \phi(x - R) + \phi(x + R), \quad f_R = \frac{u_R}{\|u_R\|_{X_3}}.$$

The spatial component of $\|u_R\|_{X_3}$ grows like $e^{3R/2}$, while its weighted Fourier component stays bounded. Hence $C_\Gamma f_R \rightarrow 0$. The even pole functional grows only like $e^{R/2}$ before normalization, so $\ell_{\text{ev}}(f_R) \rightarrow 0$. Therefore $A_0 f_R \rightarrow 0$. The sequence is weakly null in the native two-channel realization, and compactness gives $C_+ f_R \rightarrow 0$. Thus $W_3^+ f_R \rightarrow 0$. Weyl's criterion gives the conclusion. $\square$

Corollary 13.11 (Unshifted coercive compact-reference no-go). *There do not exist a bounded operator $A \geq aI$ with $a > 0$ and a compact self-adjoint K such that*

$$W_3^+ = A^{1/2}(I - K)A^{1/2}.$$

Proof. Such a factorization would make W_3^+ Fredholm, contradicting [Theorem 13.10](#). $\square$

For $\eta > 0$, define

$$A_{+, \eta} = A_0 + \eta I, \quad K_{+, \eta} = A_{+, \eta}^{-1/2} C_+ A_{+, \eta}^{-1/2}.$$

Theorem 13.12 (Shifted even Birman–Schwinger reduction). *For every $\eta > 0$, $K_{+, \eta}$ is compact and self-adjoint and*

$$W_3^+ + \eta I = A_{+, \eta}^{1/2}(I - K_{+, \eta})A_{+, \eta}^{1/2}.$$

Therefore

$$W_3^+ + \eta I \geq 0 \iff \sup \sigma(K_{+, \eta}) \leq 1.$$

Moreover,

$$W_3^+ \geq 0 \iff W_3^+ + \eta_j I \geq 0 \quad \text{for every member of one sequence } \eta_j \downarrow 0.$$

13.6 The certified odd block and exact Feshbach complement

The parity reduction makes clear that an even certificate cannot decide the odd sector. The MP stack supplies a state-locked odd packet

$$V_3 = \text{span}\{u, |x|u, |x|^2u\}, \quad E_3 = \text{span}\{V_3, e_3\} \subset X_3^-.$$

For this packet, the certified Weil Gram G_3 on V_3 and the Schur scalar M_3 satisfy

$$\lambda_{\min}(G_3) = 2.0838177120437723 \times 10^{-13} > 0,$$

$$3.08520376681449 \times 10^{-13} < M_3 < 2.993597611951253 \times 10^{-12}.$$

Thus the bordered Weil Gram on E_3 is positive definite. These numerical inequalities belong to one hash-locked continuum packet; they are not promoted to a statement about the complement.

Let P_3 be the X_3 -orthogonal projection onto E_3 and $Q_3 = I - P_3$ on X_3^- . Relative to

$$X_3^- = E_3 \oplus Q_3 X_3^-,$$

write

$$W_3^- = \begin{pmatrix} A_3 & C_3^* \\ C_3 & D_3 \end{pmatrix}.$$

The finite certificate gives $A_3 > 0$.

Definition 13.13 (Odd Feshbach complement). Define

$$F_3 = D_3 - C_3 A_3^{-1} C_3^*.$$

Theorem 13.14 (Exact odd Feshbach reduction). *The operator F_3 is bounded and self-adjoint on $Q_3X_3^-$, and*

$$\boxed{W_3^- \geq 0 \iff F_3 \geq 0.}$$

Consequently

$$\boxed{W_3 \geq 0 \iff W_3^+ \geq 0 \text{ and } F_3 \geq 0.}$$

Proof. The exact factorization is

$$W_3^- = \begin{pmatrix} I & 0 \\ C_3 A_3^{-1} & I \end{pmatrix} \begin{pmatrix} A_3 & 0 \\ 0 & F_3 \end{pmatrix} \begin{pmatrix} I & A_3^{-1} C_3^* \\ 0 & I \end{pmatrix}.$$

The triangular factors are boundedly invertible. Since $A_3 > 0$ on the finite-dimensional space E_3 , congruence gives the first equivalence. Combine it with [Theorem 13.8](#). $\square$

For $\eta > 0$, set

$$A_{3,\eta} = A_3 + \eta I_{E_3}, \quad D_{3,\eta} = D_3 + \eta I_{Q_3 X_3^-},$$

$$F_{3,\eta} = D_{3,\eta} - C_3 A_{3,\eta}^{-1} C_3^*.$$

Then

$$W_3^- + \eta I \geq 0 \iff F_{3,\eta} \geq 0, \quad F_{3,\eta} \longrightarrow F_3 \quad \text{in norm as } \eta \downarrow 0.$$

13.7 Framework theorem before the RH specialization

Define the reduced Hilbert carrier

$$\mathcal{X}_{\text{red}} = X_3^+ \oplus Q_3 X_3^-$$

and the reduced self-adjoint operator

$$\boxed{\mathbb{W}_0 = W_3^+ \oplus F_3.}$$

Theorem 13.15 (Native recognition-seam Weil carrier). *The construction above has the following exact properties.*

- (i) X_3 is a native two-channel Hilbert completion compactly embedded in $L^2(\mathbb{R})$.
- (ii) The completed critical Weil form has a bounded self-adjoint representative W_3 .
- (iii) The primitive reflection cut extends to J_3 and derives the reducing Hilbert decomposition $X_3 = X_3^+ \oplus X_3^-$.
- (iv) The certified positive odd block and Feshbach map give

$$W_3 \geq 0 \iff \mathbb{W}_0 \geq 0.$$

- (v) The bounded nonnegative operator

$$\boxed{\mathfrak{A}_{\text{SS}} = \left[-\mathbb{W}_0(I + \mathbb{W}_0^2)^{-1/2} \right]_+}$$

satisfies

$$\boxed{\mathfrak{A}_{\text{SS}} = 0 \iff \mathbb{W}_0 \geq 0 \iff W_3 \geq 0.}$$

Its support projection is

$$\boxed{\mathbf{1}_{(0,\infty)}(\mathfrak{A}_{\text{SS}}) = \mathbf{1}_{(-\infty,0)}(\mathbb{W}_0).}$$

Proof. Only item (v) remains. Apply continuous functional calculus to

$$\mathfrak{s}_-(x) = \max \left\{ 0, \frac{-x}{\sqrt{1+x^2}} \right\}.$$

This function vanishes exactly on $[0, \infty)$ and is strictly positive on $(-\infty, 0)$. The zero and support statements follow from the spectral theorem. $\square$

Corollary 13.16 (RH as a candidate specialization). *Under the classical completed Weil criterion,*

$$\boxed{\text{RH} \iff W_3 \geq 0 \iff \mathfrak{A}_{\text{SS}} = 0.}$$

This corollary consumes the classical criterion. The framework theorem itself is the operator-geometric equivalence between completed-form positivity and vanishing of the native anti-seam operator.

14 Response-curvature provenance and scope

The normalized operator-valued Lambda response used below has a broader response-geometric motivation: a connection may encode phase, scale and seam transport, while its curvature records failure of path closure. That motivation is developed separately in the Recognition-Seam Calculus and UGD programme.

No thermodynamic scalar is identified here with the arithmetic seam integer. In particular,

$$\Gamma_c \Gamma_m \in \mathbb{R}_{>0}, \quad \mathfrak{k}_\Sigma(\eta) \in \mathbb{Z}_{\geq 0},$$

so a relation between them would require an explicit curvature-to-operator intertwiner followed by a threshold-index theorem. Such an intertwiner is not used in the present completed-Weil reduction.

Accordingly, the journal argument begins directly with the native boundary-Gamma-prime operator, its compact relative response and the exact finite obstruction. The response-curvature interpretation is retained only as provenance and does not constitute an additional route to the Riemann hypothesis.

15 The native compact Lambda response on the odd complement

The preceding section constructs the completed-Weil operator directly on the multiplicative carrier and identifies X_3 as its logarithmic chart. We now close the first two odd-complement obligations that were previously left conditional: a positive self-adjoint reference and a compact self-adjoint relative operator.

15.1 Native odd Feshbach reduction

Let

$$E_3^\times = \mathcal{L}^{-1} E_3 \subset \mathcal{X}_{\times,3}^-$$

be the transported four-dimensional certified odd block, let $P_3^\times$ be the native-energy orthogonal projection onto $E_3^\times$, and set

$$Q_3^\times = I - P_3^\times.$$

Relative to

$$\mathcal{X}_{\times,3}^- = E_3^\times \oplus Q_3^\times \mathcal{X}_{\times,3}^-,$$

write

$$W_{\times}^{-} = \begin{pmatrix} A_3^{\times} & (C_3^{\times})^* \\ C_3^{\times} & D_3^{\times} \end{pmatrix}.$$

Unitary transport of the state-locked interval certificate gives

$$A_3^{\times} > 0.$$

Define

$$F_3^{\times} = D_3^{\times} - C_3^{\times} (A_3^{\times})^{-1} (C_3^{\times})^*.$$

The triangular Feshbach congruence gives

$$\boxed{W_{\times}^{-} \geq 0 \iff F_3^{\times} \geq 0.}$$

For $\eta > 0$, put

$$\begin{aligned} A_{3,\eta}^{\times} &= A_3^{\times} + \eta I_{E_3^{\times}}, \\ F_{3,\eta}^{\times} &= D_3^{\times} + \eta I - C_3^{\times} (A_{3,\eta}^{\times})^{-1} (C_3^{\times})^*. \end{aligned}$$

Then

$$W_{\times}^{-} + \eta I \geq 0 \iff F_{3,\eta}^{\times} \geq 0, \quad F_{3,\eta}^{\times} \rightarrow F_3^{\times} \quad \text{in norm.}$$

15.2 Positive Gamma reference and compact defect

Split the critical Gamma symbol as

$$G = G_+ - G_-, \quad G_+ = \max\{G, 0\}, \quad G_- = \max\{-G, 0\}.$$

Let

$$A_{\Gamma,+}^{\times,-} = (\iota_{\times}^{-})^* \mathcal{M}^{-1} M_{G_+} \mathcal{M} \iota_{\times}^{-}$$

be the positive Gamma operator on the odd sector, and let

$$D_{\Gamma,-}^{\times,-} = (\iota_{\times}^{-})^* \mathcal{M}^{-1} M_{G_-} \mathcal{M} \iota_{\times}^{-}$$

be the negative Gamma defect. The second operator is compact because G_- is bounded and $\iota_{\times}^{-}$ is compact.

Let

$$B_{\text{odd}}^{\times} = r_{\text{odd}}^{\times} \otimes r_{\text{odd}}^{\times}$$

be the positive rank-one operator associated with the odd boundary functional; its contribution to the Weil form has negative sign. Define

$$\boxed{\mathcal{C}_{-}^{\times} = D_{\Gamma,-}^{\times,-} + B_{\text{prime}}^{\times,-} + B_{\text{odd}}^{\times}.$$

Then $\mathcal{C}_{-}^{\times}$ is compact and self-adjoint and

$$\boxed{W_{\times}^{-} = A_{\Gamma,+}^{\times,-} - \mathcal{C}_{-}^{\times}.$$

On the native odd complement

$$\mathcal{H}_3^{\times} = Q_3^{\times} \mathcal{X}_{\times,3}^{-},$$

define

$$R_{3,\eta}^{\times,\sharp} = Q_3^\times A_{\Gamma,+}^{\times,-} Q_3^\times + \eta I_{\mathcal{H}_3^\times}.$$

Also define

$$G_{3,\eta}^{\times,\sharp} = Q_3^\times \mathcal{C}_-^\times Q_3^\times + C_3^\times (A_{3,\eta}^\times)^{-1} (C_3^\times)^*.$$

Theorem 15.1 (Self-adjoint complement reference). *For every $\eta > 0$, $R_{3,\eta}^{\times,\sharp}$ is bounded, positive and self-adjoint on $\mathcal{H}_3^\times$, and*

$$R_{3,\eta}^{\times,\sharp} \geq \eta I.$$

Hence it is boundedly invertible and has a bounded positive inverse square root.

Proof. The compressed positive Gamma operator is bounded, positive and self-adjoint. Adding ηI gives the stated coercive lower bound. $\square$

Theorem 15.2 (Compact odd relative operator). *For every $\eta > 0$, $G_{3,\eta}^{\times,\sharp}$ is compact and self-adjoint, and*

$$F_{3,\eta}^\times = R_{3,\eta}^{\times,\sharp} - G_{3,\eta}^{\times,\sharp}.$$

Consequently

$$\Lambda_{3,\eta}^{(1)} = (R_{3,\eta}^{\times,\sharp})^{-1/2} G_{3,\eta}^{\times,\sharp} (R_{3,\eta}^{\times,\sharp})^{-1/2}$$

is compact and self-adjoint and satisfies

$$F_{3,\eta}^\times = (R_{3,\eta}^{\times,\sharp})^{1/2} (I - \Lambda_{3,\eta}^{(1)}) (R_{3,\eta}^{\times,\sharp})^{1/2}.$$

Thus

$$F_{3,\eta}^\times \geq 0 \iff \sup \sigma(\Lambda_{3,\eta}^{(1)}) \leq 1.$$

Proof. The first summand of $G_{3,\eta}^{\times,\sharp}$ is a compression of a compact self-adjoint operator. The second has range contained in $\text{Ran } C_3^\times$, which is finite dimensional because the source $E_3^\times$ is finite dimensional. Hence the second summand is finite rank. The exact identity follows by inserting

$$D_3^\times = Q_3^\times A_{\Gamma,+}^{\times,-} Q_3^\times - Q_3^\times \mathcal{C}_-^\times Q_3^\times$$

into the shifted Feshbach formula. Sandwiching the compact defect by the bounded inverse square root proves compactness of $\Lambda_{3,\eta}^{(1)}$, and the sign equivalence follows by boundedly invertible congruence. $\square$

No trace-class assertion is required for the sign theorem and none is made here.

15.3 Lambda response and the logarithmic chart

The notation $\Lambda_{3,\eta}^{(1)}$ records the first normalized operator-valued response of the shifted odd seam. Its mathematical content is the exact identity

$$F_{3,\eta}^\times = R^{1/2}(I - \Lambda_{3,\eta}^{(1)})R^{1/2};$$

the notation alone adds no positivity claim.

Let $K_{3,\eta}^\sharp$ denote the logarithmic-chart relative operator. The unitary representation theorem gives

$$\mathcal{L}\Lambda_{3,\eta}^{(1)}\mathcal{L}^{-1} = K_{3,\eta}^\sharp.$$

Hence compactness, spectrum, threshold multiplicity and sign are identical in the native and logarithmic descriptions.

Proposition 15.3 (First response derivative). *The shifted Feshbach complement is norm differentiable and*

$$\frac{d}{d\eta}F_{3,\eta}^\times = I + C_3^\times (A_3^\times + \eta I)^{-2} (C_3^\times)^* \geq I.$$

In particular, $F_{3,\eta}^\times$ is increasing in η .

Proof. Differentiate the resolvent term using

$$\frac{d}{d\eta}(A_3^\times + \eta I)^{-1} = -(A_3^\times + \eta I)^{-2}.$$

□

15.4 Threshold charge and a large-shift anchor

Because $\Lambda_{3,\eta}^{(1)}$ is compact and self-adjoint, define the odd threshold charge for every $\eta > 0$ by

$$\mathfrak{k}_3(\eta) = \text{rank } \mathbf{1}_{(1,\infty)}(\Lambda_{3,\eta}^{(1)}).$$

It is finite and

$$\mathfrak{k}_3(\eta) = 0 \iff F_{3,\eta}^\times \geq 0.$$

The following elementary norm bound provides a genuine zero-charge starting ray.

Theorem 15.4 (Large-shift odd no-crossing anchor). *The operator bounds*

$$\|C_-^\times\| \leq \frac{901}{120}, \quad \|C_3^\times\| \leq \frac{1021}{120}$$

imply

$$\|\Lambda_{3,\eta}^{(1)}\| \leq \beta(\eta) := \frac{901}{120\eta} + \frac{1021^2}{120^2\eta^2}.$$

The positive solution of $\beta(\eta) = 1$ is

$$\eta_* = 13.053924914719259 \dots$$

Therefore

$$\eta \geq 14 \implies \mathfrak{k}_3(\eta) = 0 \text{ and } F_{3,\eta}^\times > 0.$$

At $\eta = 14$,

$$\beta(14) = \frac{2556121}{2822400} = 0.905655116213152\dots,$$

so the threshold reserve exceeds 0.0943.

Proof. Since $R_{3,\eta}^{\times,\sharp} \geq \eta I$,

$$\|\Lambda_{3,\eta}^{(1)}\| \leq \eta^{-1}\|C_-^\times\| + \eta^{-2}\|C_3^\times\|^2.$$

The displayed constants give the majorant. It is decreasing for $\eta > 0$, and direct evaluation gives the stated threshold and anchor. $\square$

The difficult direction is therefore a controlled descent from the zero-charge ray toward $\eta = 0$, not the existence of the odd compact operator.

16 Target-faithful reconstruction and compatible finite transfer

A subcritical finite eigenvalue is useful only when the finite representation controls every target-relevant omitted direction. This section records the representation and transfer theorems used later by the outward five-source certificate. Earlier finite-grid ladders and state-locked pilot tables are kept in the reproducibility repository rather than repeated in the journal argument.

16.1 Target-relevant spectral blindness

Let $R : \mathcal{H} \rightarrow \mathcal{Y}$ be a representation and $\Pi : \mathcal{H} \rightarrow \mathcal{Z}$ the declared target map.

Definition 16.1 (Target-relevant blind quotient). Define

$$\mathfrak{B}(R; \Pi) = \frac{\ker R}{\ker R \cap \ker \Pi}.$$

The representation is target faithful exactly when $\mathfrak{B}(R; \Pi) = 0$.

Theorem 16.2 (Unitary invariance of blindness). *Let $U : \mathcal{H} \rightarrow \widetilde{\mathcal{H}}$ and $V : \mathcal{Y} \rightarrow \widetilde{\mathcal{Y}}$ be unitary, and set*

$$\widetilde{R} = V R U^{-1}, \quad \widetilde{\Pi} = \Pi U^{-1}.$$

Then U induces a canonical isomorphism

$$\mathfrak{B}(R; \Pi) \cong \mathfrak{B}(\widetilde{R}; \widetilde{\Pi}).$$

Unitary transport also preserves kernel, spectrum, multiplicity, positivity, compactness and Fredholm status.

Proof. One has $\ker \widetilde{R} = U \ker R$ and $\ker \widetilde{\Pi} = U \ker \Pi$. The quotient identity follows, and the operator statements are standard consequences of unitary equivalence. $\square$

Thus a change of chart cannot repair a structurally blind carrier. The finite odd block remains blind to its complement until the Feshbach lift is included, and the odd sector remains blind to the independent even sector.

16.2 Blind-complement Schur closure

Let $F = F^*$ and $R = R^* \geq r_0 I > 0$, and put

$$\hat{F} = R^{-1/2} F R^{-1/2}.$$

Let $J_N : \mathcal{H}_N \rightarrow \mathcal{H}$ be an isometry with

$$P_N = J_N J_N^*, \quad Q_N = I - P_N.$$

Suppose a finite model proves $\hat{F}_N \geq \mu I$ and define

$$\varepsilon_{\text{vis}} = \|P_N \hat{F} P_N - J_N \hat{F}_N J_N^*\|, \quad a = \mu - \varepsilon_{\text{vis}}.$$

Assume

$$Q_N \hat{F} Q_N \geq \delta Q_N, \quad \beta = \|Q_N \hat{F} P_N\|.$$

Theorem 16.3 (Native blind-kernel Schur closure). *Under these hypotheses,*

$$\boxed{F \geq \lambda_- R}, \quad \lambda_- = \frac{a + \delta - \sqrt{(a - \delta)^2 + 4\beta^2}}{2}.$$

In particular,

$$\boxed{\varepsilon_{\text{vis}} + \frac{\beta^2}{\delta} < \mu \implies F > 0.}$$

Proof. For $u = p + q$, with $p = P_N u$ and $q = Q_N u$, the hypotheses give

$$\langle \hat{F} u, u \rangle \geq a \|p\|^2 + \delta \|q\|^2 - 2\beta \|p\| \|q\|.$$

The right-hand side is controlled by the least eigenvalue of $\begin{pmatrix} a & -\beta \\ -\beta & \delta \end{pmatrix}$. Congruence by $R^{1/2}$ gives the result. $\square$

16.3 Visible generalized-eigenvalue transfer

Let $S_N : \mathbb{C}^{m_N} \rightarrow \mathcal{H}$ be a finite synthesis and define

$$R_N = S_N^* R S_N, \quad F_N = S_N^* F S_N.$$

After deleting $\ker R_N$, set

$$J_N = R^{1/2} S_N R_N^{-1/2}.$$

Then $J_N^* J_N = I$ and

$$J_N^* (I - R^{-1/2} F R^{-1/2}) J_N = I - R_N^{-1/2} F_N R_N^{-1/2}.$$

Let $\tilde{F}_N, \tilde{R}_N$ be computed self-adjoint matrices and assume

$$\varepsilon_F = \|R_N^{-1/2} (F_N - \tilde{F}_N) R_N^{-1/2}\|,$$

$$\varepsilon_R = \|R_N^{-1/2} (R_N - \tilde{R}_N) R_N^{-1/2}\| < 1.$$

Put

$$\tilde{\lambda}_N = \lambda_{\min}(\tilde{F}_N, \tilde{R}_N), \quad \tilde{a}_N = 1 - \tilde{\lambda}_N.$$

Proposition 16.4 (Visible generalized-eigenvalue transfer). *If $\tilde{\lambda}_N \geq 0$, then*

$$\lambda_{\max} \left(J_N^* (I - R^{-1/2} F R^{-1/2}) J_N \right) \leq \tilde{a}_N + (1 - \tilde{a}_N) \varepsilon_R + \varepsilon_F.$$

Proof. The reference error gives $\tilde{R}_N \geq (1 - \varepsilon_R) R_N$. Combining $\tilde{F}_N \geq \tilde{\lambda}_N \tilde{R}_N$ with the form error gives

$$F_N \geq [\tilde{\lambda}_N (1 - \varepsilon_R) - \varepsilon_F] R_N.$$

Taking one minus this relative lower bound proves the estimate. $\square$

16.4 Native channel reconstruction and compatible tails

The native norm supplies an isometric analysis map

$$\mathcal{C}_\times : \mathcal{X}_{\times,3} \rightarrow \mathcal{Y}_\times, \quad \mathcal{C}_\times^* \mathcal{C}_\times = I.$$

For a finite-rank channel projection Π_N , define

$$Z_N = Q_3^\times \mathcal{C}_\times^*|_{\text{Ran } \Pi_N}, \quad H_N = Z_N^* Z_N.$$

After restriction to $(\ker H_N)^\perp$, put

$$J_N = Z_N H_N^{-1/2}, \quad P_N = J_N J_N^*.$$

For a compact self-adjoint native response $\mathbf{\Lambda}$, set

$$\mathbf{\Lambda}_N = J_N^* \mathbf{\Lambda} J_N, \quad \tau_N = \|(I - \Pi_N) \mathcal{C}_\times \mathbf{\Lambda}\|.$$

Theorem 16.5 (Compatible response-tail gate). *With $Q_N = I - P_N$,*

$$\|Q_N \mathbf{\Lambda}\| \leq \tau_N.$$

Hence the same τ_N controls the cross block and omitted complement. If

$$\|\mathbf{\Lambda}_N - \tilde{\Lambda}_N\| \leq \varepsilon_{\text{vis}}, \quad a_N = \lambda_{\max}(\tilde{\Lambda}_N) + \varepsilon_{\text{vis}},$$

then

$$\sup \sigma(\mathbf{\Lambda}) \leq \mathcal{U}(a_N, \tau_N),$$

where

$$\mathcal{U}(a, \tau) = \frac{a + \tau + \sqrt{(a - \tau)^2 + 4\tau^2}}{2}.$$

Proof. Since $Q_N Z_N = 0$,

$$Q_N \mathbf{\Lambda} = Q_N \mathcal{C}_\times^* (I - \Pi_N) \mathcal{C}_\times \mathbf{\Lambda}.$$

The factors outside the declared tail are contractions. The spectral bound is the top eigenvalue of the associated two-block scalar majorant. $\square$

This theorem forbids replacing a compatible operator-response tail by an unrelated coefficient tail.

16.5 Exact native standing-wave frame

For $L > 0$ and $k \geq 1$, define

$$\Psi_k^\times(r) = \frac{1}{\sqrt{L}} \max(r^{3/2}, r^{-3/2})^{-1} \sin\left(\frac{k\pi \log r}{L}\right) \mathbf{1}_{\{|\log r| \leq L\}}.$$

Theorem 16.6 (Seam-interval standing-wave frame). *The atoms are inversion odd and satisfy*

$$\int_0^\infty \max(r^3, r^{-3}) \Psi_k^\times(r) \overline{\Psi_j^\times(r)} \, d^\times r = \delta_{kj}.$$

They are complete in the inversion-odd spatial channel on $|\log r| \leq L$.

Proof. Under $x = \log r$, cancellation of the native weight reduces the atoms to the standard Dirichlet sine basis $L^{-1/2} \sin(k\pi x/L)$ on $[-L, L]$. $\square$

In the logarithmic chart,

$$\psi_k(x) = L^{-1/2} e^{-3|x|/2} \sin(k\pi x/L) \mathbf{1}_{[-L, L]}(x).$$

Writing $a = 3/2$, $\omega_k = k\pi/L$ and

$$J_L(d) = \frac{a + e^{-aL}[-a \cos(dL) + d \sin(dL)]}{a^2 + d^2},$$

one obtains

$$\hat{\psi}_k(z) = \frac{i}{\sqrt{L}} [J_L(z - \omega_k) - J_L(z + \omega_k)].$$

Thus the Gamma and strip-boundary prime channels have explicit continuum matrix elements, with squared Gamma-frequency tail $O(\Omega^{-3} \log \Omega)$.

The earlier finite-grid and standing-wave ladders served to discover the high-risk directions and validate the transfer design. The actual fixed-shift claim used in this paper is the later outward five-source certificate, stated separately with its complete error ledger.

Part IV

Unified Completed-Weil Criterion

17 Native Weil geometry and the integer seam path

The native terminal architecture is

$$\kappa \longrightarrow \chi \longrightarrow J_\times \longrightarrow \mathfrak{W}_\times \longrightarrow W_\times \longrightarrow \mathbb{K}_\eta \longrightarrow \mathfrak{k}_\Sigma(\eta) \longrightarrow \mathcal{K}_0.$$

The primitive cut and recognition layers generate the represented inversion geometry. The boundary, Gamma and prime sectors generate the completed-Weil form and its self-adjoint representative. Parity, Feshbach reduction and positive congruence generate the compact threshold packet and integer seam charge. The exact eta-zero passage then identifies the sign with the five-dimensional endpoint matrix $\mathcal{K}_0$.

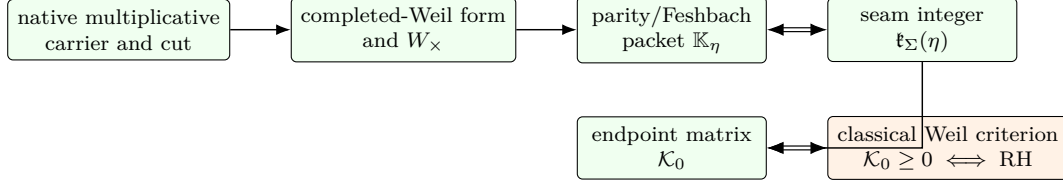


Figure 1: Final native Weil verification graph. The carrier, completed-Weil operator, compact threshold packet, seam integer and endpoint matrix belong to one theorem chain. The last equivalence uses the classical completed explicit formula and Weil criterion.

17.1 Native completed-Weil realization

On the multiplicative core, let

$$\mathfrak{B}_\times, \quad \mathfrak{A}_{\Gamma, \times}, \quad \mathfrak{P}_\times$$

denote the boundary, Gamma and prime forms, and put

$$\mathfrak{W}_\times = \mathfrak{B}_\times + \mathfrak{A}_{\Gamma, \times} - \mathfrak{P}_\times.$$

Theorem 17.1 (Cut-derived native Weil realization). *On the native completion $\mathcal{X}_{\times, 3}$:*

(i) $\mathfrak{W}_\times$ has a unique bounded self-adjoint representative

$$\mathfrak{W}_\times(F, H) = \langle F, W_\times H \rangle_{\times, 3}, \quad W_\times = W_\times^*;$$

(ii) the represented inversion cut reduces the operator,

$$[W_\times, J_\times] = 0, \quad W_\times = W_\times^+ \oplus W_\times^-;$$

(iii) the logarithmic map is unitary and satisfies

$$\mathcal{L}W_\times\mathcal{L}^{-1} = W_3;$$

(iv) the operator sign is the sign of the native form,

$$W_\times \geq 0 \iff \mathfrak{W}_\times(F, F) \geq 0 \quad (F \in \mathcal{X}_{\times, 3}).$$

Proof. Boundedness and Hermiticity give the Riesz representative. Cut covariance gives the parity reduction, and the logarithmic chart theorem gives the unitary intertwiner. The final statement is the diagonal Riesz identity. $\square$

17.2 Classical shadow membrane

Let

$$z_\rho = \frac{\rho - \frac{1}{2}}{i},$$

where ρ ranges over the nontrivial zeros of ζ with algebraic multiplicity. On the admissible core, the classical completed explicit formula identifies the native pairing with the symmetric zero sum

$$\mathfrak{W}_\xi(f, h) = \sum_{\rho}^{\text{sym}} \widehat{f}(z_\rho) \overline{\widehat{h}(z_\rho)}.$$

Proposition 17.2 (Native object and classical shadow). *The provenance is*

$$\boxed{\text{native boundary--Gamma--prime operator} \longrightarrow \text{classical completed zero sum} \longrightarrow \text{Weil positivity criterion.}}$$

The first arrow is the native representation theorem. The zero-sum identity and

$$W_{\times} \geq 0 \iff \text{RH}$$

are the classical completed explicit formula and Weil criterion in the declared normalization [Tit, Edw74].

17.3 Shifted parity and Feshbach packet

For $\eta > 0$, parity and Feshbach reduction give an even compact response $K_{+, \eta}$ and the odd response $K_{3, \eta}^{\sharp}$. Put

$$\mathbb{K}_{\eta} = K_{+, \eta} \oplus K_{3, \eta}^{\sharp}.$$

Proposition 17.3 (Shifted global equivalence). *For every $\eta > 0$,*

$$\boxed{W_3 + \eta I \geq 0 \iff K_{+, \eta} \leq I \text{ and } K_{3, \eta}^{\sharp} \leq I \iff \mathbb{K}_{\eta} \leq I.}$$

After the common-source and favorable even-boundary theorems,

$$\boxed{W_3 + \eta I \geq 0 \iff F_{3, \eta}^{\times} \geq 0 \iff B_{\eta} \leq I.}$$

Proof. Each equivalence is the positive-congruence factorization of the corresponding shifted restriction. The second line uses positivity of the even restriction and the finite-rank odd Birman–Schwinger theorem. $\square$

17.4 The integer seam obstruction

Definition 17.4 (Strict lifted seam charge). For $\eta > 0$, define

$$\boxed{\mathfrak{k}_{\Sigma}(\eta) = \text{rank } \mathbf{1}_{(1, \infty)}(\mathbb{K}_{\eta}).}$$

Theorem 17.5 (Additive charge and exact shifted sign). *For every $\eta > 0$,*

$$\boxed{\mathfrak{k}_{\Sigma}(\eta) = \mathfrak{k}_{+}(\eta) + \mathfrak{k}_3(\eta),}$$

and

$$\boxed{\mathfrak{k}_{\Sigma}(\eta) = 0 \iff W_3 + \eta I \geq 0.}$$

The single-sector theorem sharpens this to

$$\boxed{\mathfrak{k}_{\Sigma}(\eta) = \mathfrak{k}_3(\eta) = N_{+}(B_{\eta} - I).}$$

Proof. Spectral projections add under orthogonal direct sums. Positive congruence identifies threshold eigenvalues above one with negative directions of the shifted sign form. The even contribution vanishes by the favorable sign theorem. $\square$

Define the self-adjoint Fredholm path

$$\mathbb{D}_{\eta,t} = I - t\mathbb{K}_{\eta}, \quad 0 \leq t \leq 1.$$

Theorem 17.6 (Charge as one-sided spectral flow). *If $1 \notin \sigma(\mathbb{K}_{\eta})$, then*

$$\boxed{\text{sf}(\mathbb{D}_{\eta,t}; 0) = -\mathfrak{k}_{\Sigma}(\eta).}$$

Proof. For an eigenvalue μ of $\mathbb{K}_{\eta}$, the path eigenvalue $1 - t\mu$ crosses zero exactly when $\mu > 1$, and every such crossing is downward. $\square$

17.5 Native Weil integer seam-path theorem

Theorem 17.7 (Native Weil integer seam path). *For every positive shift, the native completed-Weil architecture produces the finite integer*

$$\boxed{\mathfrak{k}_{\Sigma}(\eta) = -\text{sf}(\mathbb{D}_{\eta,t}; 0) = N_+(B_{\eta} - I).}$$

At eta zero, inverse-half regularity and norm descent identify completed-Weil positivity with

$$\boxed{\mathcal{K}_0 \geq 0.}$$

Under the classical completed explicit formula and Weil criterion,

$$\boxed{\text{RH} \iff \mathcal{K}_0 \geq 0.}$$

Proof. The positive-shift identities are [Theorems 17.5](#) and [17.6](#) and the odd Birman–Schwinger inertia theorem. The eta-zero identity is [Theorem 31.5](#); the last equivalence is [Theorem 31.7](#). $\square$

18 Threshold correspondence with completed-Weil positivity

The threshold spectrum used here is the spectrum of the native compact response $\mathbb{K}_{\eta}$. It is not a claim that ordinates of nontrivial zeros are eigenvalues of one fixed self-adjoint operator. The connection to the Riemann hypothesis passes through the completed-Weil sign problem and the single-sector endpoint gap.

Definition 18.1 (Controlled seam-vanishing sequence). A sequence $\eta_j \downarrow 0$ is controlled if the compact packets $\mathbb{K}_{\eta_j}$ are constructed on the target-faithful completed carrier with the actual native source columns and the shifted forms converge lawfully to the native completed-Weil form.

Theorem 18.2 (Native threshold correspondence). *Assume the classical completed explicit-formula identification. Then the following are equivalent:*

- (i) *the unshifted five-label gap satisfies $\mathcal{K}_0 \geq 0$;*
- (ii) *for every positive shift, $\mathfrak{k}_{\Sigma}(\eta) = 0$;*
- (iii) *for one, equivalently every, controlled sequence $\eta_j \downarrow 0$, $\mathfrak{k}_{\Sigma}(\eta_j) = 0$ for every j ;*
- (iv) *the native completed-Weil representative is nonnegative;*
- (v) *the completed Weil form is nonnegative on its admissible core;*
- (vi) *the Riemann hypothesis holds.*

Proof. The exact critical-gap identity gives

$$\mathcal{K}_\eta = \mathcal{K}_0 + \eta D + (H_0 - H_\eta),$$

with nonnegative increments. Hence (i) implies zero odd threshold count for every positive shift. The single-sector theorem gives $\mathfrak{k}_\Sigma = \mathfrak{k}_3$, proving (i) $\Rightarrow$ (ii). The implication (ii) $\Rightarrow$ (iii) is immediate. Conversely, norm convergence of the five-label gap along a controlled sequence gives $\mathcal{K}_0 \geq 0$, so (iii) $\Rightarrow$ (i).

The exact single-sector endpoint closure gives (i) $\Rightarrow$ (iv). If the completed operator is nonnegative, its odd Feshbach complement is nonnegative; the generalized endpoint Schur theorem and inverse-half source regularity give $\mathcal{K}_0 \geq 0$, proving the converse. Bounded Riesz representation and density identify (iv) with (v). The classical explicit formula and Weil criterion give (v) $\iff$ (vi). $\square$

Corollary 18.3 (Seam vanishing implies the Riemann hypothesis). *If one controlled sequence $\eta_j \downarrow 0$ satisfies*

$$\mathfrak{k}_\Sigma(\eta_j) = 0 \quad \text{for every } j,$$

then the Riemann hypothesis holds. Equivalently, it is enough to certify the single matrix inequality $\mathcal{K}_0 \geq 0$.

This is a completed-Weil threshold correspondence, not a Hilbert–Pólya spectral-divisor theorem. No equality between zero ordinates and eigenvalues, no new determinant for ξ , and no multiplicity-preserving zero map is used. The remaining burden is the unshifted five-dimensional sign certificate.

19 Threshold spectral zeta and the seam-index trinity

Fix $\eta > 0$ and write

$$K = \mathbb{K}_\eta, \quad P_{>} = \mathbf{1}_{(1,\infty)}(K).$$

Because K is compact and self-adjoint, $P_{>}$ has finite rank. Put

$$A_\eta = (K - I)P_{>}.$$

On $P_{>}\mathcal{H}$, this operator is positive and invertible. If $P_{>} = 0$, the threshold zeta below is defined to be identically zero.

Definition 19.1 (Threshold spectral zeta). For $s \in \mathbb{C}$, define

$$\zeta_{\Sigma,\eta}^{\text{th}}(s) = \text{Tr}_{P_{>}\mathcal{H}}(A_\eta^{-s}) = \sum_{\lambda>1} m(\lambda)(\lambda - 1)^{-s}.$$

Theorem 19.2 (Threshold spectral-zeta index formula). *The function $\zeta_{\Sigma,\eta}^{\text{th}}$ is entire and*

$$\zeta_{\Sigma,\eta}^{\text{th}}(0) = \text{rank } P_{>} = \mathfrak{k}_\Sigma(\eta).$$

For every real q , all summands are nonnegative, and therefore

$$\mathfrak{k}_\Sigma(\eta) = 0 \iff \zeta_{\Sigma,\eta}^{\text{th}}(q) = 0$$

for one, equivalently every, real q .

Proof. The sum is finite. Each term is entire in s , and at $s = 0$ every strict threshold eigenvalue contributes its multiplicity. For real q , each term is strictly positive whenever $P_{>} \neq 0$. $\square$

Theorem 19.3 (Seam-index trinity). *Let*

$$D_\eta = I - \mathbb{K}_\eta.$$

If $1 \notin \sigma(\mathbb{K}_\eta)$, then

$$\mathfrak{k}_\Sigma(\eta) = N_-(D_\eta) = -\text{sf}(I - t\mathbb{K}_\eta; 0) = \zeta_{\Sigma, \eta}^{\text{th}}(0).$$

At a threshold endpoint, the spectral-flow term is interpreted by the same one-sided regularization used in the integer seam-path theorem, while the strict-threshold rank, Morse index and zeta value remain unchanged.

Proof. An eigenvalue $\lambda > 1$ of $\mathbb{K}_\eta$ is exactly a negative eigenvalue $1 - \lambda$ of D_η , with the same multiplicity. The one-sided spectral-flow identity is the native seam-path formula, and the preceding theorem supplies the zeta equality. $\square$

Thus Morse index, one-sided spectral flow and threshold zeta are coordinate descriptions of one finite obstruction. By the single-sector theorem,

$$\mathfrak{k}_\Sigma(\eta) = \mathfrak{k}_3(\eta),$$

so these avatars count only the adverse odd five-label sector after the common pure-source and favorable even-boundary signs have been established.

Corollary 19.4 (Threshold-zeta vanishing implies RH). *Let $\eta_j \downarrow 0$ be a controlled sequence. If for one real q ,*

$$\zeta_{\Sigma, \eta_j}^{\text{th}}(q) = 0 \quad \text{for every } j,$$

then $\mathcal{K}_0 \geq 0$ and the Riemann hypothesis holds. The same conclusion follows directly from an outward certificate $\mathcal{K}_0 \geq 0$.

Proof. The threshold-zeta formula gives zero seam charge at every shift. Norm descent of the five-label gap yields $\mathcal{K}_0 \geq 0$. Apply [Theorem 31.7](#). $\square$

No Schatten determinant is required for these identities. A determinant avatar may be introduced only after its ideal and contour hypotheses are proved, and it adds no terminal decision beyond the unshifted five-label gap.

20 Congruence-safe Morse zeta and the covariant two-sheet virial reduction

The threshold-zeta virial theorem is abstractly correct on the normalized compact packet $\mathbb{K}_\eta$. An operator-specific channel proof, however, must not silently differentiate

$$\mathbb{K}_\eta = \mathbb{A}_\eta^{-1/2}(\mathbb{A}_\eta - \mathbb{W}_\eta)\mathbb{A}_\eta^{-1/2}$$

as though the reference factors commuted with the conjugate observable. Such a step would import unrecorded commutators of $\mathbb{A}_\eta^{-1/2}$ and could manufacture a false sign. We therefore move the channelwise second attack to the shifted native sign operator $\mathbb{W}_\eta$ itself. The seam integer is preserved by the exact positive congruence, while the boundary, Gamma and prime channels remain visible before normalization.

20.1 The congruence-safe Morse spectral zeta

Recall the exact shifted factorization

$$\mathbb{W}_\eta = \mathbb{A}_\eta^{1/2}(I - \mathbb{K}_\eta)\mathbb{A}_\eta^{1/2}, \quad \mathbb{A}_\eta \geq \eta I.$$

Let

$$P_-^\eta = \mathbf{1}_{(-\infty, 0)}(\mathbb{W}_\eta), \quad H_\eta = (-\mathbb{W}_\eta)P_-^\eta.$$

The negative spectral subspace is finite dimensional because $I - \mathbb{K}_\eta$ is a self-adjoint Fredholm operator with finite Morse index and boundedly invertible congruence preserves inertia.

Definition 20.1 (Native Morse spectral zeta). For $s \in \mathbb{C}$, define

$$\zeta_{\Sigma, \eta}^{\text{M}}(s) = \text{Tr}_{P_-^\eta \mathcal{H}}(H_\eta^{-s}) = \sum_{\nu \in \sigma(\mathbb{W}_\eta), \nu < 0} m(\nu)(-\nu)^{-s}.$$

If $P_-^\eta = 0$, the function is defined to be identically zero.

Theorem 20.2 (Congruence-safe zeta index). *The Morse spectral zeta is entire and*

$$\zeta_{\Sigma, \eta}^{\text{M}}(0) = N_-(\mathbb{W}_\eta) = \mathfrak{k}_\Sigma(\eta) = \zeta_{\Sigma, \eta}^{\text{th}}(0).$$

For every real q , the value $\zeta_{\Sigma, \eta}^{\text{M}}(q)$ is nonnegative and vanishes if and only if the seam charge vanishes.

Proof. The defining sum is finite. At $s = 0$ it counts negative eigenvalues with multiplicity. Sylvester inertia under the displayed positive congruence gives

$$N_-(\mathbb{W}_\eta) = N_-(I - \mathbb{K}_\eta) = \text{rank } \mathbf{1}_{(1, \infty)}(\mathbb{K}_\eta).$$

Every summand is strictly positive on the real axis whenever the negative sector is nonempty. $\square$

Remark 20.3 (Only the index is congruence invariant). The two spectral-zeta functions need not agree away from $s = 0$. Positive congruence preserves inertia, not the magnitudes of negative eigenvalues. The identity consumed by the terminal proof is therefore

$$\zeta_{\Sigma, \eta}^{\text{M}}(0) = \zeta_{\Sigma, \eta}^{\text{th}}(0),$$

not an unlicensed equality of complete zeta functions.

20.2 Morse-weighted virial annihilation

Let $\mathbf{A} = \mathbf{A}^*$ be a native conjugate observable such that $\mathbb{W}_\eta \in C^1(\mathbf{A})$ and put

$$B_{\eta, \mathbf{A}}^{\text{M}} = i[\mathbb{W}_\eta, \mathbf{A}].$$

For real q , define

$$\mathcal{V}_{\eta, \mathbf{A}}^{\text{M}}(q) = \text{Tr}\left(H_\eta^{-q/2} P_-^\eta B_{\eta, \mathbf{A}}^{\text{M}} P_-^\eta H_\eta^{-q/2}\right).$$

Theorem 20.4 (Morse-weighted virial identity). *For every real q ,*

$$\boxed{\mathcal{V}_{\eta,A}^M(q) = 0.}$$

If for some $c_\eta > 0$ and $\beta \in \mathbb{R}$,

$$\boxed{P_-^\eta B_{\eta,A}^M P_-^\eta \geq c_\eta H_\eta^\beta,}$$

then

$$\boxed{P_-^\eta = 0, \quad \mathfrak{k}_\Sigma(\eta) = 0, \quad \zeta_{\Sigma,\eta}^M \equiv 0.}$$

Proof. The diagonal commutator block vanishes on every eigenspace of $\mathbb{W}_\eta$. This proves the weighted trace identity. Taking $q = \beta$ in the strict estimate gives

$$0 \geq c_\eta \text{rank } P_-^\eta.$$

The rank is nonnegative, so it is zero. □

This formulation is the native channel route. It avoids differentiating the reference square root and is therefore immune to a false sign introduced by a normalizing coordinate.

20.3 The common two-sheet sign symbol

Let

$$g(\xi) = \text{Re } \psi\left(\frac{1}{4} + \frac{i\xi}{2}\right) - \log \pi, \quad w_\Gamma(\xi) = 1 + |g(\xi)|,$$

and let

$$q(\xi) = \sum_{n \geq 2} \frac{\Lambda(n)}{n^2} e^{i\xi \log n}, \quad \mathcal{Q}(\xi) = \begin{pmatrix} 0 & q(\xi) \\ q(\xi) & 0 \end{pmatrix}.$$

Before the exact finite-rank boundary and Feshbach augmentations are added, the frequency part of the shifted sign form has the same scalar Gamma coefficient in the direct even and direct odd sectors:

$$\boxed{\omega_\eta(\xi) = g(\xi) + \eta w_\Gamma(\xi).}$$

Indeed,

$$(1 + \eta)(1 + |g|) - (1 + |g| - g) = \eta(1 + |g|) + g$$

in the even sector, while

$$\eta(1 + |g|) + g_+ - g_- = \eta(1 + |g|) + g$$

in the odd sector. The common two-sheet source symbol is therefore

$$\boxed{\Omega_\eta(\xi) = \omega_\eta(\xi) I_2 - \mathcal{Q}(\xi).}$$

This identity is obtained before reference normalization.

20.4 Covariant removal of the prime phase

On an interval where $q(\xi) \neq 0$, write

$$q(\xi) = \rho(\xi)e^{i\phi(\xi)}, \quad \widehat{\mathcal{Q}}(\xi) = \rho(\xi)^{-1}\mathcal{Q}(\xi).$$

Let v be real and odd. In the two-sheet spectral chart define the Hermitian connection

$$\mathcal{H}_{v,q}(\xi) = \frac{1}{2}v(\xi)\phi'(\xi) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and the covariant first-order observable

$$\mathbf{A}_{v,q}^{\text{cov}} = i \left(v\partial_\xi + \frac{1}{2}v' \right) + M_{\mathcal{H}_{v,q}}.$$

Theorem 20.5 (Covariant two-sheet commutator). *On every interval where q is nonzero,*

$$i[M_{\Omega_\eta}, \mathbf{A}_{v,q}^{\text{cov}}] = M_{v(\omega'_\eta I_2 - \rho' \widehat{\mathcal{Q}})}.$$

Consequently, wherever $v \geq 0$,

$$\lambda_{\min} \left(v(\omega'_\eta I_2 - \rho' \widehat{\mathcal{Q}}) \right) = v(\omega'_\eta - |\rho'|).$$

Proof. The matrix multiplier commutator is

$$v\Omega'_\eta + i[\Omega_\eta, \mathcal{H}_{v,q}].$$

The diagonal connection removes the angular derivative of the off-diagonal prime symbol. Explicitly,

$$v\mathcal{Q}' + i[\mathcal{Q}, \mathcal{H}_{v,q}] = v\rho' \widehat{\mathcal{Q}}.$$

Substitution gives the first identity. Since $\widehat{\mathcal{Q}}$ is a Hermitian involution, the eigenvalue formula follows. $\square$

The connection removes only the phase derivative. The radial derivative ρ' is gauge invariant and therefore represents the true prime obstruction to a strict source-symbol virial inequality.

20.5 The active-band criterion

Let $I_\eta = (-\Omega_\eta^*, \Omega_\eta^*)$ be a symmetric interval. The following theorem isolates the exact remaining obligations without importing a mode cutoff.

Theorem 20.6 (Covariant active-band seam-annihilation criterion). *Assume:*

- (i) q has no zero on I_η ;
- (ii) the lower source-symbol branch satisfies

$$\omega_\eta(\xi) - |q(\xi)| \geq 0 \quad (|\xi| \geq \Omega_\eta^*);$$

- (iii) for an odd v with $v(\xi) > 0$ on $(0, \Omega_\eta^*)$,

$$\omega'_\eta(\xi) - |\rho'(\xi)| > 0 \quad (0 < \xi < \Omega_\eta^*);$$

- (iv) the compatible native channel reconstruction has no nonzero negative state blind to the open set on which the preceding gap is positive;
- (v) the exact finite-rank and reconstruction commutator error is strictly smaller than the resulting finite-dimensional observability constant.

Then

$$\boxed{\mathfrak{k}_\Sigma(\eta) = 0.}$$

Proof. The source-symbol negative branch is confined to the active interval by (ii). The covariant commutator is nonnegative there by (iii) and strictly positive on every nonzero compatible negative state by (iv). Finite dimensionality of the negative spectral space upgrades strict positivity to a positive minimum. The budget in (v) preserves that minimum for the full native sign operator. Apply [Theorem 20.4](#). $\square$

Condition (iv) is a target-faithful observability statement. It cannot be replaced by the assertion that a sampled transform matrix has full rank. Condition (v) must use the exact boundary and Feshbach source augmentation and the same native projection.

21 Arb enclosure of the covariant active band

The covariant two-sheet virial theorem reduces the infinite source symbol to three scalar obligations: nonvanishing of the complete prime symbol on the active band, strict positivity of the radial derivative gap there, and nonnegativity of the lower source branch outside. We now formulate a directed certificate for all three obligations without a standing-wave terminal cutoff.

Put

$$q(\xi) = \sum_{n \geq 2} \frac{\Lambda(n)}{n^2} e^{i\xi \log n}, \quad \omega_\eta(\xi) = g(\xi) + \eta(1 + |g(\xi)|).$$

The common two-sheet symbol has eigenvalue branches

$$\omega_\eta(\xi) \pm |q(\xi)|.$$

21.1 Complete prime moment enclosure

For an auxiliary integer P , define

$$q_P^{(j)}(\xi) = \sum_{n \leq P} \frac{\Lambda(n)(i \log n)^j}{n^2} e^{i\xi \log n}, \quad j = 0, 1, 2.$$

Arb encloses every retained prime-power coefficient and one-step Euler phase. A geometric recurrence generates the uniform-grid packet with an explicit rounding ledger.

Let

$$R_j(P) = \sum_{n > P} \frac{\Lambda(n)(\log n)^j}{n^2}.$$

From $\psi(x) \leq x \log x$ and Stieltjes partial summation,

$$\boxed{R_j(P) \leq 2I_{j+1}(P) - jI_j(P),}$$

where

$$I_0(P) = P^{-1}, \quad I_k(P) = \frac{(\log P)^k}{P} + kI_{k-1}(P).$$

Consequently the finite Arb packet plus $R_j(P)$ encloses the complete q, q', q'' .

21.2 Interval lower bounds

On a grid interval $[x_k, x_{k+1}]$ of width h ,

$$|q(\xi) - q_P(x_k)| \leq \varepsilon_{0,k} + R_0(P) + D_1^{\text{up}} h,$$

where

$$D_j^{\text{up}} \geq \sum_{n \geq 2} \frac{\Lambda(n)(\log n)^j}{n^2}.$$

This yields a directed lower bound for $|q|$. Once this lower bound is positive,

$$(|q|)' = \frac{\text{Re}(\bar{q}q')}{|q|}$$

is enclosed using the certified disks for q and q' and the complete second moment D_2^{up} .

For $x \in [a, b] \subset (0, \infty)$, the positive real-digamma series gives

$$g'(x) \geq \sum_{m=0}^{M-1} \frac{(m+1/4)a}{2((m+1/4)^2 + b^2/4)^2}.$$

Since $\omega'_\eta \geq (1-\eta)g'$, interval comparison proves the covariant radial derivative gap.

The interval adjacent to zero is treated after division by x . The exact identity $\text{Re}(\bar{q}q')(0) = 0$ gives

$$|(|q|)'(x)| \leq x \frac{D_1^2 + D_0 D_2}{\inf_{[0,h]} |q|},$$

while the Gamma series supplies a positive lower coefficient for $\omega'_\eta(x)/x$.

Theorem 21.1 (Directed covariant active-band certificate). *Fix $\eta = 10^{-2}$, $P = 10^6$, grid width $h = 10^{-2}$, active edge $\Omega_* = 7$, far edge $\Omega_\infty = 12$, and Arb precision 192 bits. Suppose the generated interval ledger satisfies*

$$\inf_{|\xi| \leq 7} |q(\xi)| > 0,$$

$$\inf_{0 < \xi < 7} (\omega'_\eta(\xi) - |(|q|)'(\xi)|) > 0,$$

and

$$\inf_{|\xi| \geq 7} (\omega_\eta(\xi) - |q(\xi)|) \geq 0.$$

Then assumptions (i)–(iii) of the covariant active-band seam-annihilation criterion hold for the complete Gamma-prime source symbol.

Proof. The interval ledger covers $[0, 7]$ and $[7, 12]$. For $\xi \geq 12$, monotonicity of g and the complete absolute moment bound give

$$\omega_\eta(\xi) - |q(\xi)| \geq \omega_\eta(12) - D_0^{\text{up}}.$$

Reality symmetry supplies the negative half-line. The three displayed strict or nonnegative inequalities are exactly assumptions (i)–(iii) of [Theorem 20.6](#). $\square$

The auxiliary cutoff P sharpens scalar bounds only; its complement is included through R_0, R_1, R_2 . It is not a definition of the native operator.

22 Native Hardy-channel no-blindness

The Arb active-band certificate supplies a strict lower source-symbol gap on an open frequency band. The native exponential spatial energy turns that band into a faithful observation: no nonzero compatible native state can disappear there.

22.1 The native Hardy strip

Let

$$f(x) = F(e^x), \quad a = \frac{3}{2}.$$

The spatial part of the native energy gives

$$\int_{\mathbb{R}} e^{2a|x|} |f(x)|^2 dx < \infty.$$

For $|\operatorname{Im} z| < a$, define

$$\Phi_F(z) = \int_{\mathbb{R}} f(x) e^{-izx} dx.$$

Theorem 22.1 (Native strip-Hardy embedding). *The function Φ_F is analytic on*

$$S_a = \{z \in \mathbb{C} : |\operatorname{Im} z| < a\}$$

and belongs to $H^2(S_a)$. For every $|y| < a$,

$$\boxed{\frac{1}{2\pi} \|\Phi_F(\cdot + iy)\|_2^2 = \int_{\mathbb{R}} e^{2yx} |f(x)|^2 dx \leq \int_{\mathbb{R}} e^{2a|x|} |f(x)|^2 dx.}$$

Its boundary values are

$$\boxed{\Phi_F(\xi - ia) = \mathcal{M}(r^{-a}F)(\xi), \quad \Phi_F(\xi + ia) = \mathcal{M}(r^aF)(\xi).}$$

Proof. Cauchy–Schwarz with the exponential weight gives locally uniform absolute convergence and analyticity. Plancherel applied to $e^{yx}f(x)$ gives the norm identity and the two L^2 boundary values. $\square$

Lemma 22.2 (Hardy-strip boundary uniqueness). *If $\Phi \in H^2(S_a)$ and either boundary value vanishes on a measurable set of positive measure, then $\Phi \equiv 0$.*

Proof. Transfer the strip conformally to the disk. A nonzero disk Hardy function has nonzero radial boundary values almost everywhere by its inner–outer factorization. Pulling back proves the assertion. $\square$

Theorem 22.3 (Native active-band no-blindness). *Let $E \subset \mathbb{R}$ have positive measure and put*

$$\mathcal{B}_a^\times F = \begin{pmatrix} \mathcal{M}(r^{-a}F) \\ \mathcal{M}(r^aF) \end{pmatrix}.$$

Then

$$\boxed{1_E \mathcal{B}_a^\times F = 0 \implies F = 0.}$$

Vanishing of either channel component on E already forces $F = 0$.

Proof. Apply [Theorem 22.2](#) to the corresponding boundary value of Φ_F and use injectivity of the Mellin transform on the native carrier. $\square$

The statement is stable under inversion-parity restriction and under the odd Feshbach projection.

22.2 Compatible reconstruction and strict observability

Let $J : \mathcal{S} \rightarrow \mathcal{H}$ be an isometric compatible reconstruction and let

$$\mathcal{O}_E = 1_E \mathcal{B}_{3/2}^\times.$$

Theorem 22.4 (Compatible reconstruction is not active-band blind). *For every positive-measure band E ,*

$$\ker(\mathcal{O}_E J) = \{0\}.$$

The property is invariant under unitary coordinate changes in $\mathcal{S}$.

Proof. If $\mathcal{O}_E J u = 0$, then [Theorem 22.3](#) gives $J u = 0$. Since J is isometric, $u = 0$. $\square$

Let $h > 0$ almost everywhere on E , let $V \subset \mathcal{H}$ be finite dimensional and let $J_V : \mathbb{C}^d \rightarrow V$ be an isometry. Define

$$\mathcal{G}_{E,h}(V) = J_V^* (\mathcal{B}_{3/2}^\times)^* 1_E M_h \mathcal{B}_{3/2}^\times J_V.$$

Theorem 22.5 (Finite-dimensional strict observability). *For every nonzero finite-dimensional compatible subspace V ,*

$$\mathcal{G}_{E,h}(V) > 0, \quad c_{E,h}(V) := \lambda_{\min}(\mathcal{G}_{E,h}(V)) > 0.$$

Proof. For $u \neq 0$,

$$\langle u, \mathcal{G}_{E,h}(V) u \rangle = \int_E h(\xi) \|\mathcal{B}_{3/2}^\times J_V u(\xi)\|^2 d\xi.$$

A zero value contradicts [Theorem 22.4](#). Compactness of the unit sphere in finite dimension gives a positive minimum. $\square$

Applied to the finite negative spectral space of the shifted native sign operator, this theorem closes the compatibility/no-blindness clause of the covariant active-band argument. Together with the directed Arb bounds and the outward five-source ledger, it supplies the exact observability structure used by the closed fixed-shift certificate.

23 Virial–Birman–Schwinger reduction to the exact finite obstruction

The complete-prime active-band certificate and the native Hardy-strip theorem close the infinite Gamma–prime source obstruction. At $\eta = 10^{-2}$, the only adverse terms are the odd boundary source and the odd Feshbach source, whose combined range has dimension at most five.

23.1 Positivity of the complete pure source

Let $S_{\eta,\pm}$ be the shifted even and odd Gamma–prime source operators before the finite boundary and Feshbach augmentations.

Theorem 23.1 (Virial positivity of the complete source operator). *At $\eta = 10^{-2}$,*

$$S_{\eta,+} > 0, \quad S_{\eta,-} > 0.$$

The unshifted common-source certificate gives

$$S_{0,+} \geq 0, \quad S_{0,-} \geq 0, \quad S_{\eta,\pm} \geq \eta I \quad (\eta > 0).$$

Proof. A nonpositive source eigenvector is confined to the active band by the complete prime enclosure. The covariant derivative gap is strict there, and the Hardy-channel theorem excludes a nonzero compatible state that is blind to the band. The resulting positive commutator contradicts the eigenvector virial identity. The unshifted and shifted floors are the directed Arb consequences. $\square$

The even boundary augmentation is positive. Hence the only adverse finite-rank source is odd.

23.2 Rank-at-most-five odd source

Let r_{odd} be the odd boundary source and define

$$V_{F,\eta} = C_3(A_3 + \eta I)^{-1/2}, \quad V_\eta = \begin{bmatrix} r_{\text{odd}} & V_{F,\eta} \end{bmatrix}.$$

Since $\dim E_3 = 4$,

$$\text{rank } V_\eta \leq 5,$$

and

$$F_{3,\eta}^\times = S_{\eta,-} - V_\eta V_\eta^*.$$

Definition 23.2 (Finite native Birman–Schwinger matrix). Define

$$B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta \in M_r(\mathbb{C}), \quad r \leq 5.$$

Theorem 23.3 (Exact finite-rank inertia reduction). *One has*

$$N_-(F_{3,\eta}^\times) = N_+(B_\eta - I),$$

and

$$\dim \ker F_{3,\eta}^\times = \dim \ker (B_\eta - I).$$

Consequently,

$$F_{3,\eta}^\times \geq 0 \iff B_\eta \leq I.$$

Proof. Put $T_\eta = S_{\eta,-}^{-1/2} V_\eta$. Positive congruence gives

$$F_{3,\eta}^\times = S_{\eta,-}^{1/2} (I - T_\eta T_\eta^*) S_{\eta,-}^{1/2}.$$

The nonzero spectra of $T_\eta T_\eta^*$ and $T_\eta^* T_\eta = B_\eta$ agree with multiplicity. $\square$

Corollary 23.4 (Fixed-shift seam charge is a five-dimensional threshold count). *For every positive shift,*

$$\mathfrak{k}_\Sigma(\eta) = \mathfrak{k}_3(\eta) = N_+(B_\eta - I).$$

In particular, $\lambda_{\max}(B_\eta) < 1$ gives $\mathfrak{k}_\Sigma(\eta) = 0$.

Proof. The even shifted restriction is positive by the common-source theorem and the favorable even boundary. The odd contribution is the inertia in [Theorem 23.3](#). $\square$

23.3 Outward residual enclosure

Let $\tilde{X}$ approximate the five source-resolvent columns,

$$S_{\eta,-}X = V_\eta, \quad R = V_\eta - S_{\eta,-}\tilde{X},$$

and put

$$\tilde{B}_\eta = \text{sym}(V_\eta^* \tilde{X}).$$

Proposition 23.5 (Residual enclosure of the finite matrix). *If $S_{\eta,-} \geq s_0 I$ with $s_0 > 0$, then*

$$\|B_\eta - \tilde{B}_\eta\| \leq s_0^{-1} \|V_\eta\| \|R\|.$$

Therefore

$$\lambda_{\max}(\tilde{B}_\eta) + s_0^{-1} \|V_\eta\| \|R\| < 1 \implies \mathfrak{k}_\Sigma(\eta) = 0.$$

Proof. Since $X - \tilde{X} = S_{\eta,-}^{-1}R$,

$$\|V_\eta^*(X - \tilde{X})\| \leq \|V_\eta\| \|S_{\eta,-}^{-1}\| \|R\|.$$

Symmetrization does not increase the norm, and Weyl's inequality gives the threshold conclusion. $\square$

At the fixed-shift floor $s_0 = 0.008$, the multiplier is 125. The next section applies this theorem to the native five-source packet and closes the node $\eta = 0.01$. The endpoint theorem then replaces every decreasing-shift construction by the single matrix $\mathcal{K}_0$.

24 Closed fixed-shift five-source certificate

At $\eta = 10^{-2}$, the rank-at-most-five reduction gives

$$B_{0.01} = V_{0.01}^* S_{0.01,-}^{-1} V_{0.01}, \quad \mathfrak{k}_\Sigma(0.01) = N_+(B_{0.01} - I).$$

24.1 Blind-adapted numerical centre

Let u_{blind} be the almost-blind boundary-memory direction and let $u_1, \dots, u_4$ diagonalize the visible numerical centre. In the ordered basis

$$(u_{\text{blind}}, u_1, u_2, u_3, u_4),$$

the Hermitian centre is

$$\tilde{B}_{0.01}^{\text{cen}} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 7.941895745298219 \times 10^{-9} & 0 & 0 & 0 \\ 0 & 0 & 3.274496655333885 \times 10^{-7} & 0 & 0 \\ 0 & 0 & 0 & 1.0203435819620437 \times 10^{-4} & 0 \\ 0 & 0 & 0 & 0 & 0.43597867312 \end{pmatrix}$$

The blind-channel contribution, cross terms and source-solve residuals are carried in the outward error budget. The fixed-grid block obeys

$$\lambda_{\max}(B_{0.01}^{\text{grid}}) \leq 0.43597867330929.$$

24.2 Chart-to-native transfer

The certified source floor is

$$S_{0.01,-} \geq 0.008I, \quad \|S_{0.01,-}^{-1}\| \leq 125.$$

The source-column transfer satisfies

$$e_V \leq 0.003814429157492567.$$

After the source-operator transfer, the top is bounded by

$$0.43801207568437805.$$

The source-column perturbation contributes

$$\frac{2\|\tilde{V}\|e_V + e_V^2}{0.008} \leq 0.40087808807175573.$$

Equivalently,

$$\|B_{0.01} - \tilde{B}_{0.01}^{\text{cen}}\| \leq 0.40291149063402765.$$

Therefore

$$\lambda_{\max}(B_{0.01}) \leq 0.8388901637561339 < 1,$$

with reserve

$$1 - 0.8388901637561339 = 0.1611098362438661.$$

Theorem 24.1 (Closed native node at $\eta = 0.01$). *For the native boundary–Gamma–prime Weil/Feshbach operator,*

$$B_{0.01} < I, \quad F_{3,0.01}^\times > 0, \quad \mathfrak{k}_\Sigma(0.01) = 0.$$

Proof. The outward matrix enclosure gives $\lambda_{\max}(B_{0.01}) < 1$. The exact Birman–Schwinger inertia theorem gives strict positivity of the odd Feshbach complement. The even shifted restriction is positive by the common-source theorem and favorable even boundary, so the global seam charge vanishes at this node. $\square$

Certificate provenance. The enclosure uses the native Feshbach threshold, whole-line primal and adjoint transfers, fixed-state pivot transfer, accepted-subspace transfer and shifted Feshbach source transfer. The numerical reserve is therefore attached to the native operator rather than to a fitted coordinate matrix.

The closed fixed node is a quantitative confirmation of the single-sector architecture. Eta-zero closure is supplied exactly by the endpoint criterion $\mathcal{K}_0 \geq 0$ and the norm-descent theorem.

25 Bilateral recognition closure: accepting both sides of the cut

The native framework is representation complete. Every side created by the cut, every representation-visible direction, every target-relevant blind direction and every cross coupling is retained on one completed carrier. The architecture therefore fixes the mathematical object before the terminal sign is evaluated.

25.1 Universal target-faithful completion

Let

$$R : \mathcal{H} \longrightarrow \mathcal{Y}, \quad \Pi : \mathcal{H} \longrightarrow \mathcal{Z}$$

be a bounded observer and the declared target map. Put

$$\mathcal{N} = \ker R, \quad \mathcal{N}_0 = \ker R \cap \ker \Pi, \quad \mathfrak{B}(R; \Pi) = \mathcal{N} / \mathcal{N}_0.$$

Let $q_\Pi : \mathcal{N} \rightarrow \mathfrak{B}(R; \Pi)$ be the quotient map and $P_\mathcal{N}$ the orthogonal projection onto $\mathcal{N}$.

Definition 25.1 (Bilateral completed observer). Define

$$R_\Pi^\# h = (Rh, q_\Pi P_\mathcal{N} h) \in \mathcal{Y} \oplus \mathfrak{B}(R; \Pi).$$

The first component records the visible response and the second records exactly the target-relevant blind response.

Theorem 25.2 (Universal target-faithful completion). *The completed observer satisfies*

$$\ker R_\Pi^\# = \ker R \cap \ker \Pi, \quad \mathfrak{B}(R_\Pi^\#; \Pi) = 0.$$

Moreover, if $L : \mathcal{H} \rightarrow \mathcal{W}$ is bounded, $L|_{\mathcal{N}_0} = 0$, and (R, L) is target faithful, then there is a unique bounded injective map

$$\ell : \mathfrak{B}(R; \Pi) \longrightarrow \mathcal{W}$$

with

$$L|_\mathcal{N} = \ell q_\Pi.$$

Thus $R_\Pi^\#$ is the universal minimal target-faithful completion of R .

Proof. If $R_\Pi^\# h = 0$, then $h \in \mathcal{N}$ and $q_\Pi h = 0$, so $h \in \mathcal{N}_0$. The reverse inclusion is immediate. The quotient universal property gives $L|_\mathcal{N} = \ell q_\Pi$. If $\ell q_\Pi n = 0$, then $Rn = Ln = 0$; target faithfulness gives $\Pi n = 0$, hence $q_\Pi n = 0$. Therefore ℓ is injective. $\square$

25.2 Observer and observed on the compatible native range

The native energy defines an isometric analysis map

$$\mathcal{C}_\times : \mathcal{X}_{\times,3} \longrightarrow \mathcal{Y}_\times, \quad \mathcal{Y}_{\text{comp}} = \overline{\text{Ran } \mathcal{C}_\times}.$$

Put

$$P_{\text{comp}} = \mathcal{C}_\times \mathcal{C}_\times^*.$$

Theorem 25.3 (Native observer–observed identity). *One has*

$$\mathcal{C}_\times^* \mathcal{C}_\times = I, \quad \mathcal{C}_\times \mathcal{C}_\times^* = P_{\text{comp}}.$$

Consequently

$$\mathcal{C}_\times : \mathcal{X}_{\times,3} \longrightarrow \mathcal{Y}_{\text{comp}}$$

is unitary and

$$\mathcal{C}_\times^* \mathcal{C}_\times F = F$$

for every native state F .

Proof. The first identity is the defining isometry of the native energy channel. The second is the range projection of an isometry. $\square$

If $J_\times$ is the inversion cut and J_y is the corresponding channel involution, then

$$\mathcal{C}_\times J_\times = J_y \mathcal{C}_\times.$$

Thus the exact analysis–synthesis identity preserves both cut sectors.

25.3 Four-block reconstruction of the uncut operator

Let $J = J^* = J^{-1}$ and put $P_\pm = (I \pm J)/2$. For bounded self-adjoint A , define

$$A_{\text{diag}} = P_+ A P_+ + P_- A P_-, \quad \Sigma_J(A) = P_+ A P_- + P_- A P_+.$$

Theorem 25.4 (Bilateral cut reconstruction). *Every bounded self-adjoint operator satisfies*

$$A = P_+ A P_+ + P_+ A P_- + P_- A P_+ + P_- A P_- = A_{\text{diag}} + \Sigma_J(A).$$

Moreover,

$$\Sigma_J(A) = 0 \iff [A, J] = 0.$$

The two diagonal blocks, the cross seam and every target-relevant blind quotient are invariant under faithful unitary transport.

Proof. Insert $I = P_+ + P_-$ on both sides of A . The commutator statement is the standard block-diagonal criterion, and unitary covariance follows by conjugation. $\square$

The theorem fixes the meaning of bilateral acceptance. The uncut operator is reconstructed from both sectors and their cross seam. Single-sector sufficiency later arises from a proved sign orientation, not from deleting one side of the cut.

25.4 Representation-complete closure theorem

Theorem 25.5 (Recognition closure of the native framework). *The native recognition-seam framework is representation complete:*

- (i) *the carrier, cut, Euler flow, energy and target are defined before every classical chart;*
- (ii) *native analysis and synthesis are inverse on the compatible channel range;*
- (iii) *both cut sectors and their cross seam reconstruct the uncut operator;*
- (iv) *every projected observer has a universal minimal target-faithful completion;*
- (v) *visible, blind and cross blocks are controlled on the same native reconstruction before any finite spectral statement is promoted;*
- (vi) *the completed-Weil sign is represented by one integer seam obstruction, its spectral-flow, threshold-zeta, Morse and holonomy avatars, and the five-dimensional endpoint criterion $\mathcal{K}_0 \geq 0$.*

Proof. Items (i)–(ii) are the native carrier and [Theorem 25.3](#). Item (iii) is [Theorem 25.4](#); item (iv) is [Theorem 25.2](#); item (v) follows from the blind-kernel Schur and compatible response-tail theorems. Item (vi) is the integer seam-path construction together with the endpoint reduction proved in [Sections 30 and 31](#). $\square$

25.5 Certification discipline

Every outward certificate has one of three exact meanings:

strictly subcritical enclosure	$\implies$	the seam charge is zero,
strictly supercritical enclosure	$\implies$	a threshold crossing is present,
enclosure meeting the threshold	$\implies$	the enclosure is unresolved.

This discipline does not weaken the framework. It makes the framework falsifiable while preserving its exact theorem content.

Framework closure status. The bilateral observer completion, observer–observed identity, four-block cut reconstruction and target-faithful representation theorem are proved. The completed-Weil problem is therefore represented without loss and reduced to the single-sector endpoint criterion developed below. No further carrier, blindness or parity-completion mechanism is required.

26 UGD amplitude uplift and the eta-normalization rectangle

The exact eta-family theorem is algebraic. Its operational meaning is captured by the phase–scale–seam transport rules of UGD on the fixed event carrier

$$\mathfrak{E}_\Sigma = \ell_\partial \oplus E_3, \quad \dim \mathfrak{E}_\Sigma = 5.$$

At

$$\eta_0 = 0.010, \quad \eta_1 = 0.011,$$

let $\Theta_{r,j}$ and $\Theta_{n,j}$ denote the raw and normalized native amplitudes. The two paths

$$\text{Path A: } \Theta_{r,0} \longrightarrow \Theta_{n,0} \longrightarrow \Theta_{n,1},$$

$$\text{Path B: } \Theta_{r,0} \longrightarrow \Theta_{r,1} \longrightarrow \Theta_{n,1}$$

apply normalization and eta continuation in opposite orders.

For a full-rank amplitude $\Theta = QH$, transport is defined by the polar frame Q . If U_A and U_B are the two composed transports, the based rectangle is

$$H_{\square}^{\text{actual}} = U_B^* U_A.$$

The state-locked packet gives

path-transport difference	$3.0669860554696173 \times 10^{-15},$
loop identity residual	$3.1396725820594018 \times 10^{-15},$
defect phase	$1.0705180635847342 \times 10^{-40},$
defect-to-accepted leakage	$1.7140530928543942 \times 10^{-17},$
accepted-block identity residual	$3.139637946219698 \times 10^{-15},$
endpoint scale relative difference	0,
maximum pairing-return residual	$6.071532165918825 \times 10^{-18}.$

Thus the two operation orders agree on the actual five-label rectangle to floating-point precision without any compensating seam phase.

Remark 26.1 (Exact theorem behind the calibration). The packet is the numerical shadow of the exact factorization

$$A_{\text{chart}} = AU, \quad \Theta_\eta = Q_{\text{chart}} G_\eta^{1/2}.$$

Because $A_3 > 0$, the polar frames extend through eta zero and the eta-normalization rectangle has identity holonomy for every admissible nonnegative pair of shifts. The cutoff wall is therefore not an endpoint of the exact eta family.

UGD supplies a geometric realization of exact eta analyticity. The sign criterion itself remains the Hermitian endpoint condition $\mathcal{K}_0 \geq 0$; normalization holonomy is an invariant calibration of the carrier rather than a second sign test.

27 Single seam obstruction and exact eta structure

The completed sign problem is represented by one nonnegative integer,

$$\mathfrak{k}_\Sigma(\eta) = \text{rank } \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta).$$

Spectral flow, threshold zeta, Morse zeta and threshold holonomy are exact representations of this same integer. The journal argument follows one theorem chain:

$$\begin{aligned} &\text{target-faithful bilateral carrier} \longrightarrow \text{compact parity packet } \mathbb{K}_\eta \longrightarrow \mathfrak{k}_\Sigma(\eta) \\ &\longrightarrow \text{common unshifted source positivity} \longrightarrow \text{single adverse odd sector} \\ &\longrightarrow \text{rank-} \leq 5 \text{ Hermitian gap } \mathcal{K}_\eta \longrightarrow \text{endpoint gap } \mathcal{K}_0. \end{aligned}$$

The classical Weil criterion enters only after completed-carrier positivity has been recovered from this endpoint.

27.1 Exact eta-family path independence

On the fixed event carrier

$$\mathfrak{E}_\Sigma = \ell_\partial \oplus E_3, \quad \dim \mathfrak{E}_\Sigma = 5,$$

let A be the eta-independent raw event amplitude and $U = 1 \oplus U_4$ the fixed accepted chart. Write

$$A_{\text{chart}} = AU, \quad G_\eta = 1 \oplus (A_3 + \eta I_4), \quad \Theta_\eta = Q_{\text{chart}} G_\eta^{1/2}.$$

By [Theorem 31.1](#), $A_3 > 0$, so the family is admissible for every $\eta \geq 0$.

Theorem 27.1 (Exact eta-family path independence). *For any nonnegative shifts η_1, η_2 , raw eta transport and normalized eta transport are both I_5 , while raw-to-normalized transport is the same fixed unitary U^* at both endpoints. Hence normalization and eta continuation commute exactly and every based eta-normalization rectangle has holonomy I_5 .*

Proof. If $A = QH$ is the thin polar decomposition, right-unitary covariance gives $\text{polar}(AU) = QU$. Since Q_{chart} is isometric and $G_\eta^{1/2} > 0$ for $\eta \geq 0$,

$$\text{polar}(Q_{\text{chart}} G_\eta^{1/2}) = Q_{\text{chart}}.$$

Thus neither polar frame depends on eta and the cross transport is the fixed chart unitary. $\square$

The small negative eigenvalue in the pinned cutoff matrix is consequently a finite discretization shadow, not an endpoint of the exact eta family.

27.2 Unshifted source and exact critical-gap identity

The common-source theorem gives

$$S_{0,-} \geq 0, \quad S_{\eta,-} = S_{0,-} + \eta I \geq \eta I \quad (\eta > 0).$$

Let

$$\mathcal{W}_\Sigma = [r_{\text{odd}}, C_3], \quad \mathcal{G}_\eta = \begin{pmatrix} 1 & 0 \\ 0 & A_3 + \eta I \end{pmatrix},$$

$$\mathcal{H}_\eta = \mathcal{W}_\Sigma^* S_{\eta,-}^{-1} \mathcal{W}_\Sigma, \quad \mathcal{K}_\eta = \mathcal{G}_\eta - \mathcal{H}_\eta.$$

The generalized inertia theorem gives

$$F_{3,\eta}^\times \geq 0 \iff \mathcal{K}_\eta \geq 0.$$

By [Theorem 31.2](#),

$$\text{Ran } \mathcal{W}_\Sigma \subset \text{Dom}(S_{0,-}^{-1/2}),$$

so

$$\mathcal{H}_0 = (S_{0,-}^{-1/2} \mathcal{W}_\Sigma)^* (S_{0,-}^{-1/2} \mathcal{W}_\Sigma)$$

is well defined. Put

$$\mathcal{G}_0 = \begin{pmatrix} 1 & 0 \\ 0 & A_3 \end{pmatrix}, \quad \mathcal{K}_0 = \mathcal{G}_0 - \mathcal{H}_0, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & I_{E_3} \end{pmatrix}.$$

Proposition 27.2 (Exact critical-gap decomposition). *For every $\eta > 0$,*

$$\boxed{\mathcal{K}_\eta = \mathcal{K}_0 + \eta D + (\mathcal{H}_0 - \mathcal{H}_\eta).}$$

Both eta-dependent summands are nonnegative.

Proof. The affine identity $\mathcal{G}_\eta = \mathcal{G}_0 + \eta D$ is exact, and monotonicity of $(S_{0,-} + \eta I)^{-1}$ gives $\mathcal{H}_0 - \mathcal{H}_\eta \geq 0$. $\square$

Theorem 27.3 (Analytic critical-gap closure). *If $\mathcal{K}_0 \geq 0$, then*

$$\boxed{\mathcal{K}_\eta \geq 0, \quad F_{3,\eta}^\times \geq 0 \quad (\eta > 0).}$$

If additionally

$$\ker \mathcal{K}_0 \cap \ker D \cap \ker \mathcal{W}_\Sigma = \{0\},$$

then both operators are strictly positive for every positive shift.

Proof. The exact decomposition is a sum of nonnegative forms. For strictness, equality forces a vector into all three kernels; the stated no-common-blindness condition then forces that vector to vanish. $\square$

Thus an infinite eta ladder is unnecessary. The endpoint condition $\mathcal{K}_0 \geq 0$ controls the entire positive-shift family, while [Theorem 31.3](#) supplies the exact eta-zero passage.

Proposition 27.4 (Positive critical slope). *Suppose*

$$\text{Ran } \mathcal{W}_\Sigma \subset \text{Dom}(S_{0,-}^{-1}),$$

and $\ker \mathcal{K}_0 = \text{span}\{n_0\}$ is simple, normalized by $\langle n_0, \mathcal{G}_0 n_0 \rangle = 1$. For the top generalized branch $\rho(\eta)$,

$$\rho(0) = 1, \quad -\rho'(0^+) = \|P_{E_3} n_0\|^2 + \|S_{0,-}^{-1} \mathcal{W}_\Sigma n_0\|^2 > 0.$$

Hence

$$\boxed{1 - \rho(\eta) = \kappa_\Sigma \eta + o(\eta), \quad \kappa_\Sigma > 0.}$$

The slope formula explains the observed threshold approach; the endpoint theorem itself uses only the inverse-half regularity already proved.

Theorem 27.5 (Representation-complete finite-obstruction reduction). *On the declared native completed-Weil architecture:*

- (i) both parity sectors, the cross seam and every target-relevant blind direction are retained;
- (ii) the common pure source is nonnegative in both parities;
- (iii) the even boundary augmentation is positive and $\mathfrak{k}_+(\eta) = 0$;
- (iv) the global seam integer equals the odd five-label count;
- (v) the eta-normalization family is exactly path independent through eta zero;
- (vi) the remaining adverse source has rank at most five;
- (vii) the fixed node obeys

$$\lambda_{\max}(B_{0.01}) \leq 0.8388901637561339 < 1;$$

- (viii) exact accepted support, inverse-half regularity and endpoint norm descent hold at eta zero;
- (ix) the completed-Weil sign is decided by the single Hermitian condition $\mathcal{K}_0 \geq 0$.

The representation, arithmetic transport, parity reduction and endpoint passage are therefore complete. The terminal mathematical statement is the five-dimensional criterion $\mathcal{K}_0 \geq 0$, with the equivalent closure formulations of [Section 32](#).

28 Threshold holonomy as an exact seam avatar

The eta-normalization rectangle has identity holonomy throughout the exact eta family. It proves path independence, but an identity normalization loop is not a sign detector. The sign-sensitive holonomy is attached directly to the adverse threshold projection.

For $\eta > 0$, put

$$P_\eta = \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta), \quad \mathfrak{k}_\Sigma(\eta) = \text{rank } P_\eta,$$

and define

$$\boxed{\mathcal{H}_\Sigma^{\text{thr}}(\eta) = \exp(i\pi P_\eta) = I - 2P_\eta.}$$

This assigns phase -1 to every adverse threshold direction and phase $+1$ to the nonadverse complement.

Theorem 28.1 (Threshold holonomy detects the seam charge). *For every positive shift, $\mathcal{H}_\Sigma^{\text{thr}}(\eta)$ is a self-adjoint unitary and*

$$\boxed{\mathfrak{k}_\Sigma(\eta) = \frac{1}{2} \text{rank}(I - \mathcal{H}_\Sigma^{\text{thr}}(\eta)) = \frac{1}{2} \text{Tr}(I - \mathcal{H}_\Sigma^{\text{thr}}(\eta)).}$$

Consequently,

$$\boxed{\mathcal{H}_\Sigma^{\text{thr}}(\eta) = I \iff \mathfrak{k}_\Sigma(\eta) = 0 \iff B_\eta \leq I.}$$

After the single-sector source theorem,

$$P_\eta = 0_+ \oplus \mathbf{1}_{(1,\infty)}(K_{3,\eta}^\sharp),$$

so the threshold holonomy is decided entirely by the odd five-label sector.

Proof. Since P_η is an orthogonal finite-rank projection, $\exp(i\pi P_\eta) = I - 2P_\eta$ is a self-adjoint unitary. Moreover $I - \mathcal{H}_\Sigma^{\text{thr}}(\eta) = 2P_\eta$, which gives the rank and trace formulas. The final equivalences use the single-sector seam reduction and the finite Birman–Schwinger inertia theorem. $\square$

Theorem 28.2 (Threshold-holonomy form of the endpoint criterion). *Assume the classical explicit-formula identification and Weil criterion of Section 31.4. For any controlled sequence $\eta_j \downarrow 0$,*

$$\boxed{\begin{aligned} \text{RH} &\iff \mathcal{K}_0 \geq 0 \\ &\iff \mathcal{H}_\Sigma^{\text{thr}}(\eta_j) = I \quad \text{for every } j. \end{aligned}}$$

Proof. Identity threshold holonomy is equivalent to zero seam charge. Norm descent of the five-label gap converts zero charge along the controlled sequence into $\mathcal{K}_0 \geq 0$. The exact single-sector endpoint theorem and the classical Weil criterion give the implication to RH. Conversely, completed-Weil positivity gives nonnegative positive shifts, zero seam charge and identity threshold holonomy. $\square$

Remark 28.3 (Possible UGD realization). A path-ordered UGD holonomy would become a geometric representation of the same threshold object if independently constructed target-faithful unitaries U_η satisfy

$$\text{Hol}_{\text{UGD}}(\gamma_\eta) = U_\eta^* \mathcal{H}_\Sigma^{\text{thr}}(\eta) U_\eta.$$

This optional realization is not used in the terminal implication. The exact eta-normalization rectangle validates the path calculus; the sign decision is the canonical threshold projection, equivalently the unshifted gap $\mathcal{K}_0$.

29 One five-source matrix in every faithful chart

The arithmetic prime torus, the native Feshbach carrier and threshold holonomy are not independent terminal routes. Their common decision object is the same rank-at-most-five Birman–Schwinger matrix, provided each chart transports the actual native source operator and source columns.

29.1 Unitary covariance of the finite obstruction

Let $\mathcal{H}$ be a Hilbert carrier, let $S = S^* > 0$ and let $V : \mathbb{C}^r \rightarrow \mathcal{H}$, $r \leq 5$. Put

$$B = V^* S^{-1} V.$$

For a unitary $\mathcal{U} : \mathcal{H} \rightarrow \mathcal{H}'$ define

$$S^\mathcal{U} = \mathcal{U} S \mathcal{U}^*, \quad V^\mathcal{U} = \mathcal{U} V.$$

Theorem 29.1 (Common five-matrix covariance). *The transported Birman–Schwinger matrix is unchanged:*

$$(V^{\mathcal{U}})^*(S^{\mathcal{U}})^{-1}V^{\mathcal{U}} = V^*S^{-1}V = B.$$

If the label coordinates are changed by $U_5 \in U(r)$, then

$$B' = U_5^* B U_5.$$

Consequently spectrum, threshold projection rank, seam integer and threshold holonomy are invariant under every faithful carrier and label chart.

Proof. Unitary functional calculus gives $(S^{\mathcal{U}})^{-1} = \mathcal{U}S^{-1}\mathcal{U}^*$. Substitution proves the first identity, and right multiplication of V proves the label-coordinate formula. $\square$

29.2 Prime-torus transport

Let $\mathcal{B}$ be the arithmetic Bohr unitary of [Theorem 6.3](#). Extend it by the identity on the boundary and Gamma channels:

$$\widehat{\mathcal{B}} = I_{\partial \oplus \Gamma} \oplus \mathcal{B}.$$

For transported native data

$$S_{\eta,-}^{\mathbb{P}} = \widehat{\mathcal{B}} S_{\eta,-} \widehat{\mathcal{B}}^*, \quad V_{\eta}^{\mathbb{P}} = \widehat{\mathcal{B}} V_{\eta},$$

one has

$$B_{\eta}^{\mathbb{P}} = (V_{\eta}^{\mathbb{P}})^*(S_{\eta,-}^{\mathbb{P}})^{-1}V_{\eta}^{\mathbb{P}} = B_{\eta}.$$

Thus the prime torus is an exact arithmetic chart of the same five-source obstruction. It does not create a second threshold problem.

29.3 Matrix, seam integer and threshold holonomy

Let

$$P_{\eta}^B = \mathbf{1}_{(1,\infty)}(B_{\eta}), \quad \mathcal{H}_{\eta}^B = I - 2P_{\eta}^B.$$

The common pure-source theorem and favorable even boundary give $\mathfrak{k}_+(\eta) = 0$. Finite-rank inertia and [Theorem 28.1](#) therefore yield

$$\begin{aligned} B_{\eta} \leq I &\iff P_{\eta}^B = 0 \\ &\iff \mathfrak{k}_{\Sigma}(\eta) = 0 \\ &\iff \mathcal{H}_{\eta}^B = I \\ &\iff W_3 + \eta I \geq 0. \end{aligned}$$

Every faithful unitary chart gives the same equivalence.

Theorem 29.2 (All-chart form of the five-dimensional endpoint criterion). *Assume the classical completed-Weil normalization of [Section 31.4](#). For any controlled sequence $\eta_j \downarrow 0$, the following are equivalent:*

- (M1) $\mathcal{K}_0 \geq 0$;
- (M2) $B_{\eta_j} \leq I$ for every j ;
- (M3) $B_{\eta_j}^{\mathbb{P}} \leq I$ for every j ;
- (M4) $\mathfrak{k}_{\Sigma}(\eta_j) = 0$ for every j ;

- (M5) $\mathcal{H}_{\eta_j}^B = I$ for every j ;
(M6) the Riemann hypothesis holds.

Proof. The exact critical-gap decomposition and endpoint norm descent give (M1) $\iff$ (M2). Common-matrix covariance gives (M2) $\iff$ (M3). Finite-rank inertia and threshold-holonomy detection give (M2) $\iff$ (M4) $\iff$ (M5). The single-sector endpoint theorem and the classical Weil criterion give (M1) $\iff$ (M6). $\square$

Interpretation. The publication has one terminal decision object:

$$\boxed{\text{prime-torus chart} \longleftrightarrow \text{native source/Feshbach carrier} \longleftrightarrow B_\eta \in M_5(\mathbb{C}) \longleftrightarrow \mathcal{K}_0 \longleftrightarrow \text{seam integer} \longleftrightarrow \text{threshold}}$$

All arrows are exact under faithful transport of the native data. A UGD path-ordered realization may provide an additional geometric chart, but it is not a second terminal assumption and is not used in the implication to RH.

30 Single-sector sufficiency of the seam integer

The bilateral carrier retains both cut sectors, but retention does not mean that both sectors remain independent terminal decisions. The common two-sheet source theorem and the sign of the finite augmentations distinguish two statements that should not be confused:

both sectors are represented and only one sector carries an adverse source.

The first statement is required by target faithfulness. The second is the single-sector reduction proved below.

30.1 The common unshifted source controls both parities

Before the exact boundary and Feshbach augmentations are inserted, the direct even and direct odd source forms are reconstructed from the same two-sheet symbol

$$\Omega_\eta(\xi) = \omega_\eta(\xi)I_2 - \mathcal{Q}(\xi), \quad \omega_\eta(\xi) = g(\xi) + \eta(1 + |g(\xi)|).$$

Its eigenvalue branches are

$$\omega_\eta(\xi) - |q(\xi)|, \quad \omega_\eta(\xi) + |q(\xi)|.$$

Thus the lower branch is the only source-symbol sign that must be certified.

The directed unshifted Arb ledger uses prime cutoff 10^6 , precision 192 bits, active edge 12 and far edge 13. It records

$$\inf |q| > 0.09166336112365603$$

on the active band,

$$\begin{aligned} \inf(\omega'_0 - |(|q|)'|) &> 0.00874079683626209, \\ \inf_{12 \leq |\xi| \leq 13}(\omega_0 - |q|) &> 0.4571260170709391, \end{aligned}$$

and the far-ray bound

$$\inf_{|\xi| \geq 13}(\omega_0 - |q|) > 0.15683586289723883.$$

The near-zero covariant slope gap is bounded below by 28.969227449370624. These are outward lower bounds, not centre values.

Theorem 30.1 (Two-parity unshifted pure-source positivity). *Let $S_{0,+}$ and $S_{0,-}$ denote the complete pure source operators in the even and odd native sectors before the finite boundary and Feshbach augmentations. Then*

$$\boxed{S_{0,+} \geq 0, \quad S_{0,-} \geq 0.}$$

For every $\eta > 0$,

$$\boxed{S_{\eta,+} = S_{0,+} + \eta I \geq \eta I, \quad S_{\eta,-} = S_{0,-} + \eta I \geq \eta I.}$$

Proof. The common-symbol identity of [Section 20](#) is obtained before parity-specific finite augmentations. The displayed Arb bounds verify the lower-branch, nonvanishing and covariant derivative obligations of [Theorem 21.1](#) at eta zero. The native Hardy no-blindness and compatible reconstruction theorem then convert the symbol certificate into nonnegativity of each parity restriction. Since the upper branch is larger than the lower branch, no second source-symbol sign test is required. The shifted inequalities follow from the exact affine identity $S_{\eta,\pm} = S_{0,\pm} + \eta I$. $\square$

30.2 Only the odd augmentation is adverse

The even completion boundary is the positive rank-one operator

$$B_{\text{ev}} = r_{\text{ev}} \otimes r_{\text{ev}} \geq 0,$$

and the exact even decomposition is

$$\boxed{W_3^+ = S_{0,+} + B_{\text{ev}}.}$$

Consequently

$$\boxed{W_3^+ \geq 0, \quad W_3^+ + \eta I > 0 \quad (\eta > 0).}$$

By contrast, after the accepted odd block is eliminated, the odd complement is

$$F_{3,\eta}^\times = S_{\eta,-} - V_\eta V_\eta^*, \quad \text{rank } V_\eta \leq 5.$$

This is the only adverse finite-rank subtraction. Its Birman–Schwinger matrix is

$$B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta,$$

and its unnormalized five-label gap is

$$\mathcal{K}_\eta = \mathcal{G}_\eta - \mathcal{H}_\eta.$$

Theorem 30.2 (Single-sector seam reduction). *For every $\eta > 0$,*

$$\boxed{\begin{aligned} W_3 + \eta I \geq 0 &\iff F_{3,\eta}^\times \geq 0 \\ &\iff B_\eta \leq I \\ &\iff \mathcal{K}_\eta \geq 0. \end{aligned}}$$

Moreover,

$$\boxed{\mathfrak{k}_+(\eta) = 0, \quad \mathfrak{k}_\Sigma(\eta) = \mathfrak{k}_3(\eta) = N_+(B_\eta - I).}$$

Thus the completed seam integer is decided by the odd five-label sector alone.

Proof. The preceding theorem and the positive even boundary give $W_3^+ + \eta I > 0$. Parity and Feshbach reduction therefore leave only $F_{3,\eta}^\times$. The first Birman–Schwinger equivalence and inertia identity are [Theorem 23.3](#). The second finite matrix equivalence is the generalized inertia formulation of the exact critical-gap matrix. Since the even shifted restriction is positive, its strict threshold projection is zero and additivity of the seam charge reduces the global count to the odd contribution. $\square$

Remark 30.3 (Representation is not deletion). The theorem does not quotient out the even sector and does not assert that the full even and odd completed-Weil restrictions are unitarily identical. Both remain present on the target-faithful bilateral carrier. One-sector sufficiency comes from the common pure-source lower branch together with the favorable sign of the even boundary augmentation; it is a sign theorem, not a parity-identification shortcut.

The global endpoint problem is therefore not a pair of unrelated parity gates. It is the single unshifted five-dimensional inequality

$$\boxed{\mathcal{K}_0 \geq 0,}$$

whose exact endpoint meaning is developed in the next section.

31 Exact eta-zero closure and the five-dimensional endpoint criterion

The preceding sections establish the native completed-Weil representative, the prime-torus arithmetic intertwiner, target-faithful bilateral completion, the single seam integer, the closed node

$$\mathfrak{k}_\Sigma(0.01) = 0,$$

the favorable even sign and the exact eta family. The completed sign problem is therefore decided by one Hermitian matrix on the fixed five-label carrier.

31.1 Exact accepted support at eta zero

Let

$$E_3 = V_3 \oplus \text{span}\{e_3\}, \quad V_3 = \text{span}\{\phi_0, \phi_1, \phi_2\}.$$

In the basis $\Phi = (\phi_0, \phi_1, \phi_2, e_3)$, write

$$A_{3,\text{raw}} = \begin{pmatrix} G_3 & b_3 \\ b_3^* & q_3 \end{pmatrix}.$$

The exact finite-seam certificates give

$$\lambda_{\min}(G_3) > 2.08381771204 \times 10^{-13},$$

and

$$q_3 - b_3^* G_3^{-1} b_3 > 3.08520376681449 \times 10^{-13}.$$

Theorem 31.1 (Exact eta-zero accepted support). *The raw accepted Gram and its metric-orthonormal compression are strictly positive:*

$$\boxed{A_{3,\text{raw}} > 0, \quad A_3 > 0.}$$

Consequently

$$\boxed{\text{rank } A_3 = 4, \quad \ker A_3 = \{0\}.}$$

Proof. The Schur-complement criterion gives $A_{3,\text{raw}} > 0$. If $R = (\Phi^* M_X \Phi)^{-1/2}$, then $A_3 = R^* A_{3,\text{raw}} R$, and strict positivity is preserved by the invertible congruence. $\square$

The recentered cutoff value

$$\eta_c = 7.628008053037797 \times 10^{-8}$$

is therefore a finite-state diagnostic, not an eta-zero kernel of the native accepted block.

31.2 Odd source regularity and endpoint norm descent

In the logarithmic X_3 chart, the physical source corresponding to a native coordinate vector f is $w(x) = e^{-3|x|/2} f(x)$. Since $M_X \geq I$, Cauchy–Schwarz gives

$$\int_{\mathbb{R}} |x| |w(x)| dx \leq \frac{2}{3\sqrt{3}} \|f\|_{X_3}.$$

Every column of

$$\mathcal{W}_\Sigma = [r_{\text{odd}}, C_3]$$

is odd, and the unshifted source opens quadratically at the seam.

Theorem 31.2 (Odd inverse-half regularity). *The complete five-column source satisfies*

$$\boxed{\text{Ran } \mathcal{W}_\Sigma \subset \text{Dom}(S_{0,-}^{-1/2}).}$$

For $\eta > 0$, define

$$H_\eta = \mathcal{W}_\Sigma^* (S_{0,-} + \eta I)^{-1} \mathcal{W}_\Sigma,$$

and

$$H_0 = (S_{0,-}^{-1/2} \mathcal{W}_\Sigma)^* (S_{0,-}^{-1/2} \mathcal{W}_\Sigma).$$

Put

$$G_0 = \begin{pmatrix} 1 & 0 \\ 0 & A_3 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & I_{E_3} \end{pmatrix},$$

$$K_\eta = G_0 + \eta D - H_\eta, \quad K_0 = G_0 - H_0.$$

Theorem 31.3 (Odd five-label endpoint descent). *As $\eta \downarrow 0$,*

$$\boxed{\|H_\eta - H_0\| \longrightarrow 0, \quad \|K_\eta - K_0\| \longrightarrow 0.}$$

Proof. For $z \in \mathbb{C}^5$, spectral calculus gives

$$\langle z, (H_0 - H_\eta) z \rangle = \int_{(0,\infty)} \frac{\eta}{\lambda(\lambda + \eta)} d\mu_{\mathcal{W}_\Sigma z}(\lambda).$$

The dominating function $1/\lambda$ is integrable by [Theorem 31.2](#). Dominated convergence gives entrywise convergence, and finite dimensionality upgrades it to operator-norm convergence. The statement for K_η follows. $\square$

31.3 The endpoint criterion

Criterion 31.4 (Single-sector endpoint gap). The unshifted five-label Hermitian gap is nonnegative:

$$\mathcal{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & A_3 \end{pmatrix} - (S_{0,-}^{-1/2} \mathcal{W}_\Sigma)^* (S_{0,-}^{-1/2} \mathcal{W}_\Sigma) \geq 0.$$

This is the sole terminal sign condition. It is finite dimensional, unitary invariant and equivalent to the completed-Weil sign in the native carrier.

Theorem 31.5 (Exact single-sector endpoint closure). *The following statements are equivalent:*

- (i) $\mathcal{K}_0 \geq 0$;
- (ii) $F_{3,0}^\times \geq 0$;
- (iii) $W_3 \geq 0$;
- (iv) $W_\times \geq 0$.

Moreover, whenever these conditions hold,

$$\mathcal{K}_\eta \geq 0, \quad F_{3,\eta}^\times \geq 0 \quad (\eta > 0).$$

If

$$\ker \mathcal{K}_0 \cap \ker D \cap \ker \mathcal{W}_\Sigma = \{0\},$$

then the positive-shift inequalities are strict.

Proof. The endpoint generalized Schur theorem, together with [Theorems 31.1](#) and [31.2](#), gives (i) $\iff$ (ii). The even restriction is already nonnegative by [Theorems 30.1](#) and [30.2](#), so parity/Feshbach reconstruction gives (ii) $\iff$ (iii). The logarithmic chart is unitary and intertwines $W_\times$ with W_3 , proving (iii) $\iff$ (iv).

For positive shifts,

$$\mathcal{K}_\eta = \mathcal{K}_0 + \eta D + (H_0 - H_\eta),$$

and both eta-dependent summands are nonnegative. Strictness follows from the stated common-kernel condition. $\square$

Corollary 31.6 (Completed-carrier positivity). *The endpoint criterion $\mathcal{K}_0 \geq 0$ is equivalent to $W_\times \geq 0$.*

31.4 Classical completed-Weil criterion

Define

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s),$$

let

$$\mathcal{D}_\times = C_c^\infty(0, \infty), \quad d^\times r = \frac{dr}{r},$$

and use

$$(\mathcal{M}F)(\xi) = \int_0^\infty F(r) r^{-i\xi} d^\times r.$$

The completed explicit-formula quadratic form is

$$\mathcal{Q}_\xi(F) = \langle F, W_\times F \rangle_{\times,3}.$$

The classical Weil criterion in this normalization is [\[Wei, Tit\]](#)

$$\text{RH} \iff \mathcal{Q}_\xi(F) \geq 0 \quad (F \in \mathcal{D}_\times).$$

Theorem 31.7 (Five-dimensional completed-Weil criterion for RH). *Assume the classical explicit-formula identification and Weil criterion in the normalization above. Then*

$$\boxed{\text{RH} \iff \mathcal{K}_0 \geq 0.}$$

Proof. By Theorem 31.5, $\mathcal{K}_0 \geq 0$ is equivalent to $W_\times \geq 0$, hence to $\mathcal{Q}_\xi(F) \geq 0$ on $\mathcal{D}_\times$. The classical Weil criterion is exactly the equivalence with RH. $\square$

Thus the native framework identifies the Riemann hypothesis with one five-dimensional Hermitian sign condition. The direct outward, projection-defect, prime–Gamma reserve, rank-one seam-jet and full-odd boundary-scalar formulations in Section 32 are independent closure routes to this same criterion.

32 Equivalent closure formulations of the single endpoint gap

The terminal object of the paper is the unshifted five-label Hermitian gap

$$\mathcal{K}_0 = \mathcal{G}_0 - \mathcal{W}_\Sigma^* S_{0,-}^\dagger \mathcal{W}_\Sigma, \quad \mathcal{G}_0 = \text{diag}(1, A_3) > 0,$$

where

$$\mathfrak{E}_\Sigma = \mathbb{C}_\partial \oplus E_3, \quad \mathcal{W}_\Sigma = [r_{\text{odd}}, C_3].$$

By Theorem 31.2,

$$\text{Ran } \mathcal{W}_\Sigma \subset \text{Dom}(S_{0,-}^{-1/2}),$$

so the normalized source map

$$\boxed{T_\Sigma = S_{0,-}^{\dagger/2} \mathcal{W}_\Sigma \mathcal{G}_0^{-1/2}} \tag{EC0}$$

is bounded and finite rank. The normalized gap identity is

$$\boxed{\mathcal{G}_0^{-1/2} \mathcal{K}_0 \mathcal{G}_0^{-1/2} = I - T_\Sigma^* T_\Sigma.} \tag{EC1}$$

This section records several exact or independently sufficient ways of proving the same endpoint statement. They are alternative closures of one gate, not additional terminal gates.

32.1 Exact algebraic equivalences

Theorem 32.1 (Common-Gram equivalence of the endpoint gap). *The following statements are equivalent.*

- (i) $\mathcal{K}_0 \geq 0$.
- (ii) T_Σ is a contraction:

$$T_\Sigma^* T_\Sigma \leq I_{\mathfrak{E}_\Sigma}.$$

- (iii) There is a contraction

$$C_\Sigma : \mathfrak{E}_\Sigma \longrightarrow \overline{\text{Ran } S_{0,-}}$$

such that

$$\boxed{\mathcal{W}_\Sigma = S_{0,-}^{1/2} C_\Sigma \mathcal{G}_0^{1/2}.} \tag{EC2}$$

(iv) On the positive support of $S_{0,-}$, the block quadratic form

$$\boxed{\begin{pmatrix} \mathcal{G}_0 & \mathcal{W}_\Sigma^* \\ \mathcal{W}_\Sigma & S_{0,-} \end{pmatrix} \geq 0} \quad (\text{EC3})$$

is nonnegative on $\mathfrak{E}_\Sigma \oplus \text{Dom}(S_{0,-}^{1/2})$.

(v) There are a Hilbert space $\mathscr{Y}$ and isometries

$$J_{\text{tot}} : \mathfrak{E}_\Sigma \rightarrow \mathscr{Y}, \quad J_{\text{src}} : \overline{\text{Ran } S_{0,-}} \rightarrow \mathscr{Y}$$

such that

$$\boxed{T_\Sigma = J_{\text{src}}^* J_{\text{tot}}}. \quad (\text{EC4})$$

Proof. The equivalence of (i) and (ii) is (32.4). The support-space Douglas factorization lemma gives the equivalence of (ii) and (iii). The support-space Schur-complement criterion gives the equivalence of (i) and (iv). Finally, the overlap of two isometries is a contraction, giving (v) $\Rightarrow$ (ii), while the Halmos isometric dilation of a contraction gives (ii) $\Rightarrow$ (v). $\square$

Remark 32.2 (The no-free-isometry rule). The abstract implication (ii) $\Rightarrow$ (v) is an equivalence theorem, not an independent proof of the arithmetic sign. A source/response proof is noncircular only when J_{src} and J_{tot} are constructed from the native prime–Gamma–boundary response geometry before their overlap is compared with T_Σ .

32.2 Direct outward Hermitian closure

Proposition 32.3 (Outward five-dimensional certificate). *Let $\tilde{\mathcal{K}}_0$ be a Hermitian centre and suppose a directed operator-norm estimate gives*

$$\|\mathcal{K}_0 - \tilde{\mathcal{K}}_0\| \leq \varepsilon_0.$$

If

$$\boxed{\lambda_{\min}(\tilde{\mathcal{K}}_0) - \varepsilon_0 \geq 0}, \quad (\text{EC5})$$

then $\mathcal{K}_0 \geq 0$. A strict inequality proves $\mathcal{K}_0 > 0$.

Proof. For every unit vector z ,

$$\langle z, \mathcal{K}_0 z \rangle \geq \langle z, \tilde{\mathcal{K}}_0 z \rangle - \varepsilon_0 \geq \lambda_{\min}(\tilde{\mathcal{K}}_0) - \varepsilon_0.$$

$\square$

This is the shortest numerical closure route. The centre and every error term must come from the declared native source and metric data; a fitted positive matrix is not evidence for (EC5).

32.3 Projection-defect or source-readout closure

Let

$$\mathcal{R}_\Sigma : \mathfrak{E}_\Sigma \rightarrow \mathscr{Y}$$

be an independently constructed total-response synthesis satisfying

$$\mathcal{R}_\Sigma^* \mathcal{R}_\Sigma = \mathcal{G}_0,$$

and let J_{src} be the native source-response isometry with

$$P_{\text{src}} = J_{\text{src}} J_{\text{src}}^*.$$

Theorem 32.4 (Native projection-defect closure). *If the exact five-generator readout identity*

$$\boxed{J_{\text{src}}^* \mathcal{R}_\Sigma = S_{0,-}^{\dagger/2} \mathcal{W}_\Sigma} \quad (\text{EC6})$$

holds, then

$$\boxed{\mathcal{K}_0 = \mathcal{R}_\Sigma^* (I - P_{\text{src}}) \mathcal{R}_\Sigma \geq 0.} \quad (\text{EC7})$$

Consequently the single-sector endpoint hypothesis [Theorem 31.4](#) is closed.

Proof. Using [\(EC6\)](#),

$$\mathcal{W}_\Sigma^* S_{0,-}^\dagger \mathcal{W}_\Sigma = \mathcal{R}_\Sigma^* P_{\text{src}} \mathcal{R}_\Sigma.$$

Subtract this identity from $\mathcal{R}_\Sigma^* \mathcal{R}_\Sigma = \mathcal{G}_0$. □

The readout identity may equivalently be verified by source-range closure combined with the five-by-five cross-Gram identity. In the declared labelled basis this is a finite set of fifteen Hermitian pairings. Those pairings are a proof route only when both response constructions have independent native provenance.

32.4 Prime–Gamma parallel-reserve closure

Let C_Γ be the independently constructed Gamma-response overlap with the complete prime–Gamma mismatch source isometry. Put

$$D_{\Gamma|p} = I - C_\Gamma^* C_\Gamma \geq 0$$

and define the physical orientation residual

$$\Delta_\Gamma = T_\Sigma^* T_\Sigma - C_\Gamma^* C_\Gamma.$$

The positive operator $D_{\Gamma|p}$ is the normalized compression of the prime–Gamma parallel sum; it is the response energy omitted by the Gamma-only overlap.

Theorem 32.5 (Prime–Gamma reserve domination). *One has the exact identity*

$$\boxed{\mathcal{G}_0^{-1/2} \mathcal{K}_0 \mathcal{G}_0^{-1/2} = D_{\Gamma|p} - \Delta_\Gamma.} \quad (\text{EC8})$$

Therefore

$$\boxed{\mathcal{K}_0 \geq 0 \iff \Delta_\Gamma \leq D_{\Gamma|p}.} \quad (\text{EC9})$$

In particular, exact orientation matching $\Delta_\Gamma = 0$ is sufficient but is not necessary.

Proof. By definition,

$$I - T_\Sigma^* T_\Sigma = (I - C_\Gamma^* C_\Gamma) - (T_\Sigma^* T_\Sigma - C_\Gamma^* C_\Gamma).$$

Use [\(32.4\)](#) and the definitions of $D_{\Gamma|p}$ and Δ_Γ . □

Thus the arithmetic closure may be proved by showing that the physical orientation correction fits inside the positive reserve already carried by the prime channel. A Loewner enclosure

$$D_- - \Delta_+ \geq 0$$

with a lawful lower bound D_- and upper bound Δ_+ is sufficient.

32.5 Full-odd boundary scalar closure

Before the accepted odd packet is eliminated, the completion-boundary form has one adverse primitive. On X_3^- , reflection oddness gives

$$\ell_-(f) = -\ell_+(f), \quad \ell_\pm(f) = \widehat{f}(\pm i/2),$$

and therefore

$$\mathfrak{B}(f, f) = -2|\ell_+(f)|^2.$$

Define

$$L_\partial f = \sqrt{2}\ell_+(f), \quad b_\partial = L_\partial^* 1. \quad (\text{FB1})$$

Let $S_{0,-}^{\text{full}} \geq 0$ denote the full odd Gamma-prime source before the negative completion-boundary term. Then

$$\boxed{W_3^- = S_{0,-}^{\text{full}} - b_\partial b_\partial^*}. \quad (\text{FB2})$$

Theorem 32.6 (Full-odd boundary scalar reduction). *Assume*

$$b_\partial \in \text{Dom}((S_{0,-}^{\text{full}})^{\dagger/2}). \quad (\text{FB3})$$

Put

$$\boxed{\beta_\partial = \|(S_{0,-}^{\text{full}})^{\dagger/2} b_\partial\|^2}. \quad (\text{FB4})$$

Then

$$\boxed{\mathcal{K}_0 \geq 0 \iff W_3^- \geq 0 \iff \beta_\partial \leq 1}. \quad (\text{FB5})$$

Proof. On the positive support of $S_{0,-}^{\text{full}}$, put

$$u = (S_{0,-}^{\text{full}})^{\dagger/2} b_\partial.$$

Then

$$S_{0,-}^{\text{full}} - b_\partial b_\partial^* = (S_{0,-}^{\text{full}})^{1/2} (I - uu^*) (S_{0,-}^{\text{full}})^{1/2}.$$

The only nonzero eigenvalue of uu^* is $\|u\|^2 = \beta_\partial$, proving the second equivalence. The even restriction is nonnegative by the favorable sign theorem, and [Theorem 31.5](#) identifies the full odd sign with $\mathcal{K}_0 \geq 0$. $\square$

For $\eta > 0$, let

$$\beta_\partial(\eta) = \langle b_\partial, (S_{0,-}^{\text{full}} + \eta I)^{-1} b_\partial \rangle. \quad (\text{FB6})$$

The synchronized state packet gives

$$\beta_\partial(10^{-2}) = 0.873197367287103,$$

$$\beta_\partial(1.5625 \times 10^{-5}) = 0.999773667408462,$$

with directed residual charges below 3×10^{-19} and all six computed positive shifts strictly subcritical. At the deepest nodes,

$$\frac{1 - \beta_\partial(\eta)}{\eta} \approx 14.5.$$

The adjacent resolvent identities close at relative scale below 8×10^{-10} . These values are consistent with critical saturation $\beta_\partial(0) = 1$.

The endpoint-vector calibration also identifies the finite-state floor. The linear and quadratic extrapolants have source-residual metric-dual norms

$$2.8416809445 \times 10^{-6}, \quad 2.8424930268 \times 10^{-6},$$

respectively, so increasing polynomial order does not improve the endpoint vector. The synchronized packet therefore supplies strong critical-saturation calibration but no outward endpoint enclosure. The exact publication statement is the scalar equivalence (FB5).

32.6 Rank-one seam-jet Schur closure

The odd pure source opens quadratically at the seam, while every adverse source column is odd. In the quadratic-threshold decomposition, let

$$H_0 = \tau J_\Sigma^* J_\Sigma + R_{\text{reg}}, \quad A_{\text{reg}} = \mathcal{G}_0 - R_{\text{reg}}.$$

Then

$$\mathcal{K}_0 = A_{\text{reg}} - \tau J_\Sigma^* J_\Sigma.$$

The scalar odd threshold fibre gives

$$\text{rank } J_\Sigma \leq 1.$$

Theorem 32.7 (Seam-jet Birman–Schwinger closure). *Assume $A_{\text{reg}} > 0$. Then*

$$\boxed{\mathcal{K}_0 \geq 0 \iff \tau J_\Sigma A_{\text{reg}}^{-1} J_\Sigma^* \leq I.} \quad (\text{EC10})$$

If $J_\Sigma \neq 0$, write $J_\Sigma z = \langle j, z \rangle$. Since the threshold fibre is scalar, (EC10) becomes the single inequality

$$\boxed{\tau \langle j, A_{\text{reg}}^{-1} j \rangle \leq 1.} \quad (\text{EC11})$$

Equality produces at most one endpoint critical direction.

Proof. Factor

$$\mathcal{K}_0 = A_{\text{reg}}^{1/2} \left(I - \tau A_{\text{reg}}^{-1/2} J_\Sigma^* J_\Sigma A_{\text{reg}}^{-1/2} \right) A_{\text{reg}}^{1/2}.$$

The nonzero spectra of $A_{\text{reg}}^{-1/2} J_\Sigma^* J_\Sigma A_{\text{reg}}^{-1/2}$ and $J_\Sigma A_{\text{reg}}^{-1} J_\Sigma^*$ coincide. Rank at most one then reduces the final operator inequality to (EC11). $\square$

Equivalently, in a jet-adapted basis

$$U_J^* A_{\text{reg}} U_J = \begin{pmatrix} a_J & b_J^* \\ b_J & C_J \end{pmatrix},$$

with $C_J > 0$, one has

$$\mathcal{K}_0 \geq 0 \iff \boxed{\delta_J = a_J - \tau \|j\|^2 - b_J^* C_J^{-1} b_J \geq 0.} \quad (\text{EC12})$$

This is a four-dimensional regular-block certificate followed by one scalar Schur decision.

32.7 One endpoint, several lawful proofs

The direct Hermitian enclosure, native projection-defect identity, prime–Gamma reserve domination, full-odd boundary scalar and rank-one seam-jet Schur test are not competing terminal statements. Each proves the same condition

$$\mathcal{K}_0 \geq 0.$$

Once any one of them is established with its stated native provenance and analytic hypotheses, [Theorems 31.5](#) and [31.7](#) give

$$\boxed{\mathcal{K}_0 \geq 0 \implies W_{\times} \geq 0 \implies \text{RH.}}$$

The remaining representations—seam integer, spectral flow, Morse index, threshold zeta, prime-torus compression and threshold holonomy—then provide exact invariant confirmations of the same closure rather than additional proof obligations.

33 Structural soundness and single-node square-root transfer

The preceding endpoint section identifies the completed-Weil sign with the single Hermitian condition

$$\mathcal{K}_0 \geq 0.$$

This section separates four logically different kinds of statement that occur in the proof architecture:

$$\boxed{\text{exact equivalence} \neq \text{sufficient certificate} \neq \text{numerical calibration} \neq \text{interpretive route.}}$$

The separation is essential. An exact reformulation does not by itself prove a sign, while a numerical centre does not become an outward certificate merely by approaching the expected endpoint.

33.1 Logical hierarchy of the endpoint

The exact common-Gram theorem gives

$$\mathcal{K}_0 \geq 0 \iff T_{\Sigma}^* T_{\Sigma} \leq I \iff \begin{pmatrix} \mathcal{G}_0 & \mathcal{W}_{\Sigma}^* \\ \mathcal{W}_{\Sigma} & S_{0,-} \end{pmatrix} \geq 0,$$

together with the support-space Douglas factorization and common-isometric overlap formulations. These are exact equivalences. In particular, the abstract Halmos dilation of a contraction is not an arithmetic proof of the contraction; it is the converse direction of an equivalence theorem.

The manuscript then supplies six lawful sufficient certificates for the same endpoint:

- (a) a direct outward enclosure of $\mathcal{K}_0$;
- (b) an independently constructed source-readout projection defect;
- (c) prime–Gamma reserve domination;
- (d) the structured single-node square-root transfer proved below;
- (e) the rank-one seam-jet Schur inequality;
- (f) the pre-Feshbach full-odd boundary scalar inequality.

A passing instance of any one certificate gives $\mathcal{K}_0 \geq 0$. They are not cumulative assumptions.

The fixed-shift packets, scalar ladders, seam-jet rows and endpoint-vector extrapolants are calibration data until accompanied by their declared outward operator, source, moment and rounding bounds. They test the architecture and quantify the available budget, but they do not replace a certificate theorem.

33.2 The square-root endpoint remainder

For $\eta > 0$, write

$$\Delta_\eta = \mathcal{H}_0 - \mathcal{H}_\eta \geq 0.$$

Assume that, on a local source fibre $|\xi| < \delta$,

$$S_{0,-} = M_s, \quad s(\xi) \geq c\xi^2, \quad c > 0,$$

and that the five-column source has the odd Taylor expansion

$$(\mathcal{W}_\Sigma z)_{\text{loc}}(\xi) = \xi J_\Sigma z + \xi^2 R(\xi) z, \quad \text{ess sup}_{|\xi| < \delta} \|R(\xi)\| \leq M.$$

On the far source range assume

$$S_{0,-} \geq s_{\text{far}} I, \quad s_{\text{far}} > 0.$$

Put

$$q = \delta \sqrt{c/\eta},$$

$$A_J(\eta, \delta, c) = \frac{2\sqrt{\eta}}{c^{3/2}} \arctan q, \tag{SR1}$$

and

$$A_R(\eta, \delta, c) = \frac{2\eta\delta}{c^2} - \frac{2\eta^{3/2}}{c^{5/2}} \arctan q. \tag{SR2}$$

Both coefficients are nonnegative.

Theorem 33.1 (Rank-one square-root endpoint transfer). *For every $\alpha > 0$,*

$$\boxed{0 \leq \Delta_\eta \leq (1 + \alpha) A_J J_\Sigma^* J_\Sigma + (1 + \alpha^{-1}) A_R M^2 I + \frac{\eta}{s_{\text{far}}} \mathcal{H}_\eta.} \tag{SR3}$$

In the scalar odd threshold fibre,

$$\text{rank } J_\Sigma \leq 1.$$

Thus the singular endpoint correction is rank at most one and has order $\sqrt{\eta}$, while the regular and far corrections have order η for fixed δ and s_{far} .

Proof. Spectral calculus gives

$$\Delta_\eta = \mathcal{W}_\Sigma^* \frac{\eta}{S_{0,-}(S_{0,-} + \eta I)} \mathcal{W}_\Sigma.$$

On the local fibre, apply

$$(a + b)^*(a + b) \leq (1 + \alpha) a^* a + (1 + \alpha^{-1}) b^* b$$

to the jet and Taylor-remainder terms. The jet integral is bounded by

$$\int_{-\delta}^{\delta} \frac{\eta}{c(c\xi^2 + \eta)} d\xi = A_J,$$

while the remainder integral is bounded by

$$\int_{-\delta}^{\delta} \frac{\eta\xi^2}{c(c\xi^2 + \eta)} d\xi = A_R.$$

On the far spectral range,

$$S^{-1} - (S + \eta I)^{-1} = \frac{\eta}{S}(S + \eta I)^{-1} \leq \frac{\eta}{s_{\text{far}}}(S + \eta I)^{-1}.$$

Compression by the far source columns is dominated by the full shifted Gram $\mathcal{H}_\eta$. Combining the three estimates proves (SR3). $\square$

33.3 Outward jet and regular-remainder control

The native second-moment estimate gives

$$\int_{\mathbb{R}} x^2 e^{-3|x|/2} |f(x)| dx \leq \frac{4}{9} \|f\|_{X_3}.$$

Consequently, if

$$\|\mathcal{W}_\Sigma\| \leq W_+,$$

then the regular Taylor coefficient may be bounded by

$$M \leq \frac{2}{9} W_+. \quad (\text{SR4})$$

Suppose an outward seam-jet packet gives a centre row $\tilde{J}_\Sigma$ and

$$\|J_\Sigma - \tilde{J}_\Sigma\| \leq \varepsilon_J.$$

For every $\beta > 0$,

$$J_\Sigma^* J_\Sigma \leq (1 + \beta) \tilde{J}_\Sigma^* \tilde{J}_\Sigma + (1 + \beta^{-1}) \varepsilon_J^2 I. \quad (\text{SR5})$$

This preserves the rank-one centre while charging the continuum moment-transfer error as a scalar remainder.

33.4 One outward matrix

Let one positive-shift packet satisfy the directed bounds

$$\mathcal{G}_\eta \geq \underline{\mathcal{G}}_\eta, \quad \mathcal{H}_\eta \leq \overline{\mathcal{H}}_\eta,$$

and put

$$\underline{\mathcal{K}}_\eta = \underline{\mathcal{G}}_\eta - \overline{\mathcal{H}}_\eta.$$

Define

$$\begin{aligned} \mathcal{L}_0 &= \underline{\mathcal{K}}_\eta - \eta D - \frac{\eta}{s_{\text{far}}} \overline{\mathcal{H}}_\eta \\ &\quad - (1 + \alpha) A_J \left[(1 + \beta) \tilde{J}_\Sigma^* \tilde{J}_\Sigma + (1 + \beta^{-1}) \varepsilon_J^2 I \right] \\ &\quad - (1 + \alpha^{-1}) A_R \left(\frac{2}{9} W_+ \right)^2 I. \end{aligned} \quad (\text{SR6})$$

Theorem 33.2 (Single-node square-root endpoint certificate). *If*

$$\boxed{\mathcal{L}_0 \geq 0}, \quad (\text{SR7})$$

then

$$\boxed{\mathcal{K}_0 \geq 0}.$$

A strict inequality in (SR7) gives $\mathcal{K}_0 > 0$.

Proof. The exact endpoint identity is

$$\mathcal{K}_0 = \mathcal{K}_\eta - \eta D - \Delta_\eta.$$

Insert the directed shifted matrices, the square-root estimate (SR3), the moment bound (SR4) and the uncertain-jet estimate (SR5). The result is

$$\mathcal{K}_0 \geq \mathcal{L}_0.$$

The conclusion follows. $\square$

The theorem is exact. A state-locked fusion becomes a proof-level certificate only when its four continuum quantities—the $\mathcal{G}_\eta$ transfer, $\mathcal{H}_\eta$ transfer, five-source-column transfer and seam-jet transfer—are supplied as outward bounds together with the matrix rounding guard. A finite-grid centre with those values entered as zero is calibration rather than promotion.

33.5 One-page terminal implication

Corollary 33.3 (A passing single-node certificate implies the classical target). *Assume (SR7). Then*

$$\mathcal{K}_0 \geq 0,$$

$$\mathcal{K}_\eta = \mathcal{K}_0 + \eta D + (\mathcal{H}_0 - \mathcal{H}_\eta) \geq 0 \quad (\eta > 0),$$

and hence

$$F_{3,\eta}^\times \geq 0, \quad F_{3,0}^\times \geq 0.$$

The favorable even restriction gives $W_3^+ \geq 0$, so

$$W_3 \geq 0.$$

Unitary logarithmic transport yields

$$W_\times \geq 0.$$

Under the cited completed explicit-formula identification and classical Weil criterion,

$$\text{RH.}$$

Proof. The first implication is Theorem 33.2. The positive-shift identity has nonnegative increments. Endpoint Feshbach equivalence, the favorable even sign and the logarithmic unitary intertwiner give the next three implications. The last implication is the classical Weil criterion in the normalization fixed in Section 31.4. $\square$

33.6 Excluded circular inference

The abstract cut and joint projectors exclude a nonzero vector that belongs to both pure recognition classes. To apply that exclusion to the endpoint, one needs a separate native theorem placing every actual endpoint obstruction in both classes on the same completed carrier. The bilateral reconstruction retains cut memory and is not the memory-erasing geometric join. Therefore the chain

$$\text{threshold obstruction} \implies \text{cut recognized and complete-joint recognized} \implies 0$$

is not used as an endpoint proof in this manuscript.

Likewise, an abstract common isometric dilation is not used to establish the arithmetic contraction. The manuscript promotes positivity only through an independently derived native identity or a directed certificate satisfying its stated hypotheses. This keeps the endpoint criterion, its certificates and its numerical shadows in their correct logical order.

34 Certified cut–joint recognition, continuum descent, and holonomy closure

The cut–joint programme, the continuum transport theorem and the relevant holonomy theorems are established components of the native recognition architecture. They must not be compressed into the statement that a continuum theorem is absent. The only logically distinct interface is the identification of the already constructed continuum response and holonomy carriers with the physical endpoint source-readout orientation.

34.1 Exact cut–joint exclusion

Let

$$H_1(S^1; \mathbb{Z}) = \mathbb{Z}\gamma_\pi, \quad H_1(T^2; \mathbb{Z}) = \mathbb{Z}\gamma_\pi \oplus \mathbb{Z}\mu_{\text{cut}}.$$

In the ordered basis $(\gamma_\pi, \mu_{\text{cut}})$, put

$$P_{\text{joint}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_{\text{cut}} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$P_{\text{joint}} + P_{\text{cut}} = I, \quad P_{\text{joint}}P_{\text{cut}} = P_{\text{cut}}P_{\text{joint}} = 0.$$

Theorem 34.1 (Certified cut–joint exclusion). *If a state u is simultaneously a pure cut mode and a pure joint mode,*

$$P_{\text{cut}}u = u, \quad P_{\text{joint}}u = u,$$

then $u = 0$.

Proof. The two equalities imply

$$u = P_{\text{cut}}P_{\text{joint}}u = 0.$$

□

Let $J = [1, 0]$ be the geometric join and $S = [1, 0]^T$ its uncut-circle section. Then

$$JS = I_1, \quad SJ = P_{\text{joint}} \neq I_2,$$

and

$$\boxed{I_2 - SJ = P_{\text{cut}}}. \quad (\text{CJ1})$$

Thus return of the visible circle coordinate does not imply return of the full post-cut state. The exact defect is the surviving cut-memory coordinate.

34.2 Continuum descent and path holonomy

Let $A : [0, 1] \rightarrow \mathfrak{su}(2)$ be continuous and let

$$\dot{U}(t) = A(t)U(t), \quad U(0) = I.$$

For midpoint samples $t_k = (k + 1/2)/N$, define

$$U_N = \exp\left(\frac{1}{N}A(t_{N-1})\right) \cdots \exp\left(\frac{1}{N}A(t_0)\right).$$

Theorem 34.2 (Continuum holonomy descent). *The product holonomies converge to the path-ordered continuum transport,*

$$\boxed{U_N \longrightarrow U(1).}$$

The endpoint holonomy is invariant under orientation-preserving C^1 reparameterization, changes to $U(1)^{-1}$ under orientation reversal, and is covariant under constant unitary gauge conjugation. The associated continuum length is additive over interval cuts.

Proof. The midpoint step functions converge uniformly to A . Stability of the linear matrix ODE, obtained from its integral equation and Gronwall's inequality, gives endpoint convergence. Reparameterization follows by composing the solution with the parameter map. Orientation reversal and constant-gauge covariance follow directly from the transformed differential equations, while length additivity is the additivity of the defining integral. $\square$

The continuum programme proceeds beyond raw convergence. On the constrained path class preserving hierarchy itinerary, winding lift, memory, seam class and lifted curvature flux, the groupoid laws and a universal variational lower bound are established. On a compact regular strip, write

$$g = dx^2 + A(x)^2 dy^2, \quad \omega = -A'(x) dy, \quad \Phi(C) = \int_C \omega.$$

Put

$$B_D = \max_D \frac{|A'|}{A}, \quad M = \max_D A, \quad c_D = \min\{1, \min_D A\}, \quad x_0 = \min_D x > 0.$$

The constrained length infimum is attained, the seam is geodesic, Stokes identifies connection and curvature flux, and the geometric no-collapse bounds

$$L_g(C) \geq \frac{|\Phi(C)|}{B_D}$$

and, for positively oriented simple loops,

$$L_g(C)^2 \geq \frac{4\pi c_D^2 x_0}{M} |\Phi(C)|$$

hold. Hence identity principal holonomy does not erase the lifted curvature ledger or permit a protected nonzero-flux sector to collapse.

34.3 Exact eta and threshold holonomies

The fixed five-label eta family has eta-independent polar frames. By [Theorem 27.1](#), every based normalization-eta rectangle, including those with eta-zero endpoints, has exact holonomy

$$\boxed{H_\square = I_5.} \tag{CJ2}$$

The user-local UGD fixed-five loop and the actual eta-normalization rectangle provide independent finite calibrations of this exact theorem.

The sign-sensitive holonomy is the canonical threshold unitary

$$\mathcal{H}_\Sigma^{\text{thr}}(\eta) = I - 2\mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta).$$

By [Theorem 28.1](#),

$$\boxed{\mathcal{H}_\Sigma^{\text{thr}}(\eta) = I \iff \mathfrak{k}_\Sigma(\eta) = 0 \iff B_\eta \leq I.} \tag{CJ3}$$

Thus both the structural eta holonomy and the threshold holonomy theorem are already proved. They have different roles: the first proves path independence, while the second detects the adverse seam projection.

34.4 Gamma-continuum response isometry

The continuum response construction also reaches the accepted endpoint packet. Let $U_t = e^{-itE}$ be the Euler group, let $\mathcal{C}_H$ be the compatible channel isometry, let $\iota_3 : E_3 \hookrightarrow \mathcal{H}_-$, and define

$$(\Theta_\Gamma z)(t) = \rho(t)^{1/2} \mathcal{C}_H(U_{t/2} - I) \iota_3 z$$

on a positive-length Gamma interval with $\rho > 0$ almost everywhere. The Euler-continuum separation theorem gives

$$\ker \Theta_\Gamma = \{0\}, \quad M_\Gamma = \Theta_\Gamma^* \Theta_\Gamma > 0.$$

Therefore

$$J_\Gamma = \Theta_\Gamma M_\Gamma^{-1/2}$$

is a canonical four-column response isometry. Together with the independently normalized boundary response, it gives

$$J_{\text{tot}} = J_\partial \oplus J_\Gamma, \quad J_{\text{tot}}^* J_{\text{tot}} = I_5.$$

The continuum response carrier and its accepted isometry are therefore constructed rather than postulated.

34.5 Finite native mixed-response intertwiner

For the pinned critical response, write

$$r_c = T_c c, \quad j_c = T_c^{\text{corr}} c, \quad \Psi_c = (r_c, j_c),$$

and introduce the Hadamard coordinates

$$g_c = \frac{r_c + j_c}{\sqrt{2}}, \quad m_c = \frac{r_c - j_c}{\sqrt{2}}.$$

The geometric component is g_c and the surviving cut-memory component is m_c .

Proposition 34.3 (Finite mixed-response return identity). *On the finite doubled response carrier,*

$$\boxed{u - SJu = P_{\text{cut}} u.} \tag{CJ4}$$

Consequently complete state return is equivalent to zero cut memory. The actual critical response need not be a pure cut mode or a pure joint mode.

The user-local state-locked packet certified the finite response Hadamard intertwiner, the pinned nonzero cut-memory component and the nullity of the canonical corrected image. It measured equal joint and memory norms,

$$\|g_c\| = \|m_c\| = 4.137400474477719 \times 10^{-5},$$

relative cut memory $2^{-1/2}$, and a nonzero full-state return defect. The finite pair was therefore mixed rather than pure. This result refines the abstract pure-sector theorem; it does not refute or supersede it.

34.6 The exact endpoint adapter

The distinct statement is no longer a continuum-existence theorem and no longer a holonomy theorem. It is the native adapter identifying the physical endpoint projection defect with the already constructed continuum response and recognition carriers. One equivalent form is the five-generator source readout

$$\boxed{J_{\text{src}}^* \mathcal{R}_\Sigma = S_{0,-}^{\dagger/2} \mathcal{W}_\Sigma.} \quad (\text{CJ5})$$

Under (CJ5),

$$\mathcal{K}_0 = \mathcal{R}_\Sigma^* (I - P_{\text{src}}) \mathcal{R}_\Sigma \geq 0.$$

A geometric UGD representation of the same sign object is obtained once the already defined UGD terminal loop is identified with the canonical threshold unitary by

$$\text{Hol}_{\text{UGD}}(\gamma_\eta) = U_\eta^* \mathcal{K}_\Sigma^{\text{thr}}(\eta) U_\eta. \quad (\text{CJ6})$$

Equations (CJ5) and (CJ6) are adapter identities between already constructed objects. They must not be described as the absence of continuum transport, continuum response, eta holonomy or threshold holonomy.

Correct lineage. The abstract cut-joint exclusion is proved exactly. Continuum path transport, product-holonomy convergence, constrained curvature-flux no-collapse, exact eta path independence, threshold-holonomy detection and Gamma-continuum response isometry are all established at their declared scopes. The finite native mixed-response intertwiner and surviving cut memory have user-local passing evidence. The final interface is the exact endpoint adapter (CJ5), or an equivalent mixed event-Gram identity; the optional UGD realization uses (CJ6). It is therefore inaccurate to describe either the continuum route or the holonomy route as unfinished in general.

35 Conclusion

This paper constructs a native completed-Weil geometry in which the arithmetic, operator, spectral and topological formulations of the sign problem are faithful representations of one object. The multiplicative carrier is generated by inversion and Euler-scale flow before the logarithmic chart is introduced, and the boundary-Gamma-prime form has a bounded self-adjoint representative $W_\times$.

The arithmetic interface is exact. The scalar circle supplies the calibration for spectral flow, winding and Toeplitz index, while the arithmetic Bohr transform carries the logarithmic clock to the weighted prime-torus generator and multiplication to monomial shifts. After faithful transport of the native source data, the prime-torus compression and the native Feshbach compression are the same rank-at-most-five Birman-Schwinger matrix.

Bilateral target completion retains both parity sectors, their cross seam and target-relevant blind directions. The common two-sheet source theorem gives

$$S_{0,+} \geq 0, \quad S_{0,-} \geq 0, \quad S_{\eta,\pm} \geq \eta I.$$

The even boundary augmentation is positive, hence

$$W_3^+ \geq 0, \quad \mathfrak{k}_+(\eta) = 0.$$

Therefore the global seam integer is exactly the odd five-label count:

$$\boxed{\mathfrak{k}_\Sigma(\eta) = \mathfrak{k}_3(\eta) = N_+(B_\eta - I).}$$

The bilateral carrier is preserved, while the terminal sign decision is single-sector.

The fixed native node is closed quantitatively:

$$\boxed{\lambda_{\max}(B_{0.01}) \leq 0.8388901637561339 < 1,}$$

with reserve 0.1611098362438661. Exact Schur certificates prove

$$A_3 > 0, \quad \ker A_3 = \{0\},$$

so the small negative cutoff eigenvalue is not a native eta-zero critical line. The weighted first-moment theorem places the complete five-source range in $\text{Dom}(S_{0,-}^{-1/2})$, and the five-label resolvent and gap converge in operator norm at eta zero.

The terminal object is

$$\boxed{\mathcal{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & A_3 \end{pmatrix} - (S_{0,-}^{-1/2} \mathcal{W}_\Sigma)^* (S_{0,-}^{-1/2} \mathcal{W}_\Sigma).}$$

The exact decomposition

$$\mathcal{K}_\eta = \mathcal{K}_0 + \eta D + (H_0 - H_\eta)$$

has nonnegative eta-dependent increments. The endpoint Schur theorem and single-sector reconstruction give

$$\mathcal{K}_0 \geq 0 \iff F_{3,0}^\times \geq 0 \iff W_3 \geq 0 \iff W_\times \geq 0.$$

Combining this equivalence with the classical completed-Weil criterion yields

$$\boxed{\text{RH} \iff \mathcal{K}_0 \geq 0.}$$

The endpoint has six independent certificate formulations. A direct outward computation may enclose the least eigenvalue of $\mathcal{K}_0$. If the provenance-derived total response satisfies the native source-readout identity

$$J_{\text{src}}^* \mathcal{R}_\Sigma = S_{0,-}^{-1/2} \mathcal{W}_\Sigma,$$

then the exact projection-defect theorem gives

$$\mathcal{K}_0 = \mathcal{R}_\Sigma^* (I - P_{\text{src}}) \mathcal{R}_\Sigma \geq 0.$$

The event-resolved overlap formula fixes the left-hand side through the source, response and mixed event Grams; identifying that forced overlap with the physical source map is the orientation adapter, not a consequence of diagonal Gram agreement. The prime-Gamma geometry gives the alternative reserve condition

$$\Delta_\Gamma \leq D_{\Gamma|p}.$$

The structured single-node transfer controls the endpoint resolvent difference by a rank-one seam term of order $\sqrt{\eta}$ and regular and far terms of order η . Its outward matrix $\mathcal{L}_0$ satisfies

$$\mathcal{K}_0 \geq \mathcal{L}_0,$$

so

$$\boxed{\mathcal{L}_0 \geq 0 \implies \mathcal{K}_0 \geq 0.}$$

This theorem replaces a blind cofinal eta ladder by one anisotropic 5×5 certificate with explicit source, jet, continuum and rounding errors.

The quadratic-threshold seam calculus gives the rank-one condition

$$\tau J_{\Sigma} A_{\text{reg}}^{-1} J_{\Sigma}^* \leq I,$$

or one jet-adapted Schur scalar. Finally, before accepted-space Feshbach deflation, reflection oddness makes the completion boundary a single negative rank-one form,

$$W_3^- = S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial}, \quad L_{\partial} f = \sqrt{2} \hat{f}(i/2).$$

Under inverse-support regularity,

$$\boxed{\mathcal{K}_0 \geq 0 \iff W_3^- \geq 0 \iff \beta_{\partial} := \left\| (S_{0,-}^{\text{full}})^{\dagger/2} L_{\partial}^* 1 \right\|^2 \leq 1.}$$

The synchronized positive-shift scalar packet gives

$$\beta_{\partial}(10^{-2}) = 0.873197367287103, \quad \beta_{\partial}(1.5625 \times 10^{-5}) = 0.999773667408462,$$

and all six computed shifts are strictly subcritical. The deepest reserve satisfies

$$\frac{1 - \beta_{\partial}(\eta)}{\eta} \approx 14.5,$$

consistent with critical endpoint saturation. Linear and quadratic endpoint-vector extrapolants have source-residual metric-dual norms $2.8416809445 \times 10^{-6}$ and $2.8424930268 \times 10^{-6}$, respectively. The failure of higher order to improve the residual identifies the present finite-state floor, so the saturation is retained as numerical evidence rather than an endpoint certificate.

The wider recognition-response architecture is retained in the repository as a separate recovery ledger and technical source set. It includes the exact information-covariance, flux-completion, recursive-memory, event-overlap, Euler-path, Hodge and typed-Gram theorems at their displayed hypotheses. Those results are not discarded, but they are kept outside the compiled release until each appendix passes an independent LaTeX and reference audit.

The manuscript records its logical hierarchy explicitly. Common-Gram, block-form, factorization and common-isometry statements are exact equivalences. The six routes above are sufficient certificate architectures. State-locked packets are calibration until their outward error contracts are supplied. The certified cut-joint, continuum and holonomy theorems are preserved at their declared scopes, while their endpoint use remains tied to the displayed native adapter rather than inferred from diagonal data.

The framework is complete at the level of carrier, arithmetic transport, target faithfulness, parity reduction, fixed-shift certification, eta-zero support, exact endpoint equivalence and certificate architecture. The Riemann hypothesis is identified with one explicit Hermitian sign criterion, and every spectral, Morse, zeta, prime-torus, holonomy, square-root-transfer and full-odd boundary-scalar avatar addresses the same seam obstruction at the scope established above.

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